

## Project Data Page:





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## **DESIGN CALCULATION**

*In Accordance with ASME Section VIII Division 2*

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-V-310010peR2-7%.pvdb

Date of Analysis : Jul 23, 2026 2:29pm

**PV Elite 28 U2, April 2026**



### Vessel Design Summary:

#### ASME Code, Section VIII Division 2, 2025

Diameter Spec : 2656.000 x 2400.000 x 2400.000 mm. ID  
 Vessel Design Length, Tangent to Tangent 6800.00 mm.  
 Distance of Bottom Tangent above Grade 3300.00 mm.  
 Distance of Base above Grade 300.00 mm.  
 Specified Datum Line Distance 0.00 mm.  
 Internal Design Temperature 80 °C  
 Internal Design Pressure 160.000 bars  
 External Design Temperature 0 °C  
 Maximum Allowable Working Pressure 162 bars  
 Shop Test Pressure 217.000 bars  
 Required Minimum Design Metal Temperature -33.0 °C  
 Warmest Computed Minimum Design Metal Temperature -33.0 °C  
 Wind Design Code IS 875 (Part 3):2015 Amd. 1&2 2020  
 Earthquake Design Code Response Spectrum

#### Materials of Construction:

| Component Type | Material   | Class | Thickness | UNS #  | Material Treatment Options |
|----------------|------------|-------|-----------|--------|----------------------------|
| Shell          | SA-516 70  | ...   | ...       | K02700 | +IT                        |
| Head           | SA-516 70  | ...   | ...       | K02700 | +IT                        |
| Skirt          | SA-285 C   | ...   | t <= 50   | K02801 |                            |
| Skirt          | SA-516 70  | ...   | ...       | K02700 |                            |
| Nozzle         | SA-350 LF2 | 1     | ...       | K03011 | +IT                        |
| Nozzle Flg     | SA-350 LF2 | 1     | ...       | K03011 |                            |
| Basing         | SA-285 C   | ...   | t <= 50   | K02801 |                            |
| Base Bolting   | SA-36      | ...   | ...       | K02600 |                            |

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized  
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

#### Element Pressures and MAWP (bars & mm.):

| Element Description or Type | Design Pressure + Stat. head | Ext. Press. | Element M.A.W.P | Total Corrosion Allowance | Str. Flg. Gov. | In Creep Range |
|-----------------------------|------------------------------|-------------|-----------------|---------------------------|----------------|----------------|
| Skirt 3                     | ...                          | ...         | ...             | 1.5000                    | N/A            | No             |
| Skirt 2                     | ...                          | ...         | ...             | 1.5000                    | N/A            | No             |
| Top Skirt -Yring            | ...                          | ...         | ...             | 1.5000                    | N/A            | No             |
| Bottom D end                | 160.329                      | 0.00        | 162.000         | 1.5000                    | N/A            | No             |
| Shell 1                     | 160.252                      | 0.00        | 162.000         | 1.5000                    | N/A            | No             |
| Shell 2                     | 160.127                      | 0.00        | 162.000         | 1.5000                    | N/A            | No             |
| Shell 3                     | 160.002                      | 0.00        | 162.000         | 1.5000                    | N/A            | No             |
| Top D end                   | 160.002                      | 0.00        | 163.000         | 1.5000                    | N/A            | No             |

Liquid Level: 5232.00 mm. Dens.: 0.001 kg./cm<sup>3</sup> Sp. Gr.: 0.636

#### Element Types and Properties:

| Element Type | "To" Elev mm. | Element Length mm. | Nominal Thickness mm. | Finished Thickness mm. | Reqd Thk Internal mm. | Reqd Thk External mm. | Long Eff | Circ Eff |
|--------------|---------------|--------------------|-----------------------|------------------------|-----------------------|-----------------------|----------|----------|
| Skirt        | 2000.0        | 2000.0             | 30.0                  | 30.0                   | ...                   | ...                   | 0.70     | 0.70     |
| Skirt        | 2500.0        | 500.0              | 30.0                  | 30.0                   | ...                   | ...                   | 0.70     | 0.70     |



|          |        |        |       |       |       |     |      |      |
|----------|--------|--------|-------|-------|-------|-----|------|------|
| Skirt    | 3000.0 | 500.0  | 30.0  | 30.0  | ...   | ... | 0.70 | 0.55 |
| Sphere   | 3000.0 | 0.0    | 74.0  | 65.0  | 64.1  | ... | 1.00 | 1.00 |
| Cylinder | 5000.0 | 2000.0 | 128.0 | 128.0 | 126.5 | ... | 1.00 | 1.00 |
| Cylinder | 7500.0 | 2500.0 | 128.0 | 128.0 | 126.4 | ... | 1.00 | 1.00 |
| Cylinder | 9800.0 | 2300.0 | 128.0 | 128.0 | 126.3 | ... | 1.00 | 1.00 |
| Sphere   | 9800.0 | 0.0    | 74.0  | 65.0  | 64.0  | ... | 1.00 | 1.00 |

#### Local Stress Analysis Results:

| Description | Analysis Type | Max Stress Ratio | High Stress Location | Pass Fail |
|-------------|---------------|------------------|----------------------|-----------|
| N3_2"       | WRC-297       | 1.001            | Shell                | Failed    |
| N1_18"      | WRC-297       | 1.001            | Shell                | Failed    |
| N2_16"      | WRC-107/537   | 0.996            | n/a                  | Passed    |

#### Wind/Earthquake Shear, Bending:

| From | To | Distance to Support mm. | Cumulative Wind Shear Kg-f | Earthquake Shear Kg-f | Wind Bending Kg-m. | Earthquake Bending Kg-m. |
|------|----|-------------------------|----------------------------|-----------------------|--------------------|--------------------------|
| 10   | 20 | 1000                    | 10174.4                    | 17243.5               | 67883.6            | 129018                   |
| 20   | 30 | 2250                    | 9161.88                    | 17164.4               | 48546.9            | 94708.8                  |
| 30   | 40 | 2750                    | 8908.74                    | 17123.2               | 44029.2            | 86187.5                  |
| 40   | 50 | 2999.98                 | 8655.6                     | 16809.6               | 39638.3            | 77697.4                  |
| 50   | 60 | 4000                    | 8655.58                    | 15798.7               | 39638              | 77670.9                  |
| 60   | 70 | 6250                    | 6891.1                     | 12445.4               | 24091              | 46202.3                  |
| 70   | 80 | 8650                    | 4619.52                    | 6574.17               | 9702.45            | 15123.2                  |
| 80   | 90 | 9799.99                 | 2524.84                    | 1489.64               | 1486.34            | 2.36484                  |

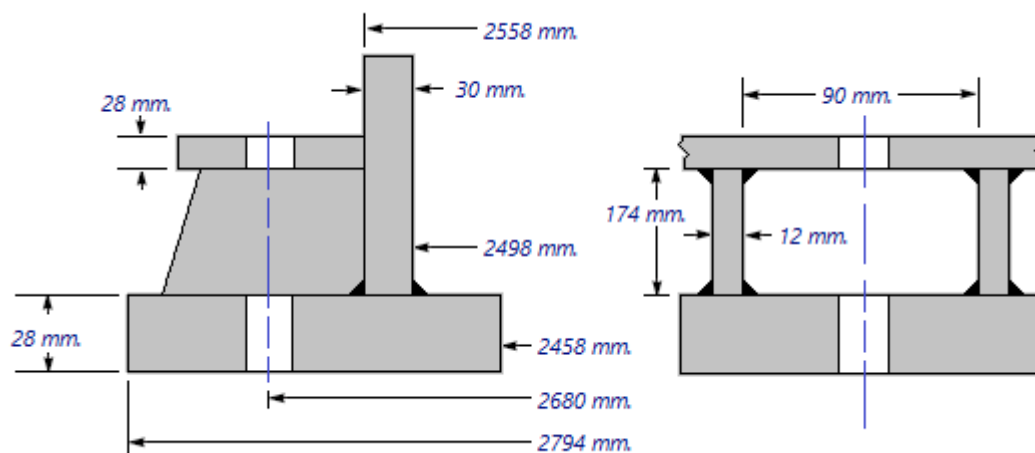
Abs Max of the all of the Stress Ratio's : 0.9418

#### Basering Data : Continuous Top Ring W/Gussets

Thickness of Basering 28.0000 mm.  
 Inside Diameter of Basering 2458.0000 mm.  
 Outside Diameter of Basering 2794.0000 mm.  
 Nominal Diameter of Bolts 36.0000 mm.  
 Diameter of Bolt Circle 2680.0000 mm.  
 Number of Bolts 20

Thickness of Gusset Plates 12.0000 mm.  
 Average Width of Gusset Plates 102.0000 mm.  
 Height of Gussets 174.0000 mm.  
 Distance between Gussets 90.0000 mm.  
 Thickness of Top Plate or Ring 28.0000 mm.  
 Radial Width of the Top Plate 130.0000 mm.  
 Number of Gussets 40

#### Basering Sketch for Cylindrical Skirts:



**Basing Cross Section View**

**Loads for Foundation/Support Design:**

|                                   |               |
|-----------------------------------|---------------|
| Total Wind Shear on Support       | 13347. Kg     |
| Total Earthquake Shear on Support | 20416. Kg     |
| Wind Moment on Support            | 118717. Kg-m. |
| Earthquake Moment on Support      | 179852. Kg-m. |
| Maximum Initial Bolt Load         | 7227. Kg      |

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

**Weights:**

|                                              |              |
|----------------------------------------------|--------------|
| Fabricated - Bare w/o removable internals    | 91392.8 kg.  |
| Shop Test - Fabricated + water ( full )      | 133702.5 kg. |
| Shipping - Fab. + rem. intls.+ shipping app. | 91392.8 kg.  |
| Erected - Fab. + rem. intls.+ insul. (etc)   | 108271.6 kg. |
| Empty - Fab. + intls. + details + wghts.     | 108385.8 kg. |
| Operating - Empty + operating liquid (no ca) | 122379.3 kg. |

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### Nozzle Calculation Summary (bars & mm.):

| Description | MAWP | Ext | MAPNC | Tbl 4.5.2 | Weld | Pass/Fail |
|-------------|------|-----|-------|-----------|------|-----------|
| N4_2"       | 162  | ... | ...   | OK        | OK   | Passed    |
| N3_2"       | 162  | ... | ...   | OK        | OK   | Passed    |
| K1B_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| K3B_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| K3D_2'      | 162  | ... | ...   | OK        | OK   | Passed    |
| K3F_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| K2B_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| K1A_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| K3A_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| K3C_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| K3E_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| K2A_2"      | 162  | ... | ...   | OK        | OK   | Passed    |
| MH1_24"     | 162  | ... | ...   | ...       | OK   | Passed    |
| N1_18"      | 162  | ... | ...   | OK        | OK   | Passed    |
| N2_16"      | 163  | ... | ...   | OK        | OK   | Passed    |

### Nozzle MAWP Summary:

Minimum MAWP Nozzles : 162 Nozzle : MH124"  
 Minimum MAWP Shells/Flanges : 162 Element : Shell 1  
 Minimum MAPnc Shells/Flanges : 177 Element : Shell 3

Computed Vessel M.A.W.P. : 162 bars

### VIII-2 Nozzle to Gross Structural Discontinuity Check:

- **Error: Nozzle MH1\_24", was too close to a gross structural discontinuity.**

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

### Check the spatial relationship between the nozzles:

| From Node | Nozzle Description | Y Coordinate<br>mm. | Layout Angle<br>deg | Inside Radius<br>mm. |
|-----------|--------------------|---------------------|---------------------|----------------------|
| 40        | N4_2"              | 0.000               | 0.000               | 26.138               |
| 50        | N3_2"              | 630.000             | 0.000               | 24.017               |
| 50        | K1B_2"             | 250.000             | 180.000             | 24.017               |
| 50        | K3B_2"             | 250.000             | 90.000              | 24.017               |
| 50        | K3D_2'             | 250.000             | 115.000             | 24.017               |
| 50        | K3F_2"             | 250.000             | 0.000               | 24.017               |
| 50        | K2B_2"             | 250.000             | 270.000             | 24.017               |
| 60        | K1A_2"             | 4149.971            | 180.000             | 24.017               |
| 60        | K3A_2"             | 4149.971            | 0.000               | 24.017               |
| 60        | K3C_2"             | 3549.970            | 90.000              | 24.017               |
| 60        | K3E_2"             | 3549.970            | 125.000             | 24.017               |
| 60        | K2A_2"             | 4149.971            | 270.000             | 24.017               |
| 70        | MH1_24"            | 5900.000            | 0.000               | 306.300              |
| 70        | N1_18"             | 5249.976            | 180.000             | 230.100              |
| 80        | N2_16"             | 0.000               | 0.000               | 204.700              |

### The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

**No interference violations have been detected!**

PV Elite 28 U2    Licensee: SL User  
FileName : 401-V-31001\_Ope\_R2-7%  
Nozzle Summary:

Step:    44    2:30pm    Jul 23,2026

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**HEXAGON**

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**Nozzle Schedule:**

| Description | Nominal or Actual Size | Schd or FVC Type | Flg Type | Nozzle O/Dia in | Wall Thk mm | Reinforcing Pad Diameter | Pad Thk mm | Cut Length mm | Flg Class |
|-------------|------------------------|------------------|----------|-----------------|-------------|--------------------------|------------|---------------|-----------|
| N4_2"       | 2.000 in               | HB               | N/A      | 4.120           | 27.686      | ...                      | ...        | 266.68        | ...       |
| N3_2"       | 2.000 in               | 160              | None     | 2.375           | 8.738       | ...                      | ...        | 279.63        | ...       |
| K1B_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | ...       |
| K3B_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| K3D_2'      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| K3F_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| K2B_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| K1A_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| K3A_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| K3C_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| K3E_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| K2A_2"      | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 530.19        | 1500      |
| N2_16"      | 16.000 in              | Actual           | None     | 18.362          | 30.000      | ...                      | ...        | 399.42        | ...       |
| N1_18"      | 18.000 in              | Actual           | None     | 24.772          | 86.000      | ...                      | ...        | 532.83        | ...       |
| MH1_24"     | 24.000 in              | Actual           | RTJWn    | 30.142          | 78.000      | ...                      | ...        | 648.84        | 1500      |

**General Notes for the above table:**

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

**Nozzle Material and Weld Fillet Leg Size Details (mm.):**

| Description | Material   | Shl Grve Weld | Noz Shl/Pad Weld | Pad OD Weld | Pad Grve Weld | Inside Weld |
|-------------|------------|---------------|------------------|-------------|---------------|-------------|
| N4_2"       | SA-350 LF2 | 66.000        | 10.000           | ...         | ...           | ...         |
| N3_2"       | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K1B_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K3B_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K3D_2'      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K3F_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K2B_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K1A_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K3A_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K3C_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K3E_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| K2A_2"      | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |
| N2_16"      | SA-350 LF2 | 38.000        | 38.000           | ...         | ...           | ...         |
| N1_18"      | SA-350 LF2 | 40.000        | 40.000           | ...         | ...           | ...         |
| MH1_24"     | SA-350 LF2 | 128.000       | 10.000           | ...         | ...           | ...         |

Note: The Outside projections below do not include the flange thickness.

**Nozzle Miscellaneous Data:**

| Description | Elev/Distance From Datum mm. | Layout Angle deg | Proj Outside mm. | Proj Inside mm. | Installed in Component |
|-------------|------------------------------|------------------|------------------|-----------------|------------------------|
| N4_2"       | ...                          | 0.0              | 200.00           | 0.00            | Bottom D end           |
| N3_2"       | 3630.000                     | 0.0              | 150.00           | 0.00            | Shell 1                |
| K1B_2"      | 3250.000                     | 180.0            | 400.00           | 0.00            | Shell 1                |
| K3B_2"      | 3250.000                     | 90.0             | 400.00           | 0.00            | Shell 1                |
| K3D_2'      | 3250.000                     | 115.0            | 400.00           | 0.00            | Shell 1                |
| K3F_2"      | 3250.000                     | 0.0              | 400.00           | 0.00            | Shell 1                |
| K2B_2"      | 3250.000                     | 270.0            | 400.00           | 0.00            | Shell 1                |



|         |          |       |        |      |           |
|---------|----------|-------|--------|------|-----------|
| K1A_2"  | 7149.970 | 180.0 | 400.00 | 0.00 | Shell 2   |
| K3A_2"  | 7149.970 | 0.0   | 400.00 | 0.00 | Shell 2   |
| K3C_2"  | 6549.970 | 90.0  | 400.00 | 0.00 | Shell 2   |
| K3E_2"  | 6549.970 | 125.0 | 400.00 | 0.00 | Shell 2   |
| K2A_2"  | 7149.970 | 270.0 | 400.00 | 0.00 | Shell 2   |
| N2_16"  | ...      | 0.0   | 300.00 | 0.00 | Top D end |
| N1_18"  | 8249.976 | 180.0 | 350.00 | 0.00 | Shell 3   |
| MH1_24" | 8900.000 | 0.0   | 400.00 | 0.00 | Shell 3   |



**Bill of Materials:**

| QTY | DESCRIPTION                                                      | MATERIAL   |
|-----|------------------------------------------------------------------|------------|
| 1   | SKIRT: 30.0mm. THK X 2498.0mm. ID X 500.0mm.                     | SA-516 70  |
| 2   | SPHERICAL HEAD: 74.0mm. THK X 2464.0mm. ID X 0.0mm.              | SA-516 70  |
| 2   | INSULATION: 1232mm X 90mm THK                                    | cold       |
| 3   | PLATFORM: 1200mm X 1500mm WIDE                                   |            |
| 1   | INSULATION: 2000mm X 90mm THK                                    | cold       |
| 1   | INSULATION: 2500mm X 90mm THK                                    | cold       |
| 1   | PACKING: 150.000mm                                               | ...        |
| 1   | INSULATION: 2300mm X 90mm THK                                    | cold       |
| 1   | PLATFORM: 1200mm X 4660mm WIDE                                   |            |
| 1   | LADDER: 11000.0mm                                                | ...        |
| 1   | CLASS 1500 GR 1.1, 24.0" BLIND FLANGE(S)                         | SA-350 LF2 |
| 1   | BASERING: 2794.0mm. OD X 2458.0 ID X 28.0mm.                     | SA-285 C   |
| 20  | BASERING BOLTS: 36.0mm. DIA                                      | SA-36      |
| 20  | BASERING BOLT NUTS: 36.0mm. DIA                                  | ...        |
| 1   | BASERING TOP RING: 2818.0mm. OD X 2558.0 ID X 28.0mm.            | ...        |
| 40  | BASERING GUSSET PLATES: 102.0mm. X 102.0mm. X 174.0mm. X 12.0mm. | ...        |
| 2   | CYLINDRICAL SKIRT SEGMENT 30.0mm. THK X 3051.8mm. X 2000.0mm.    | SA-285 C   |
| 1   | CYLINDRICAL SKIRT SEGMENT 30.0mm. THK X 1743.9mm. X 2000.0mm.    | SA-285 C   |
| 2   | CYLINDRICAL SKIRT SEGMENT 30.0mm. THK X 3095.4mm. X 500.0mm.     | SA-516 70  |
| 1   | CYLINDRICAL SKIRT SEGMENT 30.0mm. THK X 1656.7mm. X 500.0mm.     | SA-516 70  |
| 2   | CYLINDRICAL SEGMENT 128.0mm. THK X 3769.8mm. X 2000.0mm.         | SA-516 70  |
| 2   | CYLINDRICAL SEGMENT 128.0mm. THK X 3769.8mm. X 2500.0mm.         | SA-516 70  |
| 2   | CYLINDRICAL SEGMENT 128.0mm. THK X 3769.8mm. X 2300.0mm.         | SA-516 70  |
| 1   | NAMEPLATE                                                        | ...        |



### Minimum Design Metal Temperature Results Summary (°C):

| Description       | Curve | Basic MDMT | Reduced MDMT | PWHT | Thickness ratio | Gov Thk mm. | E*   |
|-------------------|-------|------------|--------------|------|-----------------|-------------|------|
| Notes             |       |            |              |      |                 |             |      |
| Bottom D end      | [9]   | !          | -33          | Yes  |                 |             |      |
| Shell 1           | [8]   | !          | -33          | Yes  |                 |             |      |
| Shell 2           | [8]   | !          | -33          | Yes  |                 |             |      |
| Shell 3           | [8]   | !          | -33          | Yes  |                 |             |      |
| Top D end         | [9]   | !          | -33          | Yes  |                 |             |      |
| N42"              | [1]   | !          | -46          | Yes  |                 |             |      |
| N32"              | [1]   | !          | -46          | Yes  |                 |             |      |
| K1B2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| K3B2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| K3D2'             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| K3F2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| K2B2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| K1A2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| K3A2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| K3C2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| K3E2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| K2A2"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -46  | Yes             |             |      |
| MH124"            | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozzle Flg        | [5]   | !          | -46          | -66  | Yes             |             |      |
| N118"             | [1]   | !          | -46          | Yes  |                 |             |      |
| N216"             | [1]   | !          | -46          | Yes  |                 |             |      |
| Nozl. Bolting[22] |       |            | -48          | No   | 1.000           | 64.000      | 1.00 |
| Warmest MDMT:     |       |            |              |      |                 |             |      |
|                   |       | -33        | -46          |      |                 |             |      |

Required Minimum Design Metal Temperature -33.0 °C  
 Warmest Computed Minimum Design Metal Temperature -33.0 °C

#### Notes:

- [ ! ] - This was an impact tested material.
- [ 1 ] - Governing Nozzle Weld.
- [ 4 ] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [ 5 ] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [ 6 ] - MDMT Calculations at the Shell/Head Joint.
- [ 7 ] - MDMT Calculations for the Straight Flange.
- [ 8 ] - Cylinder/Cone/Flange Junction MDMT.
- [ 9 ] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

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Class From To: Check of Additional Element Data

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There were no geometry errors or warnings.

*PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.*

### Information Regarding Nozzle Loads

*Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.*



*One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.*

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**Units used in this Analysis (barsUniCode):**

| Name           | System Unit | Constant   | User Unit |
|----------------|-------------|------------|-----------|
| Length         | Feet        | 304.8000   | mm.       |
| Force          | Pounds      | 0.4536     | Kgf       |
| Mass           | Pounds      | 0.4536     | kg.       |
| Area           | sq. inches  | 6.4516     | cm2       |
| Moment         | ft. lbs.    | 0.1383     | Kg-m.     |
| Stress         | lbs./sq.in. | 0.0069     | N./mm2    |
| Temperature    | Degrees F   | 0.5556     | °C        |
| Pressure       | psig        | 0.0689     | bars      |
| Elast. Modulus | lbs./sq.in. | 6.8946     | KPa.      |
| Pipe Density   | lbs./cu.in. | 0.0277     | kg./cm3   |
| Ins. Density   | lbs./cu.ft. | 0.1600E-04 | kg./cm3   |
| Fluid Density  | lbs./cu.ft. | 0.1600E-04 | kg./cm3   |
| Wind Speed     | miles/hr    | 1.6093     | Km/hr     |
| Tray Weight    | lbs./sq.ft. | 0.0005     | Kgs/cm2   |
| Inertia        | in.4        | 41.6231    | cm4       |
| G Load         | G's         | 1.0000     | g's       |
| Wind Load      | lbs./sq.ft. | 4.8825     | Kgs/m2    |
| Elevation      | Feet        | 304.8000   | mm.       |
| Volume         | in.3        | 16.3900    | cm3       |
| Diameter       | inches      | 25.4000    | mm.       |
| Thickness      | inches      | 25.4000    | mm.       |

**PV Elite Vessel Analysis Program: Input Data**

|                                          |          |
|------------------------------------------|----------|
| Design Internal Pressure (for Hydrotest) | 160 bars |
| Design Internal Temperature              | 80.0 °C  |
| Projection of Nozzle from Vessel Top     | 0 mm.    |
| Projection of Nozzle from Vessel Bottom  | 0 mm.    |
| Minimum Design Metal Temperature         | -33.0 °C |
| Type of Construction                     | Welded   |
| Special Service                          | Amine    |
| Use Higher Longitudinal Stresses (Flag)  | N        |
| Select t for Internal Pressure (Flag)    | N        |
| Select t for External Pressure (Flag)    | N        |
| Select t for Axial Stress (Flag)         | N        |
| Select Location for Stiff. Rings (Flag)  | N        |
| Consider Vortex Shedding                 | N        |

**Shop Pressure Test:**

|                                      |                    |
|--------------------------------------|--------------------|
| Type of Pressure Test                | User Defined Hydro |
| Pressure Test Position               | Horizontal         |
| Test Performed in Corroded Condition | No                 |

|              |                   |
|--------------|-------------------|
| Load Case 1  | NP+EW+1.1WI+FW+BW |
| Load Case 2  | NP+EW+EE+FS+BS    |
| Load Case 3  | NP+OW+1.1WI+FW+BW |
| Load Case 4  | NP+OW+EQ+FS+BS    |
| Load Case 5  | NP+HW+HI          |
| Load Case 6  | NP+HW+HE          |
| Load Case 7  | IP+OW+1.1WI+FW+BW |
| Load Case 8  | IP+OW+EQ+FS+BS    |
| Load Case 9  | EP+OW+1.1WI+FW+BW |
| Load Case 10 | EP+OW+EQ+FS+BS    |
| Load Case 11 | HP+HW+HI          |
| Load Case 12 | HP+HW+HE          |
| Load Case 13 | IP+WE+EW          |
| Load Case 14 | IP+WF+CW          |
| Load Case 15 | IP+VO+OW          |
| Load Case 16 | IP+VE+EW          |
| Load Case 17 | NP+VO+OW          |
| Load Case 18 | FS+BS+IP+OW       |
| Load Case 19 | FS+BS+EP+OW       |
| Load Case 20 | BL+IP+OW          |



|                                         |                           |         |
|-----------------------------------------|---------------------------|---------|
| Wind Design Code                        | IS 875 (PART 3):2015-2020 |         |
| Basic Wind Speed for IS-875             | 180                       | Km/hr   |
| Wind Zone Number                        | 5                         |         |
| Base Elevation                          | 300                       | mm.     |
| Percent Wind for Hydrotest              | 33.0                      |         |
| Risk Factor                             | [k1]                      | 1.0     |
| Terrain Category                        |                           | 2       |
| Equipment Class                         |                           | Class A |
| Topography Factor                       | [k3]                      | 1.0     |
| Importance Factor                       | [k4]                      | 1.15    |
| Damping Factor (Beta) for Wind (Ope)    |                           | 0.0100  |
| Damping Factor (Beta) for Wind (Empty)  |                           | 0.0000  |
| Damping Factor (Beta) for Wind (Filled) |                           | 0.0000  |

|                                 |              |
|---------------------------------|--------------|
| Seismic Design Code             | Spectrum     |
| Range Type                      | Period       |
| Ordinate Type                   | Acceleration |
| Include Missing Mass Components | Yes          |
| Shock Scale Factor X Direction  | 1.0000       |
| Shock Scale Factor Y Direction  | 0.6700       |
| Combination Method              | Group        |
| Spectrum Name                   | User Defined |

#### User Defined Spectrum Data Points:

| Acceleration | Period |
|--------------|--------|
| 0.0700       | 0.0100 |
| 0.1170       | 0.0500 |
| 0.1750       | 0.1000 |
| 0.1750       | 0.1300 |
| 0.1750       | 0.2000 |
| 0.1750       | 0.2700 |
| 0.1750       | 0.3000 |
| 0.1750       | 0.4000 |
| 0.1400       | 0.5000 |
| 0.1270       | 0.5500 |
| 0.1170       | 0.6000 |
| 0.1050       | 0.6700 |
| 0.0870       | 0.8000 |
| 0.0700       | 1.0000 |
| 0.0580       | 1.2000 |
| 0.0500       | 1.4000 |
| 0.0440       | 1.6000 |
| 0.0390       | 1.8000 |
| 0.0350       | 2.0000 |
| 0.0290       | 2.4000 |

|                                          |   |
|------------------------------------------|---|
| M.A.W.P + Static Head to Nozzle          | Y |
| Consider MAP New and Cold in Noz. Design | N |
| Consider External Loads for Nozzle Des.  | Y |
| Use ASME VIII-1 Appendix 1-9             | N |

|                             |                                |
|-----------------------------|--------------------------------|
| Perform Blast Load Analysis | No                             |
| Material Database Year      | Current w/Addenda or Code Year |

#### Configuration Directives:

|                                                    |     |
|----------------------------------------------------|-----|
| g0 is taken as min( g0, attached shell thk )       | Yes |
| Using Metric Material Databases, ASME II D         | Yes |
| Using ASME VIII-1 Appendix 46                      | No  |
| Calculate B31.3 type stress for Nozzles with Loads | Yes |
| Reduce the MDMT due to lower membrane stress       | Yes |
| Consider Longitudinal Stress in MDMT Calculations  | Yes |

#### Complete Listing of Vessel Elements and Details:

|                   |     |
|-------------------|-----|
| Element#          | 1/8 |
| Element From Node | 10  |



|                                         |               |         |
|-----------------------------------------|---------------|---------|
| Element To Node                         | 20            |         |
| Element Type                            | Skirt Support |         |
| Description                             | Skirt 3       |         |
| Distance "FROM" to "TO"                 | 2000          | mm.     |
| Skirt Inside Diameter                   | 2498          | mm.     |
| Diameter of Skirt at Base               | 2498          | mm.     |
| Skirt Thickness                         | 30            | mm.     |
| Internal Corrosion Allowance            | 0.75          | mm.     |
| Nominal Thickness                       | 30            | mm.     |
| External Corrosion Allowance            | 0.75          | mm.     |
| Design Temperature Internal Pressure    | 41            | °C      |
| Design Temperature External Pressure    | 0             | °C      |
| Effective Diameter Multiplier           | 1.33          |         |
| Material Name                           | SA-285 C      |         |
| Allowable Stress, Ambient               | 138           | N./mm2  |
| Allowable Stress, Operating             | 137.68        | N./mm2  |
| Allowable Stress, Hydrotest             | 194.75        | N./mm2  |
| Material Density                        | 0.00775       | kg./cm3 |
| P Number Thickness                      | 0             | mm.     |
| Yield Stress, Operating                 | 206.52        | N./mm2  |
| MDMT Curve                              | A             |         |
| External Pressure Chart Name            | CS-2          |         |
| UNS Number                              | K02801        |         |
| Class / Thickness / Grade               | ::t <= 50     |         |
| Product Form                            | Plate         |         |
| Efficiency, Longitudinal Seam           | 0.7           |         |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.7           |         |
| Weld is pre-Heated                      | No            |         |
| Element From Node                       | 10            |         |
| Detail Type                             | Weight        |         |
| Detail ID                               | skirt opening |         |
| Dist. from "FROM" Node / Offset dist    | 1000          | mm.     |
| Miscellaneous Weight                    | 325           | Kgf     |
| Offset from Element Centerline          | 0             | mm.     |

---

|                                         |               |         |
|-----------------------------------------|---------------|---------|
| Element#                                | 2/8           |         |
| Element From Node                       | 20            |         |
| Element To Node                         | 30            |         |
| Element Type                            | Skirt Support |         |
| Description                             | Skirt 2       |         |
| Distance "FROM" to "TO"                 | 500           | mm.     |
| Skirt Inside Diameter                   | 2498          | mm.     |
| Diameter of Skirt at Base               | 2498          | mm.     |
| Skirt Thickness                         | 30            | mm.     |
| Internal Corrosion Allowance            | 0.75          | mm.     |
| Nominal Thickness                       | 30            | mm.     |
| External Corrosion Allowance            | 0.75          | mm.     |
| Design Temperature Internal Pressure    | 80            | °C      |
| Design Temperature External Pressure    | 0             | °C      |
| Effective Diameter Multiplier           | 1.33          |         |
| Material Name                           | SA-516 70     |         |
| Allowable Stress, Ambient               | 175           | N./mm2  |
| Allowable Stress, Operating             | 161.86        | N./mm2  |
| Allowable Stress, Hydrotest             | 247           | N./mm2  |
| Material Density                        | 0.00775       | kg./cm3 |
| P Number Thickness                      | 0             | mm.     |
| Yield Stress, Operating                 | 243           | N./mm2  |
| MDMT Curve                              | B             |         |
| External Pressure Chart Name            | CS-2          |         |
| UNS Number                              | K02700        |         |
| Product Form                            | Plate         |         |
| Efficiency, Longitudinal Seam           | 0.7           |         |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.7           |         |
| Weld is pre-Heated                      | No            |         |

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|                                         |                  |
|-----------------------------------------|------------------|
| Element#                                | 3/8              |
| Element From Node                       | 30               |
| Element To Node                         | 40               |
| Element Type                            | Skirt Support    |
| Description                             | Top Skirt -Yring |
| Distance "FROM" to "TO"                 | 500 mm.          |
| Skirt Inside Diameter                   | 2498 mm.         |
| Diameter of Skirt at Base               | 2498 mm.         |
| Skirt Thickness                         | 30 mm.           |
| Internal Corrosion Allowance            | 0.75 mm.         |
| Nominal Thickness                       | 30 mm.           |
| External Corrosion Allowance            | 0.75 mm.         |
| Design Temperature Internal Pressure    | 80 °C            |
| Design Temperature External Pressure    | 0 °C             |
| Effective Diameter Multiplier           | 1.33             |
| Material Name                           | SA-516 70 +IT    |
| Efficiency, Longitudinal Seam           | 0.7              |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.55             |
| Weld is pre-Heated                      | No               |

|                                      |               |
|--------------------------------------|---------------|
| Element#                             | 4/8           |
| Element From Node                    | 40            |
| Element To Node                      | 50            |
| Element Type                         | Spherical     |
| Description                          | Bottom D end  |
| Distance "FROM" to "TO"              | 0 mm.         |
| Inside Diameter                      | 2464 mm.      |
| Element Thickness                    | 65 mm.        |
| Internal Corrosion Allowance         | 1.5 mm.       |
| Nominal Thickness                    | 74 mm.        |
| External Corrosion Allowance         | 0 mm.         |
| Design Internal Pressure             | 160 bars      |
| Design Temperature Internal Pressure | 80 °C         |
| Design External Pressure             | 0 bars        |
| Design Temperature External Pressure | 0 °C          |
| Effective Diameter Multiplier        | 1.33          |
| Material Name                        | SA-516 70 +IT |
| Efficiency, Longitudinal Seam        | 1.0           |
| Efficiency, Circumferential Seam     | 1.0           |
| Weld is pre-Heated                   | No            |

|                                      |                  |
|--------------------------------------|------------------|
| Element From Node                    | 40               |
| Detail Type                          | Liquid           |
| Detail ID                            | Liquid: 40       |
| Dist. from "FROM" Node / Offset dist | -1232 mm.        |
| Height/Length of Liquid              | 1232 mm.         |
| Liquid Density                       | 0.000635 kg./cm3 |

|                                      |                 |
|--------------------------------------|-----------------|
| Element From Node                    | 40              |
| Detail Type                          | Insulation      |
| Detail ID                            | Insul           |
| Dist. from "FROM" Node / Offset dist | -1232 mm.       |
| Height/Length of Insulation          | 1232 mm.        |
| Thickness of Insulation              | 90 mm.          |
| Density                              | 0.00022 kg./cm3 |

|                                      |                |
|--------------------------------------|----------------|
| Element From Node                    | 40             |
| Detail Type                          | Nozzle         |
| Detail ID                            | N42"           |
| Dist. from "FROM" Node / Offset dist | 0 mm.          |
| Nozzle Diameter                      | 1.94 in.       |
| Nozzle Schedule                      | None           |
| Nozzle Class                         | 1500           |
| Layout Angle                         | 0.0            |
| Blind Flange (Y/N)                   | N              |
| Weight of Nozzle ( Used if > 0 )     | 132.65 Kgf     |
| Grade of Attached Flange             | None           |
| Nozzle Matl                          | SA-350 LF2 +IT |



|                                      |             |
|--------------------------------------|-------------|
| Element From Node                    | 40          |
| Detail Type                          | Weight      |
| Detail ID                            | Btm dish SF |
| Dist. from "FROM" Node / Offset dist | 0 mm.       |
| Miscellaneous Weight                 | 1150 Kg     |
| Offset from Element Centerline       | 0 mm.       |

|                                      |               |
|--------------------------------------|---------------|
| Element#                             | 5/8           |
| Element From Node                    | 50            |
| Element To Node                      | 60            |
| Element Type                         | Cylinder      |
| Description                          | Shell 1       |
| Distance "FROM" to "TO"              | 2000 mm.      |
| Element Outside Diameter             | 2656 mm.      |
| Element Thickness                    | 128 mm.       |
| Internal Corrosion Allowance         | 1.5 mm.       |
| Nominal Thickness                    | 128 mm.       |
| External Corrosion Allowance         | 0 mm.         |
| Design Internal Pressure             | 160 bars      |
| Design Temperature Internal Pressure | 80 °C         |
| Design External Pressure             | 0 bars        |
| Design Temperature External Pressure | 0 °C          |
| Effective Diameter Multiplier        | 1.33          |
| Material Name                        | SA-516 70 +IT |
| Efficiency, Longitudinal Seam        | 1.0           |
| Efficiency, Circumferential Seam     | 1.0           |
| Weld is pre-Heated                   | No            |

|                                      |               |
|--------------------------------------|---------------|
| Element From Node                    | 50            |
| Detail Type                          | Platform      |
| Detail ID                            | PLAT-1        |
| Dist. from "FROM" Node / Offset dist | 0 mm.         |
| Platform Start Angle (degrees)       | 0.0           |
| Platform End Angle (degrees)         | 360.0         |
| Platform Wind Area                   | 43200 cm2     |
| Platform Weight                      | 3699.8 Kg     |
| Platform Railing Weight              | 0 Kg/mm.      |
| Platform Grating Weight              | 0.017 Kgs/cm2 |
| Platform Width                       | 1500 mm.      |
| Platform Height                      | 1200 mm.      |
| Platform Clearance or End Offset     | 230 mm.       |
| Platform Force Coefficient           | 1.2           |
| Ladder Layout Angle                  | 0.0           |
| Ladder Start Elevation               | 0 mm.         |
| Ladder End Elevation                 | 0 mm.         |
| Unit Weight of Ladder                | 0 Kg/mm.      |
| Platform Length (top head platform)  | 0 mm.         |

|                                      |                  |
|--------------------------------------|------------------|
| Element From Node                    | 50               |
| Detail Type                          | Liquid           |
| Detail ID                            | Liquid: 50       |
| Dist. from "FROM" Node / Offset dist | 0 mm.            |
| Height/Length of Liquid              | 2000 mm.         |
| Liquid Density                       | 0.000635 kg./cm3 |

|                                      |                 |
|--------------------------------------|-----------------|
| Element From Node                    | 50              |
| Detail Type                          | Insulation      |
| Detail ID                            | Insul           |
| Dist. from "FROM" Node / Offset dist | 0 mm.           |
| Height/Length of Insulation          | 2000 mm.        |
| Thickness of Insulation              | 90 mm.          |
| Density                              | 0.00022 kg./cm3 |

|                                      |         |
|--------------------------------------|---------|
| Element From Node                    | 50      |
| Detail Type                          | Nozzle  |
| Detail ID                            | N32"    |
| Dist. from "FROM" Node / Offset dist | 630 mm. |



Nozzle Diameter 2 in.  
 Nozzle Schedule 160  
 Nozzle Class None  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 15.129 Kg  
 Grade of Attached Flange None  
 Nozzle Matl SA-350 LF2 +N+IT

Element From Node 50  
 Detail Type Nozzle  
 Detail ID K1B2"  
 Dist. from "FROM" Node / Offset dist 250 mm.  
 Nozzle Diameter 2 in.  
 Nozzle Schedule 160  
 Nozzle Class 1500  
 Layout Angle 180.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 36.34 Kg  
 Grade of Attached Flange None  
 Nozzle Matl SA-350 LF2 +N+IT

Element From Node 50  
 Detail Type Nozzle  
 Detail ID K3B2"  
 Dist. from "FROM" Node / Offset dist 250 mm.  
 Nozzle Diameter 2 in.  
 Nozzle Schedule 160  
 Nozzle Class 1500  
 Layout Angle 90.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 36.34 Kg  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-350 LF2 +N+IT

Element From Node 50  
 Detail Type Nozzle  
 Detail ID K3D2'  
 Dist. from "FROM" Node / Offset dist 250 mm.  
 Nozzle Diameter 2 in.  
 Nozzle Schedule 160  
 Nozzle Class 1500  
 Layout Angle 115.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 36.34 Kg  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-350 LF2 +N+IT

Element From Node 50  
 Detail Type Nozzle  
 Detail ID K3F2"  
 Dist. from "FROM" Node / Offset dist 250 mm.  
 Nozzle Diameter 2 in.  
 Nozzle Schedule 160  
 Nozzle Class 1500  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 36.34 Kg  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-350 LF2 +N+IT

Element From Node 50  
 Detail Type Nozzle  
 Detail ID K2B2"  
 Dist. from "FROM" Node / Offset dist 250 mm.  
 Nozzle Diameter 2 in.  
 Nozzle Schedule 160  
 Nozzle Class 1500  
 Layout Angle 270.0  
 Blind Flange (Y/N) N



|                                  |            |       |
|----------------------------------|------------|-------|
| Weight of Nozzle ( Used if > 0 ) | 36.34      | Kgf   |
| Grade of Attached Flange         | GR 1.1     |       |
| Nozzle Matl                      | SA-350 LF2 | +N+IT |

|                                      |           |      |
|--------------------------------------|-----------|------|
| Element#                             | 6/8       |      |
| Element From Node                    | 60        |      |
| Element To Node                      | 70        |      |
| Element Type                         | Cylinder  |      |
| Description                          | Shell 2   |      |
| Distance "FROM" to "TO"              | 2500      | mm.  |
| Inside Diameter                      | 2400      | mm.  |
| Element Thickness                    | 128       | mm.  |
| Internal Corrosion Allowance         | 1.5       | mm.  |
| Nominal Thickness                    | 128       | mm.  |
| External Corrosion Allowance         | 0         | mm.  |
| Design Internal Pressure             | 160       | bars |
| Design Temperature Internal Pressure | 80        | °C   |
| Design External Pressure             | 0         | bars |
| Design Temperature External Pressure | 0         | °C   |
| Effective Diameter Multiplier        | 1.33      |      |
| Material Name                        | SA-516 70 | +IT  |
| Efficiency, Longitudinal Seam        | 1.0       |      |
| Efficiency, Circumferential Seam     | 1.0       |      |
| Weld is pre-Heated                   | No        |      |

|                                      |          |         |
|--------------------------------------|----------|---------|
| Element From Node                    | 60       |         |
| Detail Type                          | Platform |         |
| Detail ID                            | PLAT-2   |         |
| Dist. from "FROM" Node / Offset dist | 1500     | mm.     |
| Platform Start Angle (degrees)       | 90.0     |         |
| Platform End Angle (degrees)         | 270.0    |         |
| Platform Wind Area                   | 43200    | cm2     |
| Platform Weight                      | 1849.9   | Kgf     |
| Platform Railing Weight              | 0        | Kgf/mm. |
| Platform Grating Weight              | 0.017    | Kgs/cm2 |
| Platform Width                       | 1500     | mm.     |
| Platform Height                      | 1200     | mm.     |
| Platform Clearance or End Offset     | 230      | mm.     |
| Platform Force Coefficient           | 1.2      |         |
| Ladder Layout Angle                  | 0.0      |         |
| Ladder Start Elevation               | 0        | mm.     |
| Ladder End Elevation                 | 0        | mm.     |
| Unit Weight of Ladder                | 0        | Kgf/mm. |
| Platform Length (top head platform)  | 0        | mm.     |

|                                      |            |         |
|--------------------------------------|------------|---------|
| Element From Node                    | 60         |         |
| Detail Type                          | Liquid     |         |
| Detail ID                            | Liquid: 60 |         |
| Dist. from "FROM" Node / Offset dist | 0          | mm.     |
| Height/Length of Liquid              | 2000       | mm.     |
| Liquid Density                       | 0.000635   | kg./cm3 |

|                                      |            |         |
|--------------------------------------|------------|---------|
| Element From Node                    | 60         |         |
| Detail Type                          | Insulation |         |
| Detail ID                            | Insul      |         |
| Dist. from "FROM" Node / Offset dist | 0          | mm.     |
| Height/Length of Insulation          | 2500       | mm.     |
| Thickness of Insulation              | 90         | mm.     |
| Density                              | 0.00022    | kg./cm3 |

|                                      |        |     |
|--------------------------------------|--------|-----|
| Element From Node                    | 60     |     |
| Detail Type                          | Nozzle |     |
| Detail ID                            | K1A2"  |     |
| Dist. from "FROM" Node / Offset dist | 2150   | mm. |
| Nozzle Diameter                      | 2      | in. |
| Nozzle Schedule                      | 160    |     |
| Nozzle Class                         | 1500   |     |
| Layout Angle                         | 180.0  |     |



|                                  |                  |
|----------------------------------|------------------|
| Blind Flange (Y/N)               | N                |
| Weight of Nozzle ( Used if > 0 ) | 36.34 Kgf        |
| Grade of Attached Flange         | GR 1.1           |
| Nozzle Matl                      | SA-350 LF2 +N+IT |

|                                      |                  |
|--------------------------------------|------------------|
| Element From Node                    | 60               |
| Detail Type                          | Nozzle           |
| Detail ID                            | K3A2"            |
| Dist. from "FROM" Node / Offset dist | 2150 mm.         |
| Nozzle Diameter                      | 2 in.            |
| Nozzle Schedule                      | 160              |
| Nozzle Class                         | 1500             |
| Layout Angle                         | 0.0              |
| Blind Flange (Y/N)                   | N                |
| Weight of Nozzle ( Used if > 0 )     | 36.34 Kgf        |
| Grade of Attached Flange             | GR 1.1           |
| Nozzle Matl                          | SA-350 LF2 +N+IT |

|                                      |                  |
|--------------------------------------|------------------|
| Element From Node                    | 60               |
| Detail Type                          | Nozzle           |
| Detail ID                            | K3C2"            |
| Dist. from "FROM" Node / Offset dist | 1550 mm.         |
| Nozzle Diameter                      | 2 in.            |
| Nozzle Schedule                      | 160              |
| Nozzle Class                         | 1500             |
| Layout Angle                         | 90.0             |
| Blind Flange (Y/N)                   | N                |
| Weight of Nozzle ( Used if > 0 )     | 36.34 Kgf        |
| Grade of Attached Flange             | GR 1.1           |
| Nozzle Matl                          | SA-350 LF2 +N+IT |

|                                      |                  |
|--------------------------------------|------------------|
| Element From Node                    | 60               |
| Detail Type                          | Nozzle           |
| Detail ID                            | K3E2"            |
| Dist. from "FROM" Node / Offset dist | 1550 mm.         |
| Nozzle Diameter                      | 2 in.            |
| Nozzle Schedule                      | 160              |
| Nozzle Class                         | 1500             |
| Layout Angle                         | 125.0            |
| Blind Flange (Y/N)                   | N                |
| Weight of Nozzle ( Used if > 0 )     | 36.34 Kgf        |
| Grade of Attached Flange             | GR 1.1           |
| Nozzle Matl                          | SA-350 LF2 +N+IT |

|                                      |                  |
|--------------------------------------|------------------|
| Element From Node                    | 60               |
| Detail Type                          | Nozzle           |
| Detail ID                            | K2A2"            |
| Dist. from "FROM" Node / Offset dist | 2150 mm.         |
| Nozzle Diameter                      | 2 in.            |
| Nozzle Schedule                      | 160              |
| Nozzle Class                         | 1500             |
| Layout Angle                         | 270.0            |
| Blind Flange (Y/N)                   | N                |
| Weight of Nozzle ( Used if > 0 )     | 36.34 Kgf        |
| Grade of Attached Flange             | GR 1.1           |
| Nozzle Matl                          | SA-350 LF2 +N+IT |

|                                      |          |
|--------------------------------------|----------|
| Element From Node                    | 60       |
| Detail Type                          | Weight   |
| Detail ID                            | Cleat    |
| Dist. from "FROM" Node / Offset dist | 1261 mm. |
| Miscellaneous Weight                 | 1400 Kgf |
| Offset from Element Centerline       | 0 mm.    |

|                                      |            |
|--------------------------------------|------------|
| Element From Node                    | 60         |
| Detail Type                          | Weight     |
| Detail ID                            | Lifting wt |
| Dist. from "FROM" Node / Offset dist | 1261 mm.   |
| Miscellaneous Weight                 | 2600 Kgf   |
| Offset from Element Centerline       | 0 mm.      |



|                                      |           |
|--------------------------------------|-----------|
| Element From Node                    | 60        |
| Detail Type                          | Weight    |
| Detail ID                            | 7% margin |
| Dist. from "FROM" Node / Offset dist | 1261 mm.  |
| Miscellaneous Weight                 | 6100 Kgf  |
| Offset from Element Centerline       | 0 mm.     |

|                                      |               |
|--------------------------------------|---------------|
| Element#                             | 7/8           |
| Element From Node                    | 70            |
| Element To Node                      | 80            |
| Element Type                         | Cylinder      |
| Description                          | Shell 3       |
| Distance "FROM" to "TO"              | 2300 mm.      |
| Inside Diameter                      | 2400 mm.      |
| Element Thickness                    | 128 mm.       |
| Internal Corrosion Allowance         | 1.5 mm.       |
| Nominal Thickness                    | 128 mm.       |
| External Corrosion Allowance         | 0 mm.         |
| Design Internal Pressure             | 160 bars      |
| Design Temperature Internal Pressure | 80 °C         |
| Design External Pressure             | 0 bars        |
| Design Temperature External Pressure | 0 °C          |
| Effective Diameter Multiplier        | 1.33          |
| Material Name                        | SA-516 70 +IT |
| Efficiency, Longitudinal Seam        | 1.0           |
| Efficiency, Circumferential Seam     | 1.0           |
| Weld is pre-Heated                   | No            |

|                                      |               |
|--------------------------------------|---------------|
| Element From Node                    | 70            |
| Detail Type                          | Platform      |
| Detail ID                            | PLAT-3        |
| Dist. from "FROM" Node / Offset dist | 800 mm.       |
| Platform Start Angle (degrees)       | 270.0         |
| Platform End Angle (degrees)         | 90.0          |
| Platform Wind Area                   | 43200 cm2     |
| Platform Weight                      | 1849.9 Kgf    |
| Platform Railing Weight              | 0 Kgf/mm.     |
| Platform Grating Weight              | 0.017 Kgs/cm2 |
| Platform Width                       | 1500 mm.      |
| Platform Height                      | 1200 mm.      |
| Platform Clearance or End Offset     | 230 mm.       |
| Platform Force Coefficient           | 1.2           |
| Ladder Layout Angle                  | 0.0           |
| Ladder Start Elevation               | 0 mm.         |
| Ladder End Elevation                 | 0 mm.         |
| Unit Weight of Ladder                | 0 Kgf/mm.     |
| Platform Length (top head platform)  | 0 mm.         |

|                                      |                  |
|--------------------------------------|------------------|
| Element From Node                    | 70               |
| Detail Type                          | Packing          |
| Detail ID                            | demister         |
| Dist. from "FROM" Node / Offset dist | 2000 mm.         |
| Height of Packed Section             | 150 mm.          |
| Density                              | 0.000168 kg./cm3 |
| Percent Volume Holdup                | 0.0              |
| Specific Gravity of Packing Liquid   | 0.0              |

|                                      |                 |
|--------------------------------------|-----------------|
| Element From Node                    | 70              |
| Detail Type                          | Insulation      |
| Detail ID                            | Insul           |
| Dist. from "FROM" Node / Offset dist | 0 mm.           |
| Height/Length of Insulation          | 2300 mm.        |
| Thickness of Insulation              | 90 mm.          |
| Density                              | 0.00022 kg./cm3 |

|                   |        |
|-------------------|--------|
| Element From Node | 70     |
| Detail Type       | Nozzle |



|                                      |                |
|--------------------------------------|----------------|
| Detail ID                            | MH124"         |
| Dist. from "FROM" Node / Offset dist | 1400 mm.       |
| Nozzle Diameter                      | 24 in.         |
| Nozzle Schedule                      | None           |
| Nozzle Class                         | 1500           |
| Layout Angle                         | 0.0            |
| Blind Flange (Y/N)                   | Y              |
| Weight of Nozzle ( Used if > 0 )     | 4677 Kg        |
| Grade of Attached Flange             | GR 1.1         |
| Nozzle Matl                          | SA-350 LF2 +IT |

|                                      |                |
|--------------------------------------|----------------|
| Element From Node                    | 70             |
| Detail Type                          | Nozzle         |
| Detail ID                            | N118"          |
| Dist. from "FROM" Node / Offset dist | 749.97 mm.     |
| Nozzle Diameter                      | 18 in.         |
| Nozzle Schedule                      | None           |
| Nozzle Class                         | None           |
| Layout Angle                         | 180.0          |
| Blind Flange (Y/N)                   | N              |
| Weight of Nozzle ( Used if > 0 )     | 723.33 Kg      |
| Grade of Attached Flange             | None           |
| Nozzle Matl                          | SA-350 LF2 +IT |

|                                      |              |
|--------------------------------------|--------------|
| Element From Node                    | 70           |
| Detail Type                          | Weight       |
| Detail ID                            | FP+FLGdeadN1 |
| Dist. from "FROM" Node / Offset dist | 1200 mm.     |
| Miscellaneous Weight                 | 630 Kg       |
| Offset from Element Centerline       | 0 mm.        |

|                                      |              |
|--------------------------------------|--------------|
| Element From Node                    | 70           |
| Detail Type                          | Weight       |
| Detail ID                            | FPligN1      |
| Dist. from "FROM" Node / Offset dist | 0.004069 mm. |
| Miscellaneous Weight                 | 400 Kg       |
| Offset from Element Centerline       | 0 mm.        |

|                                      |          |
|--------------------------------------|----------|
| Element From Node                    | 70       |
| Detail Type                          | Weight   |
| Detail ID                            | Davit    |
| Dist. from "FROM" Node / Offset dist | 1800 mm. |
| Miscellaneous Weight                 | 800 Kg   |
| Offset from Element Centerline       | 1300 mm. |

|                                      |            |
|--------------------------------------|------------|
| Element From Node                    | 70         |
| Detail Type                          | Weight     |
| Detail ID                            | N10VHPP    |
| Dist. from "FROM" Node / Offset dist | 749.97 mm. |
| Miscellaneous Weight                 | 2226 Kg    |
| Offset from Element Centerline       | 2737 mm.   |

|                                      |           |
|--------------------------------------|-----------|
| Element#                             | 8/8       |
| Element From Node                    | 80        |
| Element To Node                      | 90        |
| Element Type                         | Spherical |
| Description                          | Top D end |
| Distance "FROM" to "TO"              | 0 mm.     |
| Inside Diameter                      | 2464 mm.  |
| Element Thickness                    | 65 mm.    |
| Internal Corrosion Allowance         | 1.5 mm.   |
| Nominal Thickness                    | 74 mm.    |
| External Corrosion Allowance         | 0 mm.     |
| Design Internal Pressure             | 160 bars  |
| Design Temperature Internal Pressure | 80 °C     |
| Design External Pressure             | 0 bars    |
| Design Temperature External Pressure | 0 °C      |
| Effective Diameter Multiplier        | 1.33      |



|                                      |                 |
|--------------------------------------|-----------------|
| Material Name                        | SA-516 70 +IT   |
| Efficiency, Longitudinal Seam        | 1.0             |
| Efficiency, Circumferential Seam     | 1.0             |
| Weld is pre-Heated                   | No              |
| Element From Node                    | 80              |
| Detail Type                          | Platform        |
| Detail ID                            | TOP PLT         |
| Dist. from "FROM" Node / Offset dist | 1500 mm.        |
| Platform Start Angle (degrees)       | 0.0             |
| Platform End Angle (degrees)         | 270.0           |
| Platform Wind Area                   | 88920 cm2       |
| Platform Weight                      | 4100.6 Kg       |
| Platform Railing Weight              | 0 Kg/mm.        |
| Platform Grating Weight              | 0.017 Kgs/cm2   |
| Platform Width                       | 4660 mm.        |
| Platform Height                      | 1200 mm.        |
| Platform Clearance or End Offset     | 2330 mm.        |
| Platform Force Coefficient           | 1.0             |
| Ladder Layout Angle                  | 270.0           |
| Ladder Start Elevation               | 0 mm.           |
| Ladder End Elevation                 | 11000 mm.       |
| Unit Weight of Ladder                | 0.037 Kg/mm.    |
| Platform Length (top head platform)  | 4660 mm.        |
| Element From Node                    | 80              |
| Detail Type                          | Insulation      |
| Detail ID                            | Insul           |
| Dist. from "FROM" Node / Offset dist | 0 mm.           |
| Height/Length of Insulation          | 1232 mm.        |
| Thickness of Insulation              | 90 mm.          |
| Density                              | 0.00022 kg./cm3 |
| Element From Node                    | 80              |
| Detail Type                          | Nozzle          |
| Detail ID                            | N216"           |
| Dist. from "FROM" Node / Offset dist | 0 mm.           |
| Nozzle Diameter                      | 16 in.          |
| Nozzle Schedule                      | None            |
| Nozzle Class                         | None            |
| Layout Angle                         | 0.0             |
| Blind Flange (Y/N)                   | N               |
| Weight of Nozzle ( Used if > 0 )     | 248.12 Kg       |
| Grade of Attached Flange             | None            |
| Nozzle Matl                          | SA-350 LF2 +IT  |
| Element From Node                    | 80              |
| Detail Type                          | Weight          |
| Detail ID                            | Top dish SF     |
| Dist. from "FROM" Node / Offset dist | 0 mm.           |
| Miscellaneous Weight                 | 900 Kg          |
| Offset from Element Centerline       | 0 mm.           |
| Element From Node                    | 80              |
| Detail Type                          | Weight          |
| Detail ID                            | N20VHPP         |
| Dist. from "FROM" Node / Offset dist | 0 mm.           |
| Miscellaneous Weight                 | 1105 Kg         |
| Offset from Element Centerline       | 2711 mm.        |
| Element From Node                    | 80              |
| Detail Type                          | For./Mom.       |
| Detail ID                            | N2FxFz          |
| Dist. from "FROM" Node / Offset dist | 1231 mm.        |
| Force in X Direction                 | 2243 Kg         |
| Force in Y Direction                 | -1513.7 Kg      |
| Force in Z Direction                 | 2243 Kg         |
| Moment about X Axis                  | 4484.7 Kg-m.    |
| Moment about Y Axis                  | 0 Kg-m.         |
| Moment about Z Axis                  | 3588.2 Kg-m.    |

PV Elite 28 U2    Licensee: SL User  
FileName : 401-V-31001\_Ope\_R2-7%  
Input Echo:

Step:    1    2:29pm    Jul 23,2026

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**HEXAGON**

Force/Moment Combination Method

SRSS

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**XY Coordinate Calculations:**

| From            | To | X (Horiz.)<br>mm. | Y (Vert.)<br>mm. | DX (Horiz.)<br>mm. | DY (Vert.)<br>mm. |
|-----------------|----|-------------------|------------------|--------------------|-------------------|
| Skirt 3         |    | ...               | 2000             | ...                | 2000              |
| Skirt 2         |    | ...               | 2500             | ...                | 500               |
| Top Skirt -Yrin |    | ...               | 3000             | ...                | 500               |
| Bottom D end    |    | ...               | 3000             | ...                | ...               |
| Shell 1         |    | ...               | 5000             | ...                | 2000              |
| Shell 2         |    | ...               | 7500             | ...                | 2500              |
| Shell 3         |    | ...               | 9800             | ...                | 2300              |
| Top D end       |    | ...               | 9800             | ...                | ...               |

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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### Internal Pressure Results Summary:

#### Element Thickness, Pressure, Diameter and Allowable Stress :

| From             | To | Int. Press<br>+ Liq. Hd<br>bars | Nominal<br>Thickness<br>mm. | Total Corr<br>Allowance<br>mm. | Element<br>Diameter<br>mm. | Allowable<br>Stress(SE)<br>N./mm <sup>2</sup> | Element# |
|------------------|----|---------------------------------|-----------------------------|--------------------------------|----------------------------|-----------------------------------------------|----------|
| Skirt 3          |    | ...                             | 30                          | 1.5                            | 2498                       | ...                                           | 1        |
| Skirt 2          |    | ...                             | 30                          | 1.5                            | 2498                       | ...                                           | 2        |
| Top Skirt -Yring |    | ...                             | 30                          | 1.5                            | 2498                       | ...                                           | 3        |
| Bottom D end     |    | 160.33                          | 74                          | 1.5                            | 2464                       | 161.85                                        | 4        |
| Shell 1          |    | 160.25                          | 128                         | 1.5                            | 2400                       | 161.9                                         | 5        |
| Shell 2          |    | 160.13                          | 128                         | 1.5                            | 2400                       | 161.9                                         | 6        |
| Shell 3          |    | 160                             | 128                         | 1.5                            | 2400                       | 161.9                                         | 7        |
| Top D end        |    | 160                             | 74                          | 1.5                            | 2464                       | 161.9                                         | 8        |

#### Element Required Thickness and MAWP :

| From             | To | Design<br>Pressure<br>bars | M.A.W.P.<br>Corroded<br>bars | M.A.W.P.<br>New & Cold<br>bars | Minimum<br>Thickness<br>mm. | Required<br>Thickness<br>mm. | Element# |
|------------------|----|----------------------------|------------------------------|--------------------------------|-----------------------------|------------------------------|----------|
| Skirt 3          |    | ...                        | No Calc                      | No Calc                        | 30                          | No Calc                      | 1        |
| Skirt 2          |    | ...                        | No Calc                      | No Calc                        | 30                          | No Calc                      | 2        |
| Top Skirt -Yring |    | ...                        | No Calc                      | No Calc                        | 30                          | No Calc                      | 3        |
| Bottom D end     |    | 160                        | 162                          | 180                            | 65                          | 64.1353                      | 4        |
| Shell 1          |    | 160                        | 162                          | 177                            | 128                         | 126.519                      | 5        |
| Shell 2          |    | 160                        | 162                          | 177                            | 128                         | 126.417                      | 6        |
| Shell 3          |    | 160                        | 162                          | 177                            | 128                         | 126.315                      | 7        |
| Top D end        |    | 160                        | 163                          | 180                            | 65                          | 63.9869                      | 8        |

Minimum 162 177

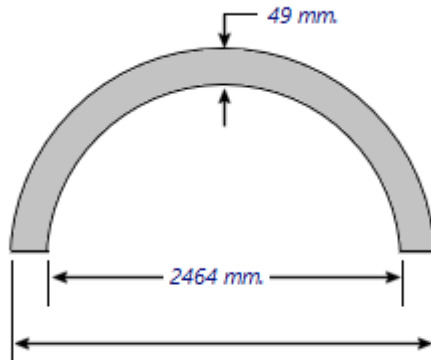
MAWP: 162 bars, limited by: Shell 1.

Elements Suitable for Design Internal Pressure.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 2, 2025



#### Internal Pressure Results for: Bottom D end

| Spherical Head Design Information: |    |               |                         |    |               |
|------------------------------------|----|---------------|-------------------------|----|---------------|
| Design Pressure                    | P  | 160.329 bars  | Int. Design Temperature |    | 80 °C         |
| Material                           |    | SA-516 70     | Ext. Design Temperature |    | 0 °C          |
| Length                             |    | 0.000 mm.     | External Pressure Chart |    | CS-2          |
|                                    |    |               | UNS Number              |    | K02700        |
| Int. Corr. All.                    | c  | 1.5000 mm.    | Allowable Stress (ope)  | S  | 161.9 N./mm^2 |
| Ext. Corr. All.                    | ce | 0.0000 mm.    | Allowable Stress (amb)  | Sa | 175.0 N./mm^2 |
| Inside Diameter                    | Di | 2464.000 mm.  | Actual Stress           |    | 151.8 N./mm^2 |
| Spec. Min. Thk t,e, or sa          |    | 65.0000 mm.   | Spec. Nominal Thk.      |    | 74.000 mm.    |
| Surface Area                       |    | 107168.3 cm^2 | SG of Contents          |    | 0.636         |
| Element Volume                     |    | 3917134.8 cm3 | Weight of Contents      |    | 2489.8 Kgf    |
| Empty Weight                       |    | 5804.8 Kgf    | Operating Weight        |    | 8294.6 Kgf    |
|                                    |    |               | Comp. All. Ope. Stress  |    | 121.5 N./mm^2 |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| ...                      |       | ...              |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.800 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.400 mm.

#### Material treatment options: +IT

#### Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:  
 $= D/2( \exp( 0.5P/(S \cdot E) ) - 1 ) + ci + co$   
 $= 2467/2( \exp( 0.5 \cdot 160/(162 \cdot 1) ) - 1 ) + 1.5 + 0$   
 $= 62.6 + 1.5 + 0 = 64.1 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:  
 $= 2 \cdot S \cdot E \cdot \ln( ( 2 \cdot t + D ) / D ) - P_{liq}$   
 $= 2 \cdot 162 \cdot 1 \cdot \ln( ( 2 \cdot 63.5 + 2467 ) / 2467 ) - 0.33$   
 $= 162 - 0.33 = 162 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:  
 $= 2 \cdot S \cdot E \cdot \ln( ( 2 \cdot t + D ) / D )$   
 $= 2 \cdot 175 \cdot 1 \cdot \ln( ( 2 \cdot 65 + 2464 ) / 2464 )$   
 $= 180 \text{ bars}$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / ( E ( D_o^2 - D^2 ) ) \\ &= 160 \cdot 2467^2 / ( 1 ( 2594^2 - 2467^2 ) ) \\ &= 151.822 \text{ N./mm}^2 \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / ( E \cdot \ln( ( 2 \cdot t + D ) / D ) ) \\ &= 0.5 \cdot 160 / ( 1 \cdot \ln( ( 2 \cdot 63.5 + 2467 ) / 2467 ) ) \\ &= 159.705 \text{ N./mm}^2 \end{aligned}$$

#### Percent Fiber Elongation per Table 6.1

$$\begin{aligned} \text{For Heads formed from Segments } &(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) && 4.374 \% \\ \text{For one Piece doubly Curved Heads } &100 \cdot \ln(D_b / (D_f - 2t)) && 40.490 \% \end{aligned}$$

Where  $D_b$  is the blank diameter for one piece doubly curved surfaces [ $D_b$ ]:

$$\begin{aligned} &= \text{square root}( 4 \cdot \text{Head Surface Area} / 3.14159 ) \\ &= \text{square root}( 4 \cdot 107168 / 3.14159 ) \\ &= 3693.926 \text{ mm.} \end{aligned}$$

#### MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

#### Tolerance for Formed Heads per 4.3.2.2:

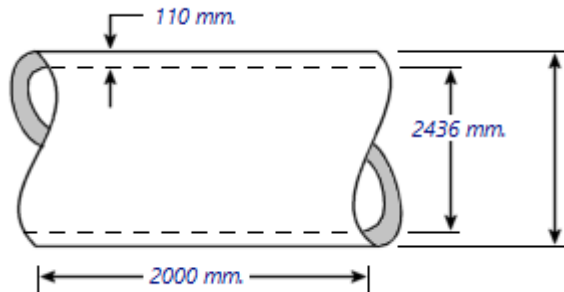
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.800 mm.  
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.400 mm.

Material treatment options: +IT

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 2, 2025



#### Internal Pressure Results for: Shell 1

| Cylindrical Shell Design Information:             |    |               |                         |    |               |
|---------------------------------------------------|----|---------------|-------------------------|----|---------------|
| Design Pressure                                   | P  | 160.252 bars  | Int. Design Temperature |    | 80 °C         |
| Material                                          |    | SA-516 70     | Ext. Design Temperature |    | 0 °C          |
| Length                                            |    | 2000.000 mm.  | External Pressure Chart |    | CS-2          |
|                                                   |    |               | UNS Number              |    | K02700        |
| Int. Corr. All.                                   | c  | 1.5000 mm.    | Allowable Stress (ope)  | S  | 161.9 N./mm^2 |
| Ext. Corr. All.                                   | ce | 0.0000 mm.    | Allowable Stress (amb)  | Sa | 175.0 N./mm^2 |
| Inside Diameter                                   | Di | 2400.000 mm.  | Actual Stress           |    | 152.2 N./mm^2 |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 128.0000 mm.  | Spec. Nominal Thk.      |    | 128.000 mm.   |
| Surface Area                                      |    | 166881.4 cm^2 | SG of Contents          |    | 0.636         |
| Element Volume                                    |    | 9049409.0 cm3 | Weight of Contents      |    | 5751.9 Kg     |
| Empty Weight                                      |    | 15757.0 Kg    | Operating Weight        |    | 21508.9 Kg    |
|                                                   |    |               | Comp. All. Ope. Stress  |    | 122.7 N./mm^2 |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| ...                      |       | ...              |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +IT

#### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$   
 $= 0.5 \cdot 2403 \cdot (\exp(160/(162 \cdot 1)) - 1) + 1.5 + 0$   
 $= 125 + 1.5 + 0 = 127 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:  
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$   
 $= 162 \cdot 1 \cdot \ln((2 \cdot 126 + 2403) / 2403) - 0.25$   
 $= 162 - 0.25 = 162 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:  
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$   
 $= 175 \cdot 1 \cdot \ln((2 \cdot 128 + 2400) / 2400)$   
 $= 177 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:  
 $= P \cdot D / (E (D_o - D))$   
 $= 160 \cdot 2403 / (1 (2656 - 2403))$   
 $= 152.2164 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:  
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 160 / ( 1 \cdot \ln( ( 2 \cdot 126 + 2403 ) / 2403 ) )$$
$$= 160.0958 \text{ N./mm}^2$$

#### Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate     $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$     5.063 %

#### Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

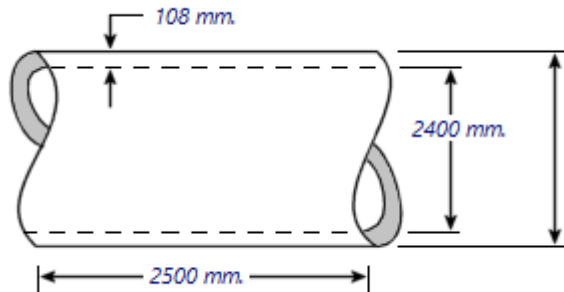
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 2, 2025



#### Internal Pressure Results for: Shell 2

| Cylindrical Shell Design Information: |    |                |                         |    |               |
|---------------------------------------|----|----------------|-------------------------|----|---------------|
| Design Pressure                       | P  | 160.127 bars   | Int. Design Temperature |    | 80 °C         |
| Material                              |    | SA-516 70      | Ext. Design Temperature |    | 0 °C          |
| Length                                |    | 2500.000 mm.   | External Pressure Chart |    | CS-2          |
|                                       |    |                | UNS Number              |    | K02700        |
| Int. Corr. All.                       | c  | 1.5000 mm.     | Allowable Stress (ope)  | S  | 161.9 N./mm^2 |
| Ext. Corr. All.                       | ce | 0.0000 mm.     | Allowable Stress (amb)  | Sa | 175.0 N./mm^2 |
| Inside Diameter                       | Di | 2400.000 mm.   | Actual Stress           |    | 152.1 N./mm^2 |
| Spec. Min. Thk t <sub>e</sub> , or sa |    | 128.0000 mm.   | Spec. Nominal Thk.      |    | 128.000 mm.   |
| Surface Area                          |    | 208601.8 cm^2  | SG of Contents          |    | 0.636         |
| Element Volume                        |    | 11311760.0 cm3 | Weight of Contents      |    | 5751.9 Kgf    |
| Empty Weight                          |    | 19696.3 Kgf    | Operating Weight        |    | 25448.2 Kgf   |
|                                       |    |                | Comp. All. Ope. Stress  |    | 122.7 N./mm^2 |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| ...                      |       | ...              |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +IT

#### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2403 \cdot (\exp(160/(162 \cdot 1)) - 1) + 1.5 + 0$$

$$= 125 + 1.5 + 0 = 126 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 162 \cdot 1 \cdot \ln((2 \cdot 126 + 2403) / 2403) - 0.13$$

$$= 162 - 0.13 = 162 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175 \cdot 1 \cdot \ln((2 \cdot 128 + 2400) / 2400)$$

$$= 177 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:  

$$= P \cdot D / (E (D_o - D))$$

$$= 160 \cdot 2403 / (1 (2656 - 2403))$$

$$= 152.0979 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:  

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 160 / ( 1 \cdot \ln( ( 2 \cdot 126 + 2403 ) / 2403 ) )$$
$$= 159.9713 \text{ N./mm}^2$$

#### Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate     $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$     5.063 %

#### Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

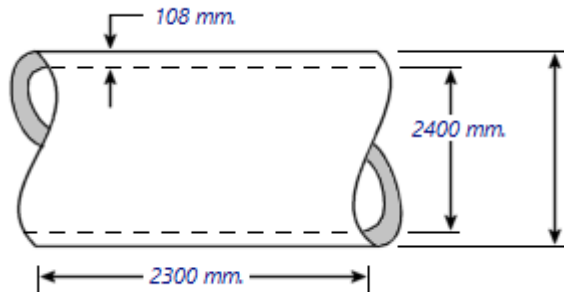
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 2, 2025



#### Internal Pressure Results for: Shell 3

| Cylindrical Shell Design Information: |    |                |                         |    |               |
|---------------------------------------|----|----------------|-------------------------|----|---------------|
| Design Pressure                       | P  | 160.002 bars   | Int. Design Temperature |    | 80 °C         |
| Material                              |    | SA-516 70      | Ext. Design Temperature |    | 0 °C          |
| Length                                |    | 2300.000 mm.   | External Pressure Chart |    | CS-2          |
|                                       |    |                | UNS Number              |    | K02700        |
| Int. Corr. All.                       | c  | 1.5000 mm.     | Allowable Stress (ope)  | S  | 161.9 N./mm^2 |
| Ext. Corr. All.                       | ce | 0.0000 mm.     | Allowable Stress (amb)  | Sa | 175.0 N./mm^2 |
| Inside Diameter                       | Di | 2400.000 mm.   | Actual Stress           |    | 152.0 N./mm^2 |
| Spec. Min. Thk t,e, or sa             |    | 128.0000 mm.   | Spec. Nominal Thk.      |    | 128.000 mm.   |
| Surface Area                          |    | 191913.7 cm^2  | SG of Contents          |    | 0.000         |
| Element Volume                        |    | 10406821.0 cm3 | Weight of Contents      |    | 0.0 Kgf       |
| Empty Weight                          |    | 18120.6 Kgf    | Operating Weight        |    | 18120.6 Kgf   |
|                                       |    |                | Comp. All. Ope. Stress  |    | 122.7 N./mm^2 |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| ...                      |       | ...              |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +IT

#### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$   
 $= 0.5 \cdot 2403 \cdot (\exp(160/(162 \cdot 1)) - 1) + 1.5 + 0$   
 $= 125 + 1.5 + 0 = 126 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:  
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$   
 $= 162 \cdot 1 \cdot \ln((2 \cdot 126 + 2403) / 2403) - 0.0025$   
 $= 162 - 0.0025 = 162 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:  
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$   
 $= 175 \cdot 1 \cdot \ln((2 \cdot 128 + 2400) / 2400)$   
 $= 177 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:  
 $= P \cdot D / (E (D_o - D))$   
 $= 160 \cdot 2403 / (1 (2656 - 2403))$   
 $= 151.9795 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:  
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 160 / ( 1 \cdot \ln( ( 2 \cdot 126 + 2403 ) / 2403 ) )$$
$$= 159.8467 \text{ N./mm}^2$$

#### Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate     $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$     5.063 %

#### Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

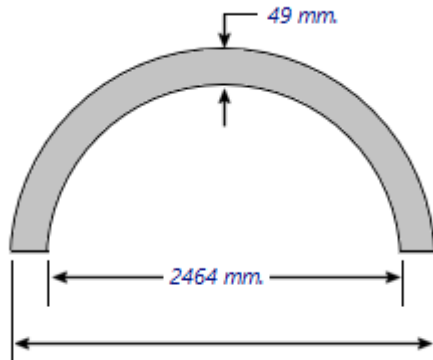
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 2, 2025



#### Internal Pressure Results for: Top D end

| Spherical Head Design Information:                |    |               |                         |        |               |
|---------------------------------------------------|----|---------------|-------------------------|--------|---------------|
| Design Pressure                                   | P  | 160.002 bars  | Int. Design Temperature |        | 80 °C         |
| Material                                          |    | SA-516 70     | Ext. Design Temperature |        | 0 °C          |
| Length                                            |    | 0.000 mm.     | External Pressure Chart |        | CS-2          |
|                                                   |    |               | UNS Number              |        | K02700        |
| Int. Corr. All.                                   | c  | 1.5000 mm.    | Allowable Stress (ope)  | S      | 161.9 N./mm^2 |
| Ext. Corr. All.                                   | ce | 0.0000 mm.    | Allowable Stress (amb)  | Sa     | 175.0 N./mm^2 |
| Inside Diameter                                   | Di | 2464.000 mm.  | Actual Stress           |        | 151.5 N./mm^2 |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 65.0000 mm.   | Spec. Nominal Thk.      |        | 74.000 mm.    |
| Surface Area                                      |    | 107168.3 cm^2 | SG of Contents          |        | 0.000         |
| Element Volume                                    |    | 3917134.8 cm3 | Weight of Contents      |        | 0.0 Kgf       |
| Empty Weight                                      |    | 5804.5 Kgf    | Operating Weight        |        | 5804.5 Kgf    |
|                                                   |    |               | Comp. All. Ope. Stress  |        | 121.5 N./mm^2 |
|                                                   |    |               |                         |        |               |
| Radiography Information:                          |    |               |                         |        |               |
|                                                   |    |               |                         |        |               |
| ...                                               |    |               | ...                     |        |               |
| Circ. Joint Efficiency                            |    | 100 %         | Long. Joint Eff.        | E or v | 100 %         |

#### Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.800 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.400 mm.

#### Material treatment options: +IT

#### Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:  
 $= D/2( \exp( 0.5P/(S \cdot E) ) - 1 ) + c_i + c_o$   
 $= 2467/2( \exp( 0.5 \cdot 160/(162 \cdot 1) ) - 1 ) + 1.5 + 0$   
 $= 62.5 + 1.5 + 0 = 64 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:  
 $= 2 \cdot S \cdot E \cdot \ln( ( 2 \cdot t + D ) / D ) - P_{liq}$   
 $= 2 \cdot 162 \cdot 1 \cdot \ln( ( 2 \cdot 63.5 + 2467 ) / 2467 ) - 0.0025$   
 $= 163 - 0.0025 = 163 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:  
 $= 2 \cdot S \cdot E \cdot \ln( ( 2 \cdot t + D ) / D )$   
 $= 2 \cdot 175 \cdot 1 \cdot \ln( ( 2 \cdot 65 + 2464 ) / 2464 )$   
 $= 180 \text{ bars}$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / ( E ( D_o^2 - D^2 ) ) \\ &= 160 \cdot 2467^2 / ( 1 ( 2594^2 - 2467^2 ) ) \\ &= 151.513 \text{ N./mm}^2 \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / ( E \cdot \ln( ( 2 \cdot t + D ) / D ) ) \\ &= 0.5 \cdot 160 / ( 1 \cdot \ln( ( 2 \cdot 63.5 + 2467 ) / 2467 ) ) \\ &= 159.380 \text{ N./mm}^2 \end{aligned}$$

#### Percent Fiber Elongation per Table 6.1

$$\begin{aligned} \text{For Heads formed from Segments } &(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) && 4.374 \% \\ \text{For one Piece doubly Curved Heads } &100 \cdot \ln(D_b / (D_f - 2t)) && 40.490 \% \end{aligned}$$

Where  $D_b$  is the blank diameter for one piece doubly curved surfaces [ $D_b$ ]:

$$\begin{aligned} &= \text{square root}( 4 \cdot \text{Head Surface Area} / 3.14159 ) \\ &= \text{square root}( 4 \cdot 107168 / 3.14159 ) \\ &= 3693.926 \text{ mm.} \end{aligned}$$

#### MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

#### Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.800 mm.  
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.400 mm.

Material treatment options: +IT

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### Hydrostatic Test Pressure Results:

User Defined Hydrostatic Test Pressure at High Point 217.000 bars

### Horizontal Test performed per: User Hydro Pressure

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

### Stresses on Elements due to Test Pressure (N./mm<sup>2</sup> & bars):

| From To      | Stress | Allowable | Ratio | Pressure |
|--------------|--------|-----------|-------|----------|
| Bottom D end | 211.3  | 247.0     | 0.855 | 217.24   |
| Shell 1      | 214.3  | 247.0     | 0.868 | 217.24   |
| Shell 2      | 214.3  | 247.0     | 0.868 | 217.24   |
| Shell 3      | 214.3  | 247.0     | 0.868 | 217.24   |
| Top D end    | 211.3  | 247.0     | 0.855 | 217.24   |

### Stress ratios for Nozzle and Pad Materials (N./mm<sup>2</sup>):

| Description | Pad/Nozzle | Ambient | Operating | Ratio |
|-------------|------------|---------|-----------|-------|
| N4_2"       | Nozzle     | 165.00  | 153.86    | 1.072 |
| N3_2"       | Nozzle     | 165.00  | 153.90    | 1.072 |
| K1B_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K3B_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K3D_2'      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K3F_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K2B_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K1A_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K3A_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K3C_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K3E_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| K2A_2"      | Nozzle     | 165.00  | 153.90    | 1.072 |
| MH1_24"     | Nozzle     | 165.00  | 153.86    | 1.072 |
| N1_18"      | Nozzle     | 165.00  | 153.86    | 1.072 |
| N2_16"      | Nozzle     | 165.00  | 153.86    | 1.072 |

Minimum

1.072

### Stress ratios for Pressurized Vessel Elements (N./mm<sup>2</sup>):

| Description  | Ambient | Operating | Ratio |
|--------------|---------|-----------|-------|
| Bottom D end | 175.00  | 161.85    | 1.081 |
| Shell 1      | 175.00  | 161.90    | 1.081 |
| Shell 2      | 175.00  | 161.90    | 1.081 |
| Shell 3      | 175.00  | 161.90    | 1.081 |
| Top D end    | 175.00  | 161.90    | 1.081 |

Minimum

1.081

### Hoop Stress in Nozzle Wall during Pressure Test (N./mm<sup>2</sup>):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| N4_2"       | 28.85   | 237.50    | 0.121 |
| N3_2"       | 74.32   | 237.50    | 0.313 |
| K1B_2"      | 74.32   | 237.50    | 0.313 |
| K3B_2"      | 74.32   | 237.50    | 0.313 |
| K3D_2'      | 74.32   | 237.50    | 0.313 |
| K3F_2"      | 74.32   | 237.50    | 0.313 |
| K2B_2"      | 74.32   | 237.50    | 0.313 |
| K1A_2"      | 74.32   | 237.50    | 0.313 |
| K3A_2"      | 74.32   | 237.50    | 0.313 |
| K3C_2"      | 74.32   | 237.50    | 0.313 |



|         |        |        |       |
|---------|--------|--------|-------|
| K3E_2"  | 74.32  | 237.50 | 0.313 |
| K2A_2"  | 74.32  | 237.50 | 0.313 |
| MH1_24" | 95.34  | 237.50 | 0.401 |
| N1_18"  | 68.03  | 237.50 | 0.286 |
| N2_16"  | 157.77 | 237.50 | 0.664 |

Elements Suitable for Test Pressure.

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## External Pressure Calculation Results Summary:

### ASME Code, Section VIII Division 2, 2025

#### External Pressure Calculations:

| From | To | Section Length<br>mm. | Outside Diameter<br>mm. | Corroded Thickness<br>mm. | Factor A | Factor B<br>N./mm <sup>2</sup> |
|------|----|-----------------------|-------------------------|---------------------------|----------|--------------------------------|
| 10   | 20 | 3000                  | ...                     | ...                       | No Calc  | No Calc                        |
| 20   | 30 | 3000                  | ...                     | ...                       | No Calc  | No Calc                        |
| 30   | 40 | 3000                  | ...                     | ...                       | No Calc  | No Calc                        |
| 40   | 50 | No Calc               | 2594                    | 63.5                      | No Calc  | No Calc                        |
| 50   | 60 | 7621.33               | 2656                    | 126.5                     | No Calc  | No Calc                        |
| 60   | 70 | 7621.33               | 2656                    | 126.5                     | No Calc  | No Calc                        |
| 70   | 80 | 7621.33               | 2656                    | 126.5                     | No Calc  | No Calc                        |
| 80   | 90 | No Calc               | 2594                    | 63.5                      | No Calc  | No Calc                        |

#### External Pressure Calculations:

| From | To | External Actual T.<br>mm. | External Required T.<br>mm. | External Design Pressure<br>bars | External M.A.W.P.<br>bars |
|------|----|---------------------------|-----------------------------|----------------------------------|---------------------------|
| 10   | 20 | ...                       | No Calc                     | ...                              | No Calc                   |
| 20   | 30 | ...                       | No Calc                     | ...                              | No Calc                   |
| 30   | 40 | ...                       | No Calc                     | ...                              | No Calc                   |
| 40   | 50 | 65                        | No Calc                     | ...                              | 112                       |
| 50   | 60 | 128                       | No Calc                     | ...                              | 110                       |
| 60   | 70 | 128                       | No Calc                     | ...                              | 110                       |
| 70   | 80 | 128                       | No Calc                     | ...                              | 110                       |
| 80   | 90 | 65                        | No Calc                     | ...                              | 112                       |

Minimum

110

#### External Pressure Calculations:

| From | To | Actual Length<br>Bet. Stiffeners<br>mm. | Allowable Length<br>Bet. Stiffeners<br>mm. | Ring Inertia<br>Required<br>cm <sup>4</sup> | Ring Inertia<br>Available<br>cm <sup>4</sup> |
|------|----|-----------------------------------------|--------------------------------------------|---------------------------------------------|----------------------------------------------|
| 10   | 20 | 3000                                    | No Calc                                    | No Calc                                     | No Calc                                      |
| 20   | 30 | 3000                                    | No Calc                                    | No Calc                                     | No Calc                                      |
| 30   | 40 | 3000                                    | No Calc                                    | No Calc                                     | No Calc                                      |
| 40   | 50 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 50   | 60 | 7621.33                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 60   | 70 | 7621.33                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 70   | 80 | 7621.33                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 80   | 90 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |

Elements Suitable for External Pressure.

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### Element and Detail Weights:

| From  | To | Element<br>Metal Wgt.<br>kg. | Element<br>ID Volume<br>cm3 | Corroded<br>Metal Wgt.<br>kg. | Corroded<br>ID Volume<br>cm3 | Extra due<br>Misc %<br>kg. |
|-------|----|------------------------------|-----------------------------|-------------------------------|------------------------------|----------------------------|
| 10    | 20 | 4298.28                      | ...                         | 4113.64                       | ...                          | ...                        |
| 20    | 30 | 923.31                       | ...                         | 877.151                       | ...                          | ...                        |
| 30    | 40 | 923.31                       | ...                         | 877.151                       | ...                          | ...                        |
| 40    | 50 | 5804.81                      | 3917134                     | 5693.8                        | 3931460                      | ...                        |
| 50    | 60 | 15757                        | 9049409                     | 15581.6                       | 9072047                      | ...                        |
| 60    | 70 | 19696.3                      | 11311760                    | 19477                         | 11340058                     | ...                        |
| 70    | 80 | 18120.6                      | 10406821                    | 17918.9                       | 10432854                     | ...                        |
| 80    | 90 | 5804.51                      | 3917134                     | 5693.51                       | 3931460                      | ...                        |
| Total |    | 71328                        | 38602264.00                 | 70232                         | 38707880.00                  | 0                          |

### Weight of Details:

| From | Type | Weight of<br>Detail<br>kg. | X Offset,<br>Dtl. Cent.<br>mm. | Y Offset,<br>Dtl. Cent.<br>mm. | Z Offset<br>Dtl. Cent.<br>mm. | Description    |
|------|------|----------------------------|--------------------------------|--------------------------------|-------------------------------|----------------|
| 10   | Wght | 325                        | ...                            | 1000                           | ...                           | skirt opening  |
| 40   | Liqd | 2489.78                    | ...                            | -616                           | ...                           | Liquid: 40     |
| 40   | Insl | 242.411                    | ...                            | -616                           | ...                           | Insul          |
| 40   | Nozl | 132.647                    | ...                            | -1299.5                        | ...                           | N4_2"          |
| 40   | Wght | 1150                       | ...                            | ...                            | ...                           | Btm dish SF    |
| 50   | Plat | 3699.81                    | 0.20065E-04                    | ...                            | ...                           | PLAT-1         |
| 50   | Liqd | 5751.91                    | ...                            | 1000                           | ...                           | Liquid: 50     |
| 50   | Insl | 342.022                    | ...                            | 1000                           | ...                           | Insul          |
| 50   | Nozl | 15.129                     | 1275.01                        | 630                            | ...                           | N3_2"          |
| 50   | Nozl | 36.34                      | -1400                          | 250                            | ...                           | K1B_2"         |
| 50   | Nozl | 36.34                      | ...                            | 250                            | -1400                         | K3B_2"         |
| 50   | Nozl | 36.34                      | -591.665                       | 250                            | -1268.83                      | K3D_2'         |
| 50   | Nozl | 36.34                      | 1400                           | 250                            | ...                           | K3F_2"         |
| 50   | Nozl | 36.34                      | ...                            | 250                            | 1400                          | K2B_2"         |
| 60   | Plat | 1849.9                     | -1442.59                       | 1500                           | ...                           | PLAT-2         |
| 60   | Liqd | 5751.91                    | ...                            | 1000                           | ...                           | Liquid: 60     |
| 60   | Insl | 427.527                    | ...                            | 1250                           | ...                           | Insul          |
| 60   | Nozl | 36.34                      | -1400                          | 2149.97                        | ...                           | K1A_2"         |
| 60   | Nozl | 36.34                      | 1400                           | 2149.97                        | ...                           | K3A_2"         |
| 60   | Nozl | 36.34                      | ...                            | 1549.97                        | -1400                         | K3C_2"         |
| 60   | Nozl | 36.34                      | -803.007                       | 1549.97                        | -1146.81                      | K3E_2"         |
| 60   | Nozl | 36.34                      | ...                            | 2149.97                        | 1400                          | K2A_2"         |
| 60   | Wght | 1400                       | ...                            | 1260.97                        | ...                           | Cleat          |
| 60   | Wght | 2600                       | ...                            | 1260.97                        | ...                           | Lifting wt     |
| 60   | Wght | 6100                       | ...                            | 1260.97                        | ...                           | 7% margin      |
| 70   | Plat | 1849.9                     | 1442.59                        | 800                            | ...                           | PLAT-3         |
| 70   | Pack | 114.136                    | ...                            | 2075                           | ...                           | demister       |
| 70   | Insl | 393.325                    | ...                            | 1150                           | ...                           | Insul          |
| 70   | Nozl | 4676.97                    | 1400                           | 1400                           | ...                           | MH1_24"        |
| 70   | Nozl | 723.326                    | -1375.01                       | 749.975                        | ...                           | N1_18"         |
| 70   | Wght | 630                        | ...                            | 1200                           | ...                           | FP+FLG_dead_N1 |
| 70   | Wght | 400                        | ...                            | 0.0040691                      | ...                           | FP_liq_N1      |
| 70   | Wght | 800                        | -1300                          | 1800                           | ...                           | Davit          |
| 70   | Wght | 2226                       | -2737                          | 749.975                        | ...                           | N1_OVH_PP      |
| 80   | Plat | 4100.55                    | ...                            | 1500                           | ...                           | TOP PLT        |
| 80   | Insl | 242.411                    | ...                            | 616                            | ...                           | Insul          |
| 80   | Nozl | 248.121                    | ...                            | 1349.5                         | ...                           | N2_16"         |
| 80   | Wght | 900                        | ...                            | ...                            | ...                           | Top dish SF    |
| 80   | Wght | 1105                       | -2711                          | ...                            | ...                           | N2_OVH_PP      |
| 80   | Forc | ...                        | ...                            | 1231                           | ...                           | N2_FxFz        |

### Total Weight of Each Detail Type:

|           |         |
|-----------|---------|
| Platforms | 11500.2 |
| Packing   | 114.1   |



Liquid 13993.6  
 Insulation 1647.7  
 Nozzles 6159.6  
 Weights 17636.0

Sum of the Detail Weights 51051.2 kg.

#### Weight Summation Results: (kg.)

|               | Fabricated | Shop Test | Shipping | Erected  | Empty    | Operating |
|---------------|------------|-----------|----------|----------|----------|-----------|
| Main Elements | 71328.2    | 71328.2   | 71328.2  | 71328.2  | 71328.2  | 71328.2   |
| Nozzles       | 6159.6     | 6159.6    | 6159.6   | 6159.6   | 6159.6   | 6159.6    |
| Wld Weights   | 13905.0    | 13905.0   | 13905.0  | 13905.0  | 13905.0  | 13905.0   |
| Platforms     | ...        | ...       | ...      | 11500.2  | 11500.2  | 11500.2   |
| Packing       | ...        | ...       | ...      | ...      | 114.1    | 114.1     |
| Insulation    | ...        | ...       | ...      | 1647.7   | 1647.7   | 1647.7    |
| Empty Weights | ...        | ...       | ...      | 3731.0   | 3731.0   | ...       |
| Test Weights  | ...        | 3731.0    | ...      | ...      | ...      | ...       |
| Ope Weights   | ...        | ...       | ...      | ...      | ...      | 3731.0    |
| Ope. Liquid   | ...        | ...       | ...      | ...      | ...      | 13993.6   |
| Test Liquid   | ...        | 38578.7   | ...      | ...      | ...      | ...       |
| Totals        | 91392.8    | 133702.5  | 91392.8  | 108271.6 | 108385.8 | 122379.3  |

#### Field Installation Options:

Platforms installed before lifting.

\* Packing installed after lifting.

Insulation installed before lifting.

#### Weight Summary:

Fabricated Wt. - Bare weight without removable internals 91392.8 kg.  
 Shop Test Wt. - Fabricated weight + water ( full ) 133702.5 kg.  
 Shipping Wt. - Fab. weight + removable intls.+ shipping app. 91392.8 kg.  
 Erected Wt. - Fab. wt + or - loose items (trays,platforms etc.) 108271.6 kg.  
 Ope. Wt. no Liq - Fab. weight + internals. + details + weights 108385.8 kg.  
 Operating Wt. - Empty weight + operating liq. uncorroded 122379.3 kg.

#### Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating) 51242.4 kg.  
 Mass of the Upper 1/3 of the Vertical Vessel (filled w/water) 61378.7 kg.  
 Mass of the Upper 1/3 of the Vertical Vessel (empty) 50297.8 kg.

#### Surface Areas of Elements:

| From | To | Outside Surface Area<br>cm^2 | Inside Surface Area<br>cm^2 |
|------|----|------------------------------|-----------------------------|
| 10   | 20 | 160724                       | 156954                      |
| 20   | 30 | 40181                        | 39238.5                     |
| 30   | 40 | 40181                        | 39238.5                     |
| 40   | 50 | 107168                       | 95367.7                     |
| 50   | 60 | 166881                       | 150796                      |
| 60   | 70 | 208602                       | 188496                      |
| 70   | 80 | 191914                       | 173416                      |
| 80   | 90 | 107168                       | 95367.7                     |

Total 1022819.3 938874.3 cm2  
 Total 1101.0 1010.6 ft<sup>2</sup>

#### Element and Detail Weights:

| From | To | Total Ele.<br>Empty Wgt.<br>kg. | Total. Ele.<br>Oper. Wgt.<br>kg. | Total. Ele.<br>Hydro. Wgt.<br>kg. | Total Dtl.<br>Offset Mom.<br>Kg-m. | Oper. Wgt.<br>No Liquid<br>kg. |
|------|----|---------------------------------|----------------------------------|-----------------------------------|------------------------------------|--------------------------------|
| 10   | 20 | 4623.28                         | 4623.28                          | 4623.28                           | ...                                | 4623.28                        |



|    |    |         |         |         |         |         |
|----|----|---------|---------|---------|---------|---------|
| 20 | 30 | 923.31  | 923.31  | 923.31  | ...     | 923.31  |
| 30 | 40 | 923.31  | 923.31  | 923.31  | ...     | 923.31  |
| 40 | 50 | 7087.46 | 9819.65 | 11244.6 | ...     | 7329.87 |
| 50 | 60 | 15953.9 | 25747.6 | 29039.6 | 2.21152 | 19995.7 |
| 60 | 70 | 29978   | 38007.3 | 43560.3 | 2697.87 | 32255.4 |
| 70 | 80 | 27576.9 | 29934.3 | 40220.6 | 1089.28 | 29934.3 |
| 80 | 90 | 8057.63 | 12400.6 | 16315.3 | 2995.71 | 12400.6 |

Empty Support Force + the Sum of the Y forces 96637.45 Kg  
 Operating Support Force + the Sum of the Y forces 123893.02 Kg  
 Hydro Support Force + the Sum of the Y forces 148364.02 Kg

#### Cumulative Vessel Weight

| From | To | Cumulative Ope<br>Wgt. No Liquid<br>kg. | Cumulative<br>Oper. Wgt.<br>kg. | Cumulative<br>Hydro. Wgt.<br>kg. |
|------|----|-----------------------------------------|---------------------------------|----------------------------------|
| 10   | 20 | 108386                                  | 122379                          | 146850                           |
| 20   | 30 | 103762                                  | 117756                          | 142227                           |
| 30   | 40 | 102839                                  | 116833                          | 141304                           |
| 40   | 50 | 101916                                  | 115909                          | 140380                           |
| 50   | 60 | 94586                                   | 106090                          | 129136                           |
| 60   | 70 | 74590.3                                 | 80342.2                         | 100096                           |
| 70   | 80 | 42334.8                                 | 42334.8                         | 56535.9                          |
| 80   | 90 | 12400.6                                 | 12400.6                         | 16315.3                          |

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

#### Cumulative Vessel Moment

| From | To | Cumulative<br>Empty Mom.<br>Kg-m. | Cumulative<br>Oper. Mom.<br>Kg-m. | Cumulative<br>Hydro. Mom.<br>Kg-m. |
|------|----|-----------------------------------|-----------------------------------|------------------------------------|
| 10   | 20 | 6785.08                           | 6785.08                           | 6785.08                            |
| 20   | 30 | 6785.08                           | 6785.08                           | 6785.08                            |
| 30   | 40 | 6785.08                           | 6785.08                           | 6785.08                            |
| 40   | 50 | 6785.08                           | 6785.08                           | 6785.08                            |
| 50   | 60 | 6785.08                           | 6785.08                           | 6785.08                            |
| 60   | 70 | 6782.87                           | 6782.87                           | 6782.87                            |
| 70   | 80 | 4084.99                           | 4084.99                           | 4084.99                            |
| 80   | 90 | 2995.71                           | 2995.71                           | 2995.71                            |



### Nozzle Flange MAWP Results: (bars & °C)

| Nozzle Description | Flange Rating |                              | Des. Temp | Class | Grade/Group | Equiv. Press | Max Pressure |     |     |
|--------------------|---------------|------------------------------|-----------|-------|-------------|--------------|--------------|-----|-----|
|                    | Ope.          | Ambient                      |           |       |             |              | UG-44        | 50% | DNV |
| K3B_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| K3D_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| K3F_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| K2B_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| K1A_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| K3A_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| K3C_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| K3E_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| K2A_2"             | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| MH1_24"            | 239.66        | 255.10                       | 80        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| Min Rating         | 239.663       | 255.100, [for Core Elements] |           |       |             |              | ...          | ... | ... |

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

#### Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 8.59496 Hz.

The Natural Frequency for the Vessel (Ope...) is 8.35247 Hz.

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**Forces/Moments Applied to Vessel (Combined w/Wind Loads)**

| From | To | X and Z Dir<br>Force Res.<br>Kgf | X,Z Moment<br>and For Res<br>Kg-m. |
|------|----|----------------------------------|------------------------------------|
| 10   | 20 | ...                              | ...                                |
| 20   | 30 | ...                              | ...                                |
| 30   | 40 | ...                              | ...                                |
| 40   | 50 | ...                              | ...                                |
| 50   | 60 | ...                              | ...                                |
| 60   | 70 | ...                              | ...                                |
| 70   | 80 | ...                              | ...                                |
| 80   | 90 | 3172.08                          | 9648.4                             |

**Forces/Moments Applied to Vessel (Combined w/Seismic Loads)**

| From | To | X and Z Dir<br>Force Res.<br>Kgf | X,Z Moment<br>and For Res<br>Kg-m. |
|------|----|----------------------------------|------------------------------------|
| 10   | 20 | ...                              | ...                                |
| 20   | 30 | ...                              | ...                                |
| 30   | 40 | ...                              | ...                                |
| 40   | 50 | ...                              | ...                                |
| 50   | 60 | ...                              | ...                                |
| 60   | 70 | ...                              | ...                                |
| 70   | 80 | ...                              | ...                                |
| 80   | 90 | 3172.08                          | 9648.4                             |

**User Input Forces and Moments:**

| From<br>Node | Distance<br>From | - - -<br>Fx | Forces - - -<br>Fy | - - -<br>Fz | - - -<br>Mx | Moments - - -<br>My | - - -<br>Mz |
|--------------|------------------|-------------|--------------------|-------------|-------------|---------------------|-------------|
| 80           | 1231.00          | 2243.       | -1514.             | 2243.       | 4485.       |                     | 3588.       |



## Wind Load Calculation Results:

### Wind Load Calculations per India Std. IS 875 (Part 3):2015, Amd. 1&2 (2020):

#### Input Values:

|                                             |                           |         |
|---------------------------------------------|---------------------------|---------|
| Wind Design Code                            | IS 875 (PART 3):2015-2020 |         |
| Basic Wind Speed for IS-875                 | 180                       | Km/hr   |
| Wind Zone Number                            | 5                         |         |
| Base Elevation                              | 300                       | mm.     |
| Percent Wind for Hydrotest                  | 33.0                      |         |
| Risk Factor                                 | [k1]                      | 1.0     |
| Terrain Category                            |                           | 2       |
| Equipment Class                             |                           | Class A |
| Topography Factor                           | [k3]                      | 1.0     |
| Importance Factor                           | [k4]                      | 1.15    |
| Damping Factor (Beta) for Wind (Ope)        |                           | 0.0100  |
| Damping Factor (Beta) for Wind (Empty)      |                           | 0.0000  |
| Damping Factor (Beta) for Wind (Filled)     |                           | 0.0000  |
| Use Gust Response Factor (Dynamic Analysis) |                           | No      |

#### Calculated Values:

Vessel Height to Diameter ratio: 4.067

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 8.352 Hz

Note: Vessel Natural frequency is  $\geq 1$  Hz.

Note: Vessel height to diameter ratio (4.067) is  $\leq 5$ .

Note: Dynamic wind analysis is not required.

Design Wind Speed [Vz]:

= Basic Wind Speed  $\cdot k_1 \cdot k_2 \cdot k_3 \cdot k_4$  (for the 1st element)

=  $50 \cdot 1 \cdot 1 \cdot 1 \cdot 1.15$

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

=  $\max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$

=  $\max(0.7 \cdot 1984, 1 \cdot 0.87 \cdot 1 \cdot 0.6 \cdot 57.5^2)$

= 1725.3 N/m<sup>2</sup>

Element Wind Load (operating condition):

= Wind Area  $\cdot C_f \cdot p_d$

=  $6.8 \cdot 0.77 \cdot 1725$

= 9026.5 N

| From | Height<br>mm. | k1     | k2     | k3     | Vz<br>m/sec | Cf     |
|------|---------------|--------|--------|--------|-------------|--------|
| 10   | 1300.00       | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.7689 |
| 20   | 2550.00       | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.7689 |
| 30   | 3050.00       | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.7689 |
| 40   | 3299.98       | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.7689 |
| 50   | 4300.00       | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.7689 |
| 60   | 6550.00       | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.7689 |
| 70   | 8950.00       | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.7689 |
| 80   | 10688.64      | 1.0000 | 1.0069 | 1.0000 | 57.89       | 0.7689 |

#### Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

#### Nomenclature



Cf - Correction factor for natural frequency  
 D - Average outer insulated diameter of vessel, mm.  
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable  
 Dr - Average outer insulated of top half of vessel, mm.  
 f - Natural frequency of vibration (Hertz)  
 f1 - Natural frequency of bare vessel based on a unit value of (D/L<sup>2</sup>)(10<sup>4</sup>)  
 L - Total height of structure mm.  
 Lc - Total length of conical section(s) of vessel, mm.  
 tb - Uncorroded plate thickness at bottom of vessel, mm.  
 V30 - Design Wind Speed provided by user, Km/hr  
 Vc - Critical wind velocity, Km/hr  
 Vw - Maximum wind speed at top of structure, Km/hr  
 W - Total corroded weight of structure, Kgf  
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf  
 Z - Maximum amplitude of vibration at top of vessel, mm.  
 D1 - Logarithmic decrement ( taken as 0.03 for Welded Structures )  
 Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)  
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: 0/9800 = 0

:: [ D / L<sup>2</sup> ] • 10<sup>4</sup> < 8 (English Units, ft.)

:: [ 9.03/32.2<sup>2</sup> ] • 10<sup>4</sup> = 87.3 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.40047E-06 no chance. [Vp]:

= W / ( L • Dr<sup>2</sup> )

= 121284/( 9800 • 2776<sup>2</sup> )

= 0.16060E-05

Since Vp is > 0.40047E-06 no further vibration analysis is required !

#### **Platform Load Calculations**

| ID      | Wind Area<br>cm <sup>2</sup> | Elevation<br>mm. | Pressure<br>Kgs/m <sup>2</sup> | Force<br>Kgf | Cf   |
|---------|------------------------------|------------------|--------------------------------|--------------|------|
| PLAT-1  | 43200.00                     | 3300.00          | 176.01                         | 760.36       | 1.20 |
| PLAT-2  | 43200.00                     | 6800.00          | 176.01                         | 760.36       | 1.20 |
| PLAT-3  | 43200.00                     | 8600.00          | 176.01                         | 760.36       | 1.20 |
| TOP PLT | 88920.00                     | 11599.97         | 180.73                         | 1607.05      | 1.00 |

#### **Wind Loads on Masses/Equipment/Piping**

| ID            | Wind Area<br>cm <sup>2</sup> | Elevation<br>mm. | Pressure<br>Kgs/m <sup>2</sup> | Force<br>Kgf |
|---------------|------------------------------|------------------|--------------------------------|--------------|
| skirt opening | 0.00                         | 1300.00          | 176.01                         | 0.00         |
| Btm dish SF   | 0.00                         | 3299.97          | 176.01                         | 0.00         |
| Cleat         | 0.00                         | 6560.97          | 176.01                         | 0.00         |
| Lifting wt    | 0.00                         | 6560.97          | 176.01                         | 0.00         |
| 7% margin     | 0.00                         | 6560.97          | 176.01                         | 0.00         |
| FP+FLGdeadN   | 0.00                         | 9000.00          | 176.08                         | 0.00         |
| FPliqN1       | 0.00                         | 7800.00          | 176.01                         | 0.00         |
| Davit         | 0.00                         | 9600.00          | 176.92                         | 0.00         |
| N10VHPP       | 0.00                         | 8549.97          | 176.01                         | 0.00         |
| Top dish SF   | 0.00                         | 10099.97         | 177.62                         | 0.00         |
| N20VHPP       | 0.00                         | 10099.97         | 177.62                         | 0.00         |

The Natural Frequency for the Vessel (Ope...) is 8.35247 Hz.

#### **Wind Load Calculation:**

| From | To | Wind<br>Height<br>mm. | Wind<br>Diameter<br>mm. | Wind<br>Area<br>cm <sup>2</sup> | Wind<br>Pressure<br>Kgs/m <sup>2</sup> | Element<br>Wind Load<br>Kgf |
|------|----|-----------------------|-------------------------|---------------------------------|----------------------------------------|-----------------------------|
|------|----|-----------------------|-------------------------|---------------------------------|----------------------------------------|-----------------------------|



|    |    |         |         |         |         |          |
|----|----|---------|---------|---------|---------|----------|
| 10 | 20 | 1300    | 3402.14 | 68042.8 | 176.01  | 1012.56  |
| 20 | 30 | 2550    | 3402.14 | 17010.7 | 176.01  | 253.139  |
| 30 | 40 | 3050    | 3402.14 | 17010.7 | 176.01  | 253.139  |
| 40 | 50 | 3299.98 | 3689.42 | 1.12454 | 176.01  | 0.016734 |
| 50 | 60 | 4300    | 3771.88 | 75437.6 | 176.01  | 1764.49  |
| 60 | 70 | 6550    | 3771.88 | 94297   | 176.01  | 2271.57  |
| 70 | 80 | 8950    | 3771.88 | 86753.3 | 176.01  | 2094.68  |
| 80 | 90 | 10688.6 | 3689.42 | 40191.7 | 178.442 | 2524.84  |

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

*Be sure the wind speed is in accordance with the specified wind design code.*

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## Earthquake Load Calculation Results:

### Input Values:

|                                 |              |
|---------------------------------|--------------|
| Seismic Design Code             | Spectrum     |
| Range Type                      | Period       |
| Ordinate Type                   | Acceleration |
| Include Missing Mass Components | Yes          |
| Shock Scale Factor X Direction  | 1.0000       |
| Shock Scale Factor Y Direction  | 0.6700       |
| Combination Method              | Group        |
| Spectrum Name                   | User Defined |

### User Defined Spectrum Data Points:

| Acceleration | Period |
|--------------|--------|
| 0.0700       | 0.0100 |
| 0.1170       | 0.0500 |
| 0.1750       | 0.1000 |
| 0.1750       | 0.1300 |
| 0.1750       | 0.2000 |
| 0.1750       | 0.2700 |
| 0.1750       | 0.3000 |
| 0.1750       | 0.4000 |
| 0.1400       | 0.5000 |
| 0.1270       | 0.5500 |
| 0.1170       | 0.6000 |
| 0.1050       | 0.6700 |
| 0.0870       | 0.8000 |
| 0.0700       | 1.0000 |
| 0.0580       | 1.2000 |
| 0.0500       | 1.4000 |
| 0.0440       | 1.6000 |
| 0.0390       | 1.8000 |
| 0.0350       | 2.0000 |
| 0.0290       | 2.4000 |

### Computed Natural Frequencies (OPE): 4

| Mode | Freq (Hz) | Freq (Rad/Sec) | Period (Sec) |
|------|-----------|----------------|--------------|
| 1    | 8.4522    | 53.1066        | 0.1183       |
| 2    | 47.2098   | 296.6279       | 0.0212       |
| 3    | 53.4543   | 335.8632       | 0.0187       |
| 4    | 133.4496  | 838.4888       | 0.0075       |

### Mass Participation Factors: 4

| Mode | X           | Y           |
|------|-------------|-------------|
| 1    | 0.56702197  | 0.00000000  |
| 2    | 0.00399429  | -0.00000000 |
| 3    | -0.00000000 | 0.00478660  |
| 4    | 0.00006615  | 0.00000000  |

### Computed EigenVectors, Mass Normalized: 4

| Node     | Dx                 | Dy        | Rz        |
|----------|--------------------|-----------|-----------|
| Mode # 1 | Freq (HZ) = 8.4522 |           |           |
| 10       | 0.000000           | 0.000000  | -0.000000 |
| 20       | 0.006880           | 0.000000  | -0.000104 |
| 30       | 0.009781           | 0.000000  | -0.000125 |
| 40       | 0.013076           | 0.000000  | -0.000144 |
| 50       | 0.013086           | 0.000000  | -0.000144 |
| 60       | 0.025542           | -0.000000 | -0.000157 |
| 70       | 0.042035           | -0.000000 | -0.000165 |
| 80       | 0.057390           | 0.000000  | -0.000167 |
| 90       | 0.057400           | 0.000000  | -0.000167 |

Mode # 2 Freq (HZ) = 47.2098



|    |           |           |           |
|----|-----------|-----------|-----------|
| 10 | 0.000001  | 0.000000  | -0.000001 |
| 20 | 0.040770  | 0.000000  | -0.000014 |
| 30 | 0.049323  | -0.000000 | 0.000054  |
| 40 | 0.056044  | -0.000000 | 0.000150  |
| 50 | 0.056045  | -0.000000 | 0.000150  |
| 60 | 0.042095  | -0.000000 | 0.000263  |
| 70 | 0.004729  | 0.000000  | 0.000383  |
| 80 | -0.037649 | -0.000000 | 0.000421  |
| 90 | -0.037677 | -0.000000 | 0.000421  |

Mode # 3 Freq (HZ) = 53.4543

|    |           |          |           |
|----|-----------|----------|-----------|
| 10 | -0.000000 | 0.000003 | 0.000000  |
| 20 | -0.000000 | 0.021931 | -0.000000 |
| 30 | -0.000000 | 0.027409 | -0.000000 |
| 40 | -0.000000 | 0.032855 | -0.000000 |
| 50 | -0.000000 | 0.032863 | -0.000000 |
| 60 | 0.000000  | 0.037042 | -0.000000 |
| 70 | -0.000000 | 0.040605 | -0.000000 |
| 80 | 0.000000  | 0.042097 | -0.000000 |
| 90 | 0.000000  | 0.042097 | -0.000000 |

Mode # 4 Freq (HZ) = 133.4496

|    |           |           |           |
|----|-----------|-----------|-----------|
| 10 | 0.000001  | 0.000000  | -0.000000 |
| 20 | 0.045279  | 0.000000  | 0.000133  |
| 30 | 0.048969  | 0.000000  | 0.000250  |
| 40 | 0.048362  | -0.000000 | 0.000388  |
| 50 | 0.048331  | -0.000000 | 0.000388  |
| 60 | -0.017177 | -0.000000 | 0.000307  |
| 70 | -0.043459 | 0.000000  | -0.000215 |
| 80 | 0.038308  | -0.000000 | -0.000524 |
| 90 | 0.038355  | -0.000000 | -0.000524 |

Mass Percentages:

| Included(X) | Included(Y) | Added(X) | Added(Y) |
|-------------|-------------|----------|----------|
| 98.08       | 97.19       | 1.92     | 2.81     |

NOTE: In the following reports, "Max Contrib" displays the contribution of the Mode/Load combination having the maximum impact on the total (and names that Mode/Load combination).

Restraint Loads:

| Node    | Fx(Kgf) | Fy(Kgf) | Mz(Kg-m. ) |
|---------|---------|---------|------------|
| 10      | 17244.3 | 6368.8  | 129018.4   |
| Max     | 17155.0 | 6366.8  | 129005.2   |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)      |

Element Forces and Moments:

| Node    | Fx(Kgf) | Fy(Kgf) | Mz(Kg-m. ) |
|---------|---------|---------|------------|
| 10      | 17243.5 | 6367.1  | 129018.4   |
| Max     | 17155.0 | 6366.8  | 129005.2   |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)      |
| 20      | 17243.5 | 6367.1  | 94708.8    |
| Max     | 17155.0 | 6366.8  | 94694.5    |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)      |
| 20      | 17164.4 | 6291.0  | 94708.8    |
| Max     | 17084.6 | 6291.0  | 94694.5    |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)      |
| 30      | 17164.4 | 6291.0  | 86187.5    |
| Max     | 17084.6 | 6291.0  | 86152.0    |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)      |
| 30      | 17123.2 | 6255.6  | 86187.5    |
| Max     | 17047.2 | 6255.6  | 86152.0    |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)      |



|         |         |        |         |
|---------|---------|--------|---------|
| 40      | 17123.2 | 6255.6 | 77697.4 |
| Max     | 17047.2 | 6255.6 | 77628.3 |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 40      | 16809.6 | 6008.9 | 77697.4 |
| Max     | 16756.3 | 6008.8 | 77628.3 |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 50      | 16809.6 | 6008.9 | 77670.9 |
| Max     | 16756.3 | 6008.8 | 77601.7 |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 50      | 15798.7 | 5193.8 | 77670.9 |
| Max     | 15792.4 | 5191.5 | 77601.7 |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 60      | 15798.7 | 5193.8 | 46202.3 |
| Max     | 15792.4 | 5191.5 | 46016.3 |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 60      | 12445.4 | 3546.4 | 46202.3 |
| Max     | 12420.0 | 3540.2 | 46016.3 |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 70      | 12445.4 | 3546.4 | 15123.2 |
| Max     | 12420.0 | 3540.2 | 14965.6 |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 70      | 6574.2  | 1615.7 | 15123.2 |
| Max     | 6505.6  | 1611.2 | 14965.6 |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 80      | 6574.2  | 1615.7 | 2.4     |
| Max     | 6505.6  | 1611.2 | 2.3     |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 80      | 1489.6  | 366.0  | 2.4     |
| Max     | 1474.1  | 365.0  | 2.3     |
| Contrib | 1 (X)   | 3 (Y)  | 1 (X)   |
| 90      | 1489.6  | 366.0  | 0.0     |
| Max     | 1474.1  | 365.0  | 0.0     |
| Contrib | 1 (X)   | 3 (Y)  |         |

Displacements:

| Node    | Dx(mm.) | Dy(mm.) | Rz(Deg) |
|---------|---------|---------|---------|
| 10      | 0.0000  | 0.0000  | 0.0000  |
| Max     | 0.0000  | 0.0000  | 0.0000  |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)   |
| 20      | 0.0992  | 0.0027  | 0.0034  |
| Max     | 0.0991  | 0.0027  | 0.0034  |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)   |
| 30      | 0.1410  | 0.0033  | 0.0041  |
| Max     | 0.1409  | 0.0033  | 0.0041  |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)   |
| 40      | 0.1884  | 0.0040  | 0.0047  |
| Max     | 0.1883  | 0.0040  | 0.0047  |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)   |
| 50      | 0.1886  | 0.0040  | 0.0047  |
| Max     | 0.1885  | 0.0040  | 0.0047  |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)   |
| 60      | 0.3679  | 0.0045  | 0.0051  |
| Max     | 0.3679  | 0.0045  | 0.0051  |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)   |
| 70      | 0.6054  | 0.0049  | 0.0054  |
| Max     | 0.6054  | 0.0049  | 0.0054  |
| Contrib | 1 (X)   | 3 (Y)   | 1 (X)   |



|         |        |        |        |
|---------|--------|--------|--------|
| 80      | 0.8266 | 0.0051 | 0.0054 |
| Max     | 0.8266 | 0.0051 | 0.0054 |
| Contrib | 1 (X)  | 3 (Y)  | 1 (X)  |
| 90      | 0.8267 | 0.0051 | 0.0054 |
| Max     | 0.8267 | 0.0051 | 0.0054 |
| Contrib | 1 (X)  | 3 (Y)  | 1 (X)  |



**Bending Moments due to user defined forces and moments.**

**User Force/Moment Shear and Bending**

| From | To | Distance to Support mm. | Cumulative Shr Wind Cas Kg-f | Cumulative Shr Eqk Cas Kg-f | Wind Bending Kg-m. | Earthquake Bending Kg-m. |
|------|----|-------------------------|------------------------------|-----------------------------|--------------------|--------------------------|
| 10   | 20 | 1000                    | 3172.08                      | 17243.5                     | 40705.2            | 40705.2                  |
| 20   | 30 | 2250                    | 3172.08                      | 17164.4                     | 34361.8            | 34361.8                  |
| 30   | 40 | 2750                    | 3172.08                      | 17123.2                     | 32776              | 32776                    |
| 40   | 50 | 2999.98                 | 3172.08                      | 16809.6                     | 31190.3            | 31190.3                  |
| 50   | 60 | 4000                    | 3172.08                      | 15798.7                     | 31190.2            | 31190.2                  |
| 60   | 70 | 6250                    | 3172.08                      | 12445.4                     | 24847.6            | 24847.6                  |
| 70   | 80 | 8650                    | 3172.08                      | 6574.17                     | 16921              | 16921                    |
| 80   | 90 | 9799.99                 | 3172.08                      | 1489.64                     | 9634.19            | 9634.19                  |

**Note:**

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;  
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

**Wind/Earthquake Shear, Bending:**

| From | To | Distance to Support mm. | Cumulative Wind Shear Kgf | Earthquake Shear Kgf | Wind Bending Kg-m. | Earthquake Bending Kg-m. |
|------|----|-------------------------|---------------------------|----------------------|--------------------|--------------------------|
| 10   | 20 | 1000                    | 10174.4                   | 17243.5              | 67883.6            | 129018                   |
| 20   | 30 | 2250                    | 9161.88                   | 17164.4              | 48546.9            | 94708.8                  |
| 30   | 40 | 2750                    | 8908.74                   | 17123.2              | 44029.2            | 86187.5                  |
| 40   | 50 | 2999.98                 | 8655.6                    | 16809.6              | 39638.3            | 77697.4                  |
| 50   | 60 | 4000                    | 8655.58                   | 15798.7              | 39638              | 77670.9                  |
| 60   | 70 | 6250                    | 6891.1                    | 12445.4              | 24091              | 46202.3                  |
| 70   | 80 | 8650                    | 4619.52                   | 6574.17              | 9702.45            | 15123.2                  |
| 80   | 90 | 9799.99                 | 2524.84                   | 1489.64              | 1486.34            | 2.36484                  |

**Note:**

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;  
 Wind: 1.100; Earthquake: 1.000.

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### Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

#### Wind Load Deflection:

| From | To | Cumulative Shear Kgf | Centroid Deflection mm. | Elem. End Deflection mm. | Elem. Ang. Rotation |
|------|----|----------------------|-------------------------|--------------------------|---------------------|
| 10   | 20 | 3172.08              | 0.0051917               | 0.020213                 | 0.19659E-04         |
| 20   | 30 | 3172.08              | 0.025407                | 0.03115                  | 0.24055E-04         |
| 30   | 40 | 3172.08              | 0.03743                 | 0.044233                 | 0.28243E-04         |
| 40   | 50 | 3172.08              | 0.044234                | 0.044234                 | 0.28243E-04         |
| 50   | 60 | 3172.08              | 0.073386                | 0.10423                  | 0.31622E-04         |
| 60   | 70 | 3172.08              | 0.14486                 | 0.18747                  | 0.34771E-04         |
| 70   | 80 | 3172.08              | 0.22808                 | 0.26976                  | 0.36614E-04         |
| 80   | 90 | 3172.08              | 0.26976                 | 0.26976                  | 0.36614E-04         |

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)  
 Allowable deflection : 49.000 Actual deflection : 0.270 mm.

The following table is for Wind Deflection; Operating Case.

#### Wind Load Deflection:

| From | To | Cumulative Shear Kgf | Centroid Deflection mm. | Elem. End Deflection mm. | Elem. Ang. Rotation |
|------|----|----------------------|-------------------------|--------------------------|---------------------|
| 10   | 20 | 10174.4              | 0.0084581               | 0.032121                 | 0.30431E-04         |
| 20   | 30 | 9161.88              | 0.040124                | 0.048895                 | 0.36564E-04         |
| 30   | 40 | 8908.74              | 0.058394                | 0.068587                 | 0.42107E-04         |
| 40   | 50 | 8655.6               | 0.068587                | 0.068588                 | 0.42107E-04         |
| 50   | 60 | 8655.58              | 0.11182                 | 0.15697                  | 0.45961E-04         |
| 60   | 70 | 6891.1               | 0.21544                 | 0.27547                  | 0.48467E-04         |
| 70   | 80 | 4619.52              | 0.33153                 | 0.388                    | 0.49196E-04         |
| 80   | 90 | 2524.84              | 0.388                   | 0.388                    | 0.49196E-04         |

#### Critical Wind Velocity for Tower Vibration:

| From | To | 1st Crit. Wind Speed Km/hr | 2nd Crit. Wind Speed Km/hr |
|------|----|----------------------------|----------------------------|
| 10   | 20 | 510.115                    | 3188.22                    |
| 20   | 30 | 510.115                    | 3188.22                    |
| 30   | 40 | 510.115                    | 3188.22                    |
| 40   | 50 | 553.19                     | 3457.44                    |
| 50   | 60 | 565.554                    | 3534.71                    |
| 60   | 70 | 565.554                    | 3534.71                    |
| 70   | 80 | 565.554                    | 3534.71                    |
| 80   | 90 | 553.19                     | 3457.44                    |

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)  
 Allowable deflection : 49.000 Actual deflection : 0.388 mm.

Total Deflection in the Operating Condition + Applied Forces:  
 Allowable deflection : 49.000 Actual deflection : 0.658 mm.



**Longitudinal Stress Constants:**

| From | To | Metal Area<br>New<br>cm <sup>2</sup> | Metal Area<br>Corroded<br>cm <sup>2</sup> | Section Modulus<br>New<br>mm. <sup>3</sup> | Section Modulus<br>Corroded<br>mm. <sup>3</sup> |
|------|----|--------------------------------------|-------------------------------------------|--------------------------------------------|-------------------------------------------------|
| 10   | 20 | 2382.58                              | 2263.46                                   | 148834224                                  | 141474096                                       |
| 20   | 30 | 2382.58                              | 2263.46                                   | 148834224                                  | 141474080                                       |
| 30   | 40 | 2382.58                              | 2263.46                                   | 148834224                                  | 141474080                                       |
| 40   | 50 | 5164.31                              | 5048.12                                   | 318541856                                  | 311735456                                       |
| 50   | 60 | 10165.7                              | 10052.5                                   | 613077184                                  | 606933888                                       |
| 60   | 70 | 10165.7                              | 10052.5                                   | 613077120                                  | 606933760                                       |
| 70   | 80 | 10165.7                              | 10052.5                                   | 613077184                                  | 606933888                                       |
| 80   | 90 | 5164.31                              | 5048.12                                   | 318541856                                  | 311735456                                       |

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**Longitudinal Allowable Stresses:**

| From | To | Tensile<br>N./mm <sup>2</sup> | Hydrotest<br>Tensile<br>N./mm <sup>2</sup> | Compressive<br>N./mm <sup>2</sup> | Hydrotest<br>Compressive<br>N./mm <sup>2</sup> |
|------|----|-------------------------------|--------------------------------------------|-----------------------------------|------------------------------------------------|
| 10   | 20 | 96.3753                       | 194.746                                    | -110.571                          | -111.497                                       |
| 20   | 30 | 113.299                       | 246.996                                    | -110.571                          | -111.497                                       |
| 30   | 40 | 89.0202                       | 246.996                                    | -110.571                          | -111.497                                       |
| 40   | 50 | 161.855                       | 246.996                                    | -121.498                          | -121.71                                        |
| 50   | 60 | 161.9                         | 247                                        | -122.731                          | -122.731                                       |
| 60   | 70 | 161.9                         | 247                                        | -122.731                          | -122.731                                       |
| 70   | 80 | 161.9                         | 247                                        | -122.731                          | -122.731                                       |
| 80   | 90 | 161.9                         | 247                                        | -121.498                          | -121.71                                        |

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## Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

### Longitudinal Pressure Stresses due to:

| From | To | Longitudinal Stress<br>Internal Pressure<br>N./mm <sup>2</sup> | Longitudinal Stress<br>External Pressure<br>N./mm <sup>2</sup> | Longitudinal Stress<br>Hydrotest Pressure<br>N./mm <sup>2</sup> |
|------|----|----------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------|
| 10   | 20 | ...                                                            | ...                                                            | ...                                                             |
| 20   | 30 | ...                                                            | ...                                                            | ...                                                             |
| 30   | 40 | ...                                                            | ...                                                            | ...                                                             |
| 40   | 50 | 151.513                                                        | ...                                                            | 200.375                                                         |
| 50   | 60 | 72.1895                                                        | ...                                                            | 96.574                                                          |
| 60   | 70 | 72.1895                                                        | ...                                                            | 96.574                                                          |
| 70   | 80 | 72.1895                                                        | ...                                                            | 96.574                                                          |
| 80   | 90 | 151.513                                                        | ...                                                            | 200.375                                                         |

### Longitudinal Stresses due to Weight Loads for these Conditions:

| From | To | Wght. Str.<br>Empty<br>N./mm <sup>2</sup> | Wght. Str.<br>Operating<br>N./mm <sup>2</sup> | Wght. Str.<br>Hydrotest<br>N./mm <sup>2</sup> | Wght. Str.<br>Emp. Mom.<br>N./mm <sup>2</sup> | Wght. Str.<br>Opr. Mom.<br>N./mm <sup>2</sup> |
|------|----|-------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| 10   | 20 | -4.69599                                  | -5.30229                                      | ...                                           | 0.47032                                       | 0.47032                                       |
| 20   | 30 | -4.49568                                  | -5.10198                                      | ...                                           | 0.47033                                       | 0.47033                                       |
| 30   | 40 | -4.45568                                  | -5.06197                                      | ...                                           | 0.47033                                       | 0.47033                                       |
| 40   | 50 | -1.97989                                  | -1.97989                                      | ...                                           | 0.21345                                       | 0.21345                                       |
| 50   | 60 | -0.92274                                  | -0.92274                                      | ...                                           | 0.10963                                       | 0.10963                                       |
| 60   | 70 | -0.72767                                  | -0.72767                                      | ...                                           | 0.1096                                        | 0.1096                                        |
| 70   | 80 | -0.413                                    | -0.413                                        | ...                                           | 0.066004                                      | 0.066004                                      |
| 80   | 90 | -0.2409                                   | -0.2409                                       | ...                                           | 0.09424                                       | 0.09424                                       |

### Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

| From | To | Wght. Str.<br>Hyd. Mom.<br>N./mm <sup>2</sup> | Bend. Str.<br>Oper. Wind<br>N./mm <sup>2</sup> | Bend. Str.<br>Oper. Equ.<br>N./mm <sup>2</sup> | Bend. Str.<br>Hyd. Wind<br>N./mm <sup>2</sup> | Bend. Str.<br>Hyd. Equ.<br>N./mm <sup>2</sup> |
|------|----|-----------------------------------------------|------------------------------------------------|------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| 10   | 20 | ...                                           | 4.70552                                        | 8.94324                                        | ...                                           | ...                                           |
| 20   | 30 | ...                                           | 3.36515                                        | 6.56498                                        | ...                                           | ...                                           |
| 30   | 40 | ...                                           | 3.05199                                        | 5.9743                                         | ...                                           | ...                                           |
| 40   | 50 | ...                                           | 1.24695                                        | 2.44422                                        | ...                                           | ...                                           |
| 50   | 60 | ...                                           | 0.64046                                        | 1.25498                                        | ...                                           | ...                                           |
| 60   | 70 | ...                                           | 0.38925                                        | 0.74652                                        | ...                                           | ...                                           |
| 70   | 80 | ...                                           | 0.15677                                        | 0.24436                                        | ...                                           | ...                                           |
| 80   | 90 | ...                                           | 0.046758                                       | 0.74394E-04                                    | ...                                           | ...                                           |

### Longitudinal Stresses due to these Conditions:

| From | To | Vortex Shedding<br>Operating Case<br>N./mm <sup>2</sup> | Vortex Shedding<br>Empty Case<br>N./mm <sup>2</sup> | Vortex Shedding<br>Test Case<br>N./mm <sup>2</sup> | Earthquake<br>Empty Case<br>N./mm <sup>2</sup> |
|------|----|---------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|------------------------------------------------|
| 10   | 20 | ...                                                     | ...                                                 | ...                                                | 7.06302                                        |
| 20   | 30 | ...                                                     | ...                                                 | ...                                                | 5.2083                                         |
| 30   | 40 | ...                                                     | ...                                                 | ...                                                | 4.74788                                        |
| 40   | 50 | ...                                                     | ...                                                 | ...                                                | 1.94657                                        |
| 50   | 60 | ...                                                     | ...                                                 | ...                                                | 0.99947                                        |
| 60   | 70 | ...                                                     | ...                                                 | ...                                                | 0.59842                                        |
| 70   | 80 | ...                                                     | ...                                                 | ...                                                | 0.19206                                        |



80 90 ... 0.47599E-04

#### Longitudinal Stresses due to Applied Axial Forces:

| From | To | Longitudinal Stress<br>Y Forces Wind<br>N./mm <sup>2</sup> | Longitudinal Stress<br>Y Forces Seismic<br>N./mm <sup>2</sup> |
|------|----|------------------------------------------------------------|---------------------------------------------------------------|
| 10   | 20 | -0.065584                                                  | -0.34145                                                      |
| 20   | 30 | -0.065584                                                  | -0.33815                                                      |
| 30   | 40 | -0.065584                                                  | -0.33662                                                      |
| 40   | 50 | -0.029406                                                  | -0.14614                                                      |
| 50   | 60 | -0.014767                                                  | -0.065436                                                     |
| 60   | 70 | -0.014767                                                  | -0.049364                                                     |
| 70   | 80 | -0.014767                                                  | -0.030529                                                     |
| 80   | 90 | -0.029406                                                  | -0.036517                                                     |

#### Longitudinal Stresses due to User Forces and Moments:

| From | To | Wind For/Mom<br>Corroded<br>N./mm <sup>2</sup> | Earthquake For/Mom<br>Corroded<br>N./mm <sup>2</sup> | Wind For/Mom<br>No Corrosion<br>N./mm <sup>2</sup> | Earthquake For/Mom<br>No Corrosion<br>N./mm <sup>2</sup> |
|------|----|------------------------------------------------|------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------|
| 10   | 20 | 2.82158                                        | 2.82158                                              | 2.68205                                            | 2.68205                                                  |
| 20   | 30 | 2.38187                                        | 2.38187                                              | 2.26409                                            | 2.26409                                                  |
| 30   | 40 | 2.27195                                        | 2.27195                                              | 2.1596                                             | 2.1596                                                   |
| 40   | 50 | 0.98119                                        | 0.98119                                              | 0.96023                                            | 0.96023                                                  |
| 50   | 60 | 0.50396                                        | 0.50396                                              | 0.49891                                            | 0.49891                                                  |
| 60   | 70 | 0.40148                                        | 0.40148                                              | 0.39746                                            | 0.39746                                                  |
| 70   | 80 | 0.2734                                         | 0.2734                                               | 0.27066                                            | 0.27066                                                  |
| 80   | 90 | 0.30307                                        | 0.30307                                              | 0.2966                                             | 0.2966                                                   |

#### Bending Stresses due to Blast Load:

| From | To | Bending<br>Stress<br>N./mm <sup>2</sup> |
|------|----|-----------------------------------------|
| 10   | 20 | ...                                     |
| 20   | 30 | ...                                     |
| 30   | 40 | ...                                     |
| 40   | 50 | ...                                     |
| 50   | 60 | ...                                     |
| 60   | 70 | ...                                     |
| 70   | 80 | ...                                     |
| 80   | 90 | ...                                     |



## Stress Combination Load Cases for Vertical Vessels:

### Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure  
 EP = Longitudinal Stress due to External Pressure  
 HP = Longitudinal Stress due to Hydrotest Pressure  
 NP = No Pressure  
 EW = Longitudinal Stress due to Weight (No Liquid)  
 OW = Longitudinal Stress due to Weight (Operating)  
 HW = Longitudinal Stress due to Weight (Hydrotest)  
 WI = Bending Stress due to Wind Moment (Operating)  
 EQ = Bending Stress due to Earthquake Moment (Operating)  
 EE = Bending Stress due to Earthquake Moment (Empty)  
 HI = Bending Stress due to Wind Moment (Hydrotest)  
 HE = Bending Stress due to Earthquake Moment (Hydrotest)  
 WE = Bending Stress due to Wind Moment (Empty) (no CA)  
 WF = Bending Stress due to Wind Moment (Filled) (no CA)  
 CW = Longitudinal Stress due to Weight (Empty) (no CA)  
 VO = Bending Stress due to Vortex Shedding Loads ( Ope )  
 VE = Bending Stress due to Vortex Shedding Loads ( Emp )  
 VF = Bending Stress due to Vortex Shedding Loads ( Test No CA. )  
 FW = Axial Stress due to Vertical Forces for the Wind Case  
 FS = Axial Stress due to Vertical Forces for the Seismic Case  
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded  
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded  
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded  
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded  
 BL = Bending Stress due to Blast Loads, Corroded

### General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads. The stresses are combined per equation 4.3.44 to determine the overall stress intensity. The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (\*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

#### Analysis of Load Case 1 : NP+EW+WI+FW+BW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 12.76            | 96.38                 | -12.76             | 135.67                | 0.1324      | 0.0940      | 0.0588      |
| 20        | 10.78            | 113.30                | -10.78             | 152.68                | 0.0951      | 0.0706      | 0.0406      |
| 30        | 10.32            | 89.02                 | -10.32             | 152.68                | 0.1159      | 0.0676      | 0.0378      |
| 40        | 4.45             | 161.85                | -4.45              | 145.77                | 0.0275      | 0.0305      | 0.0690      |
| 50        | 2.19             | 161.90                | -2.19              | 193.26                | 0.0135      | 0.0113      | 0.0065      |
| 60        | 1.64             | 161.90                | -1.64              | 193.26                | 0.0101      | 0.0085      | 0.0047      |
| 70        | 0.92             | 161.90                | -0.92              | 193.26                | 0.0057      | 0.0048      | 0.0026      |
| 80        | 0.71             | 161.90                | -0.71              | 145.77                | 0.0044      | 0.0049      | 0.0065      |

#### Analysis of Load Case 2 : NP+EW+EE+FS+BS

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 15.39            | 96.38                 | -15.39             | 135.67                | 0.1597      | 0.1135      | 0.0761      |
| 20        | 12.89            | 113.30                | -12.89             | 152.68                | 0.1138      | 0.0845      | 0.0526      |
| 30        | 12.28            | 89.02                 | -12.28             | 152.68                | 0.1380      | 0.0804      | 0.0489      |
| 40        | 5.27             | 161.85                | -5.27              | 145.77                | 0.0325      | 0.0361      | 0.0825      |



|    |      |        |       |        |        |        |        |
|----|------|--------|-------|--------|--------|--------|--------|
| 50 | 2.60 | 161.90 | -2.60 | 193.26 | 0.0161 | 0.0135 | 0.0122 |
| 60 | 1.89 | 161.90 | -1.89 | 193.26 | 0.0117 | 0.0098 | 0.0140 |
| 70 | 0.97 | 161.90 | -0.97 | 193.26 | 0.0060 | 0.0050 | 0.0149 |
| 80 | 0.67 | 161.90 | -0.67 | 145.77 | 0.0042 | 0.0046 | 0.0069 |

## Analysis of Load Case 3 : NP+OW+WI+FW+BW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 13.37            | 96.38                 | -13.37             | 135.67                | 0.1387      | 0.0985      | 0.0588      |
| 20        | 11.38            | 113.30                | -11.38             | 152.68                | 0.1005      | 0.0746      | 0.0406      |
| 30        | 10.92            | 89.02                 | -10.92             | 152.68                | 0.1227      | 0.0715      | 0.0378      |
| 40        | 4.45             | 161.85                | -4.45              | 145.77                | 0.0275      | 0.0305      | 0.0732      |
| 50        | 2.19             | 161.90                | -2.19              | 193.26                | 0.0135      | 0.0113      | 0.0065      |
| 60        | 1.64             | 161.90                | -1.64              | 193.26                | 0.0101      | 0.0085      | 0.0047      |
| 70        | 0.92             | 161.90                | -0.92              | 193.26                | 0.0057      | 0.0048      | 0.0026      |
| 80        | 0.71             | 161.90                | -0.71              | 145.77                | 0.0044      | 0.0049      | 0.0065      |

## Analysis of Load Case 4 : NP+OW+EQ+FS+BS

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 17.88            | 96.38                 | -17.88             | 135.67                | 0.1855      | 0.1318      | 0.0899      |
| 20        | 14.86            | 113.30                | -14.86             | 152.68                | 0.1311      | 0.0973      | 0.0615      |
| 30        | 14.12            | 89.02                 | -14.12             | 152.68                | 0.1586      | 0.0924      | 0.0569      |
| 40        | 5.76             | 161.85                | -5.76              | 145.77                | 0.0356      | 0.0395      | 0.0951      |
| 50        | 2.86             | 161.90                | -2.86              | 193.26                | 0.0176      | 0.0148      | 0.0122      |
| 60        | 2.03             | 161.90                | -2.03              | 193.26                | 0.0126      | 0.0105      | 0.0140      |
| 70        | 1.03             | 161.90                | -1.03              | 193.26                | 0.0063      | 0.0053      | 0.0149      |
| 80        | 0.67             | 161.90                | -0.67              | 145.77                | 0.0042      | 0.0046      | 0.0072      |

## Analysis of Load Case 5 : NP+HW+HI

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 0.00             | 194.75                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0038      |
| 20        | 0.00             | 247.00                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0035      |
| 30        | 0.00             | 247.00                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0037      |
| 40        | 0.00             | 247.00                | 0.00               | 121.71                | 0.0000      | -0.0000     | ...         |
| 50        | 0.00             | 247.00                | 0.00               | 122.73                | 0.0000      | -0.0000     | ...         |
| 60        | 0.00             | 247.00                | 0.00               | 122.73                | 0.0000      | -0.0000     | 0.0012      |
| 70        | 0.00             | 247.00                | 0.00               | 122.73                | 0.0000      | -0.0000     | 0.0015      |
| 80        | 0.00             | 247.00                | 0.00               | 121.71                | 0.0000      | -0.0000     | ...         |

## Analysis of Load Case 6 : NP+HW+HE

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 0.00             | 194.75                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0029      |
| 20        | 0.00             | 247.00                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0025      |
| 30        | 0.00             | 247.00                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0025      |
| 40        | 0.00             | 247.00                | 0.00               | 121.71                | 0.0000      | -0.0000     | ...         |
| 50        | 0.00             | 247.00                | 0.00               | 122.73                | 0.0000      | -0.0000     | ...         |
| 60        | 0.00             | 247.00                | 0.00               | 122.73                | 0.0000      | -0.0000     | ...         |
| 70        | 0.00             | 247.00                | 0.00               | 122.73                | 0.0000      | -0.0000     | ...         |
| 80        | 0.00             | 247.00                | 0.00               | 121.71                | 0.0000      | -0.0000     | ...         |

## Analysis of Load Case 7 : IP+OW+WI+FW+BW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 13.37            | 96.38                 | -13.37             | 135.67                | 0.1387      | 0.0985      | 0.0588      |
| 20        | 11.38            | 113.30                | -11.38             | 152.68                | 0.1005      | 0.0746      | 0.0406      |
| 30        | 10.92            | 89.02                 | -10.92             | 152.68                | 0.1227      | 0.0715      | 0.0378      |
| 40        | 151.88           | 161.85                |                    | 145.77                | 0.9384      |             | 0.0732      |
| 50        | 131.96           | 161.90                |                    | 193.26                | 0.8151      |             | 0.0065      |
| 60        | 131.84           | 161.90                |                    | 193.26                | 0.8143      |             | 0.0047      |
| 70        | 131.70           | 161.90                |                    | 193.26                | 0.8135      |             | 0.0026      |
| 80        | 151.60           | 161.90                |                    | 121.50                | 0.9364      |             | ...         |



Analysis of Load Case 8 : IP+OW+EQ+FS+BS

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 17.88            | 96.38                 | -17.88             | 135.67                | 0.1855      | 0.1318      | 0.0899      |
| 20        | 14.86            | 113.30                | -14.86             | 152.68                | 0.1311      | 0.0973      | 0.0615      |
| 30        | 14.12            | 89.02                 | -14.12             | 152.68                | 0.1586      | 0.0924      | 0.0569      |
| 40        | 152.43           | 161.85                |                    | 145.77                | 0.9418      |             | 0.0951      |
| 50        | 132.00           | 161.90                |                    | 193.26                | 0.8153      |             | 0.0122      |
| 60        | 131.85           | 161.90                |                    | 193.26                | 0.8144      |             | 0.0140      |
| 70        | 131.71           | 161.90                |                    | 193.26                | 0.8135      |             | 0.0149      |
| 80        | 151.57           | 161.90                |                    | 121.50                | 0.9362      |             | ...         |

Analysis of Load Case 9 : EP+OW+WI+FW+BW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 13.37            | 96.38                 | -13.37             | 136.08                | 0.1387      | 0.0982      | 0.0586      |
| 20        | 11.38            | 113.30                | -11.38             | 153.75                | 0.1005      | 0.0741      | 0.0403      |
| 30        | 10.92            | 89.02                 | -10.92             | 153.75                | 0.1227      | 0.0710      | 0.0376      |
| 40        | 4.45             | 161.85                | -4.45              | 145.77                | 0.0275      | 0.0305      | 0.0732      |
| 50        | 2.19             | 161.90                | -2.19              | 193.51                | 0.0135      | 0.0113      | 0.0065      |
| 60        | 1.64             | 161.90                | -1.64              | 193.51                | 0.0101      | 0.0085      | 0.0047      |
| 70        | 0.92             | 161.90                | -0.92              | 193.51                | 0.0057      | 0.0048      | 0.0026      |
| 80        | 0.71             | 161.90                | -0.71              | 145.77                | 0.0044      | 0.0049      | 0.0065      |

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 17.88            | 96.38                 | -17.88             | 136.08                | 0.1855      | 0.1314      | 0.0897      |
| 20        | 14.86            | 113.30                | -14.86             | 153.75                | 0.1311      | 0.0966      | 0.0611      |
| 30        | 14.12            | 89.02                 | -14.12             | 153.75                | 0.1586      | 0.0918      | 0.0565      |
| 40        | 5.76             | 161.85                | -5.76              | 145.77                | 0.0356      | 0.0395      | 0.0951      |
| 50        | 2.86             | 161.90                | -2.86              | 193.51                | 0.0176      | 0.0148      | 0.0122      |
| 60        | 2.03             | 161.90                | -2.03              | 193.51                | 0.0126      | 0.0105      | 0.0140      |
| 70        | 1.03             | 161.90                | -1.03              | 193.51                | 0.0063      | 0.0053      | 0.0149      |
| 80        | 0.67             | 161.90                | -0.67              | 145.77                | 0.0042      | 0.0046      | 0.0072      |

Analysis of Load Case 11 : HP+HW+HI

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 0.00             | 194.75                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0038      |
| 20        | 0.00             | 247.00                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0035      |
| 30        | 0.00             | 247.00                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0037      |
| 40        | 206.04           | 247.00                |                    | 121.71                | 0.8342      |             | ...         |
| 50        | 185.93           | 247.00                |                    | 122.73                | 0.7528      |             | ...         |
| 60        | 185.93           | 247.00                |                    | 122.73                | 0.7528      |             | 0.0012      |
| 70        | 185.93           | 247.00                |                    | 122.73                | 0.7528      |             | 0.0015      |
| 80        | 206.04           | 247.00                |                    | 121.71                | 0.8342      |             | ...         |

Analysis of Load Case 12 : HP+HW+HE

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 0.00             | 194.75                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0029      |
| 20        | 0.00             | 247.00                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0025      |
| 30        | 0.00             | 247.00                | 0.00               | 111.50                | 0.0000      | -0.0000     | 0.0025      |
| 40        | 206.04           | 247.00                |                    | 121.71                | 0.8342      |             | ...         |
| 50        | 185.93           | 247.00                |                    | 122.73                | 0.7528      |             | ...         |
| 60        | 185.93           | 247.00                |                    | 122.73                | 0.7528      |             | ...         |
| 70        | 185.93           | 247.00                |                    | 122.73                | 0.7528      |             | ...         |
| 80        | 206.04           | 247.00                |                    | 121.71                | 0.8342      |             | ...         |

Analysis of Load Case 13 : IP+WE+EW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 5.17             | 96.38                 | -5.17              | 135.67                | 0.0536      | 0.0381      | 0.0354      |



|    |        |        |       |        |        |        |        |
|----|--------|--------|-------|--------|--------|--------|--------|
| 20 | 4.97   | 113.30 | -4.97 | 152.68 | 0.0438 | 0.0325 | 0.0303 |
| 30 | 4.93   | 89.02  | -4.93 | 152.68 | 0.0553 | 0.0323 | 0.0300 |
| 40 | 150.80 | 161.85 |       | 104.64 | 0.9317 |        | 0.0446 |
| 50 | 131.92 | 161.90 |       | 193.26 | 0.8148 |        | 0.0049 |
| 60 | 131.80 | 161.90 |       | 193.26 | 0.8141 |        | 0.0039 |
| 70 | 131.69 | 161.90 |       | 193.26 | 0.8134 |        | 0.0022 |
| 80 | 151.44 | 161.90 |       | 121.50 | 0.9354 |        | ...    |

Analysis of Load Case 14 : IP+WF+CW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 5.04             | 96.38                 | -5.04              | 135.67                | 0.0523      | 0.0371      | 0.0380      |
| 20        | 4.85             | 113.30                | -4.85              | 152.68                | 0.0428      | 0.0317      | 0.0326      |
| 30        | 4.81             | 89.02                 | -4.81              | 152.68                | 0.0540      | 0.0315      | 0.0324      |
| 40        | 150.71           | 161.85                |                    | 104.64                | 0.9312      |             | 0.0460      |
| 50        | 131.91           | 161.90                |                    | 193.26                | 0.8148      |             | 0.0049      |
| 60        | 131.80           | 161.90                |                    | 193.26                | 0.8141      |             | 0.0039      |
| 70        | 131.69           | 161.90                |                    | 193.26                | 0.8134      |             | 0.0022      |
| 80        | 151.40           | 161.90                |                    | 121.50                | 0.9351      |             | ...         |

Analysis of Load Case 15 : IP+VO+OW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 5.77             | 96.38                 | -5.77              | 135.67                | 0.0599      | 0.0425      | 0.0400      |
| 20        | 5.57             | 113.30                | -5.57              | 152.68                | 0.0492      | 0.0365      | 0.0343      |
| 30        | 5.53             | 89.02                 | -5.53              | 152.68                | 0.0621      | 0.0362      | 0.0341      |
| 40        | 150.80           | 161.85                |                    | 104.64                | 0.9317      |             | 0.0504      |
| 50        | 131.92           | 161.90                |                    | 193.26                | 0.8148      |             | 0.0049      |
| 60        | 131.80           | 161.90                |                    | 193.26                | 0.8141      |             | 0.0039      |
| 70        | 131.69           | 161.90                |                    | 193.26                | 0.8134      |             | 0.0022      |
| 80        | 151.44           | 161.90                |                    | 121.50                | 0.9354      |             | ...         |

Analysis of Load Case 16 : IP+VE+EW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 5.17             | 96.38                 | -5.17              | 135.67                | 0.0536      | 0.0381      | 0.0354      |
| 20        | 4.97             | 113.30                | -4.97              | 152.68                | 0.0438      | 0.0325      | 0.0303      |
| 30        | 4.93             | 89.02                 | -4.93              | 152.68                | 0.0553      | 0.0323      | 0.0300      |
| 40        | 150.80           | 161.85                |                    | 104.64                | 0.9317      |             | 0.0446      |
| 50        | 131.92           | 161.90                |                    | 193.26                | 0.8148      |             | 0.0049      |
| 60        | 131.80           | 161.90                |                    | 193.26                | 0.8141      |             | 0.0039      |
| 70        | 131.69           | 161.90                |                    | 193.26                | 0.8134      |             | 0.0022      |
| 80        | 151.44           | 161.90                |                    | 121.50                | 0.9354      |             | ...         |

Analysis of Load Case 17 : NP+VO+OW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 5.77             | 96.38                 | -5.77              | 135.67                | 0.0599      | 0.0425      | 0.0400      |
| 20        | 5.57             | 113.30                | -5.57              | 152.68                | 0.0492      | 0.0365      | 0.0343      |
| 30        | 5.53             | 89.02                 | -5.53              | 152.68                | 0.0621      | 0.0362      | 0.0341      |
| 40        | 2.19             | 161.85                | -2.19              | 104.64                | 0.0136      | 0.0210      | 0.0504      |
| 50        | 1.03             | 161.90                | -1.03              | 193.26                | 0.0064      | 0.0053      | 0.0049      |
| 60        | 0.84             | 161.90                | -0.84              | 193.26                | 0.0052      | 0.0043      | 0.0039      |
| 70        | 0.48             | 161.90                | -0.48              | 193.26                | 0.0030      | 0.0025      | 0.0022      |
| 80        | 0.34             | 161.90                | -0.34              | 104.64                | 0.0021      | 0.0032      | 0.0048      |

Analysis of Load Case 18 : FS+BS+IP+OW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 8.94             | 96.38                 | -8.94              | 135.67                | 0.0927      | 0.0659      | 0.0426      |
| 20        | 8.29             | 113.30                | -8.29              | 152.68                | 0.0732      | 0.0543      | 0.0366      |
| 30        | 8.14             | 89.02                 | -8.14              | 152.68                | 0.0914      | 0.0533      | 0.0363      |
| 40        | 151.21           | 161.85                |                    | 104.64                | 0.9342      |             | 0.0753      |
| 50        | 131.94           | 161.90                |                    | 193.26                | 0.8149      |             | 0.0053      |
| 60        | 131.82           | 161.90                |                    | 193.26                | 0.8142      |             | 0.0042      |



|    |        |        |        |        |        |
|----|--------|--------|--------|--------|--------|
| 70 | 131.70 | 161.90 | 193.26 | 0.8135 | 0.0024 |
| 80 | 151.57 | 161.90 | 121.50 | 0.9362 | ...    |

Analysis of Load Case 19 : FS+BS+EP+OW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 8.94             | 96.38                 | -8.94              | 136.08                | 0.0927      | 0.0657      | 0.0424      |
| 20        | 8.29             | 113.30                | -8.29              | 153.75                | 0.0732      | 0.0539      | 0.0363      |
| 30        | 8.14             | 89.02                 | -8.14              | 153.75                | 0.0914      | 0.0530      | 0.0361      |
| 40        | 3.32             | 161.85                | -3.32              | 104.79                | 0.0205      | 0.0317      | 0.0752      |
| 50        | 1.60             | 161.90                | -1.60              | 193.51                | 0.0099      | 0.0083      | 0.0053      |
| 60        | 1.29             | 161.90                | -1.29              | 193.51                | 0.0080      | 0.0067      | 0.0042      |
| 70        | 0.78             | 161.90                | -0.78              | 193.51                | 0.0048      | 0.0040      | 0.0024      |
| 80        | 0.67             | 161.90                | -0.67              | 104.80                | 0.0042      | 0.0064      | 0.0077      |

Analysis of Load Case 20 : BL+IP+OW

| From Node | Stress Intensity | All. Stress Intensity | Compressive Stress | All. Comp. Stress Fxa | S. I. Ratio | Comp. Ratio | Unity Check |
|-----------|------------------|-----------------------|--------------------|-----------------------|-------------|-------------|-------------|
| 10        | 5.77             | 96.38                 | -5.77              | 135.67                | 0.0599      | 0.0425      | 0.0400      |
| 20        | 5.57             | 113.30                | -5.57              | 152.68                | 0.0492      | 0.0365      | 0.0343      |
| 30        | 5.53             | 89.02                 | -5.53              | 152.68                | 0.0621      | 0.0362      | 0.0341      |
| 40        | 150.80           | 161.85                |                    | 104.64                | 0.9317      |             | 0.0504      |
| 50        | 131.92           | 161.90                |                    | 193.26                | 0.8148      |             | 0.0049      |
| 60        | 131.80           | 161.90                |                    | 193.26                | 0.8141      |             | 0.0039      |
| 70        | 131.69           | 161.90                |                    | 193.26                | 0.8134      |             | 0.0022      |
| 80        | 151.44           | 161.90                |                    | 121.50                | 0.9354      |             | ...         |

Absolute Maximum of the all of the Stress Ratio's 0.9418

Governing Element: Bottom D end  
 Governing Load Case 8 : IP+OW+EQ+FS+BS

| Element Description | Long Pressure Stress | Weight + Axial Load Stress | Eccentric Moment Stress | Wind or Earthquake Stress | Combined Stress | Which Side |
|---------------------|----------------------|----------------------------|-------------------------|---------------------------|-----------------|------------|
| Skirt 3             | ...                  | -5.6                       | 0.5                     | 11.8                      | 6.6             | + Tens     |
| Skirt 3             | ...                  | -5.6                       | -0.5                    | -11.8                     | -17.9           | - Comp     |
| Skirt 2             | ...                  | -5.4                       | 0.5                     | 8.9                       | 4.0             | + Tens     |
| Skirt 2             | ...                  | -5.4                       | -0.5                    | -8.9                      | -14.9           | - Comp     |
| Top Skirt -Yring    | ...                  | -5.4                       | 0.5                     | 8.2                       | 3.3             | + Tens     |
| Top Skirt -Yring    | ...                  | -5.4                       | -0.5                    | -8.2                      | -14.1           | - Comp     |
| Bottom D end        | 151.5                | -2.1                       | 0.2                     | 3.4                       | 153.0           | + Tens     |
| Bottom D end        | 151.5                | -2.1                       | -0.2                    | -3.4                      | ...             | - Comp     |
| Shell 1             | 72.2                 | -1.0                       | 0.1                     | 1.8                       | 73.1            | + Tens     |
| Shell 1             | 72.2                 | -1.0                       | -0.1                    | -1.8                      | ...             | - Comp     |
| Shell 2             | 72.2                 | -0.8                       | 0.1                     | 1.1                       | 72.7            | + Tens     |
| Shell 2             | 72.2                 | -0.8                       | -0.1                    | -1.1                      | ...             | - Comp     |
| Shell 3             | 72.2                 | -0.4                       | 0.1                     | 0.5                       | 72.3            | + Tens     |
| Shell 3             | 72.2                 | -0.4                       | -0.1                    | -0.5                      | ...             | - Comp     |
| Top D end           | 151.5                | -0.3                       | 0.1                     | 0.3                       | 151.6           | + Tens     |
| Top D end           | 151.5                | -0.3                       | -0.1                    | -0.3                      | ...             | - Comp     |

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm.
- Fxa - Allowable comp mem stress due to compression with lamc <= 0.15, N./mm<sup>2</sup>
- Fba - Allowable all comp mem stress due to bending moment, N./mm<sup>2</sup>
- Fva - Allowable shear stress when only shear is present, N./mm<sup>2</sup>



Fha - Allowable hoop comp memb stress under ext pressure alone, N./mm2  
 Fxha - Allowable axial comp w/hoop compression and lamc <= 0.15, N./mm2  
 Fhxa - Allowable hoop comp w/axial compression and lamc <= 0.15, N./mm2  
 Faha - Allowable axial comp w/hoop compression and lamc > 0.15, N./mm2  
 Fbha - Allowable axial comp from bending w/hoop compression, N./mm2  
 Fhba - Allowable hoop comp w/long compression due to moment, N./mm2

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 5 : NP+HW+HI

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.251 | 3000 | 137 | 138 | 90  |     | 6895 | 137  | 6895 | 6895 | 6895 |
| 20   | 0.267 | 3000 | 155 | 156 | 104 |     | 6895 | 155  | 6895 | 6895 | 6895 |
| 30   | 0.267 | 3000 | 155 | 156 | 104 |     | 6895 | 155  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 6 : NP+HW+HE

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.251 | 3000 | 137 | 138 | 90  |     | 6895 | 137  | 6895 | 6895 | 6895 |
| 20   | 0.267 | 3000 | 155 | 156 | 104 |     | 6895 | 155  | 6895 | 6895 | 6895 |
| 30   | 0.267 | 3000 | 155 | 156 | 104 |     | 6895 | 155  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |



|    |       |      |     |     |     |     |      |     |      |      |      |
|----|-------|------|-----|-----|-----|-----|------|-----|------|------|------|
| 70 | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193 | 6895 | 6895 | 6895 |
| 80 |       |      |     |     |     |     | 6895 |     | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.249 | 3000 | 136 | 136 | 89  |     | 6895 | 136  | 6895 | 6895 | 6895 |
| 20   | 0.264 | 3000 | 153 | 154 | 102 |     | 6895 | 153  | 6895 | 6895 | 6895 |
| 30   | 0.264 | 3000 | 153 | 154 | 102 |     | 6895 | 153  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.249 | 3000 | 136 | 136 | 89  |     | 6895 | 136  | 6895 | 6895 | 6895 |
| 20   | 0.264 | 3000 | 153 | 154 | 102 |     | 6895 | 153  | 6895 | 6895 | 6895 |
| 30   | 0.264 | 3000 | 153 | 154 | 102 |     | 6895 | 153  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     | 145 | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 11 : HP+HW+HI

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.251 | 3000 | 137 | 138 | 90  |     | 6895 | 137  | 6895 | 6895 | 6895 |
| 20   | 0.267 | 3000 | 155 | 156 | 104 |     | 6895 | 155  | 6895 | 6895 | 6895 |
| 30   | 0.267 | 3000 | 155 | 156 | 104 |     | 6895 | 155  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 12 : HP+HW+HE

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.251 | 3000 | 137 | 138 | 90  |     | 6895 | 137  | 6895 | 6895 | 6895 |
| 20   | 0.267 | 3000 | 155 | 156 | 104 |     | 6895 | 155  | 6895 | 6895 | 6895 |
| 30   | 0.267 | 3000 | 155 | 156 | 104 |     | 6895 | 155  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.293 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 13 : IP+WE+EW



| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 104 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 14 : IP+WF+CW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 104 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 15 : IP+VO+OW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 104 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 16 : IP+VE+EW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 104 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 17 : NP+VO+OW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 104 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     | 104 | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 18 : FS+BS+IP+OW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 104 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

Allowable Stress Results for Case 19 : FS+BS+EP+OW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.249 | 3000 | 136 | 136 | 89  |     | 6895 | 136  | 6895 | 6895 | 6895 |
| 20   | 0.264 | 3000 | 153 | 154 | 102 |     | 6895 | 153  | 6895 | 6895 | 6895 |
| 30   | 0.264 | 3000 | 153 | 154 | 102 |     | 6895 | 153  | 6895 | 6895 | 6895 |



|    |       |      |     |     |     |     |      |     |      |      |      |
|----|-------|------|-----|-----|-----|-----|------|-----|------|------|------|
| 40 |       |      |     |     |     | 104 | 6895 |     | 6895 | 6895 | 6895 |
| 50 | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193 | 6895 | 6895 | 6895 |
| 60 | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193 | 6895 | 6895 | 6895 |
| 70 | 0.291 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193 | 6895 | 6895 | 6895 |
| 80 |       |      |     |     |     | 104 | 6895 |     | 6895 | 6895 | 6895 |

## Allowable Stress Results for Case 20 : BL+IP+OW

| From | Lamc  | L    | Fxa | Fba | Fva | Fha | Fxha | Fhxa | Faha | Fbha | Fhba |
|------|-------|------|-----|-----|-----|-----|------|------|------|------|------|
| 10   | 0.250 | 3000 | 135 | 136 | 89  |     | 6895 | 135  | 6895 | 6895 | 6895 |
| 20   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 30   | 0.266 | 3000 | 152 | 153 | 102 |     | 6895 | 152  | 6895 | 6895 | 6895 |
| 40   |       |      |     |     |     | 104 | 6895 |      | 6895 | 6895 | 6895 |
| 50   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 60   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 70   | 0.294 | 7621 | 193 | 193 | 139 | 106 | 6895 | 193  | 6895 | 6895 | 6895 |
| 80   |       |      |     |     |     |     | 6895 |      | 6895 | 6895 | 6895 |

## Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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### Skirt Support Analysis Per EN 13445-3 Section 16.12:

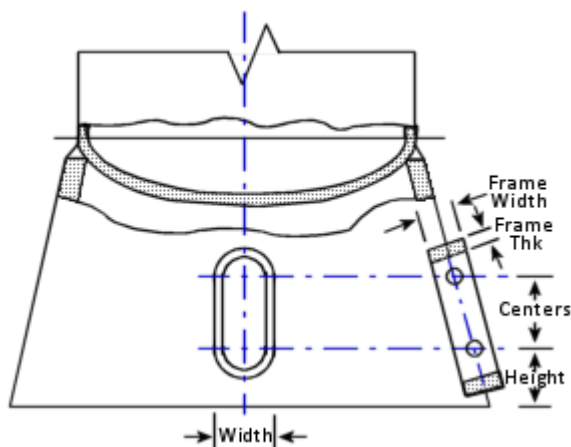
*The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.*

Item: Top Skirt -Yring

|                                       |         |                  |
|---------------------------------------|---------|------------------|
| Total vessel (empty) weight           | Fg      | 102240.836 Kg    |
| Head Weight below section 2-2         | DeltaFg | 7329.869 Kg      |
| Weight of vessel contents (liquid)    | Ff      | 13993.594 Kg     |
| Vessel Gross Weight                   | Fg + Ff | 116234.430 Kg    |
| Skirt Mean Diameter                   | Dz      | 2528.000 mm.     |
| Skirt Analysis Thickness (Corroded)   | ez      | 28.500 mm.       |
| Skirt Height                          | H       | 500.000 mm.      |
| Additional Axial Force at section 1-1 | F1      | -1513.710 Kg     |
| Moment at section 1-1                 | M1      | 108887.734 Kg-m. |
| Applied Moment - Bottom of Skirt      | Mb      | 118963.461 Kg-m. |
| Weight at Bottom of Skirt             | W       | 116832.734 Kg    |

### Structural Type B Skirt per EN 13445 Figure 16.12-2:

|                                 |    |              |
|---------------------------------|----|--------------|
| Type of Head:                   |    | Spherical    |
| Head Mean Diameter at Skirt Top | Db | 2530.500 mm. |
| Head Corroded Thickness         | eb | 63.500 mm.   |



There are no Access Openings in the Skirt - Plain Skirt Considered

|                    |       |                                |
|--------------------|-------|--------------------------------|
| Moment of Inertia  | Ixx = | 18081544.000 cm <sup>4</sup>   |
| Moment of Inertia  | Iyy = | 18081544.000 cm <sup>4</sup>   |
| Section Modulus    | W =   | 143050320.000 mm. <sup>3</sup> |
| Section Metal Area | A =   | 2263.455 cm <sup>2</sup>       |

### Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:  
 $= \sqrt{\frac{\min(Ixx, Iyy)}{Area}}$   
 $= \sqrt{\frac{\min(18081544, 18081544)}{2263}}$   
 $= 894 \text{ mm.}$

Skirt Slenderness Ratio [rg/L]:  
 $= rg / L$   
 $= 894/500$   
 $= 1.788$

Correct the Moment [M4]:  
 top of skirt is the critical end  
 Controlling Moment [M4] = 108888 Kg-m.



Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -108888/572201280 - 116833/2263$$

$$= -6.93 \text{ N./mm}^2$$

Upwind Side:

$$= +( M ) / Z - W / A$$

$$= +108888/572201280 - 116833/2263$$

$$= -3.2 \text{ N./mm}^2$$

#### 16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= -1514 - 102241 - 13994 + 4 \cdot 108888/2528$$

$$= 54539 \text{ Kgf}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= -1514 - 102241 - 13994 - 4 \cdot 108888/2528$$

$$= -290035 \text{ Kgf}$$

##### 16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [ $\sigma_{1pm}$ ]:

$$= ( Fzp + \Delta Fg ) / ( \pi \cdot Db \cdot eb ) + P \cdot Db / ( 4 \cdot eb )$$

$$= ( 54539 + 7330 + 13994 ) / ( \pi \cdot 2530 \cdot 63.5 ) + 160 \cdot 2530 / ( 4 \cdot 63.5 )$$

$$= 160.88 \text{ N./mm}^2$$

Stress [ $\sigma_{1qm}$ ]:

$$= ( Fzq + \Delta Fg + Ff ) / ( \pi \cdot Db \cdot eb ) + P \cdot Db / ( 4 \cdot eb )$$

$$= ( -290035 + 7330 + 13994 ) / ( \pi \cdot 2530 \cdot 63.5 ) + 160 \cdot 2530 / ( 4 \cdot 63.5 )$$

$$= 154.19 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [ $\sigma_{2pm}$  &  $\sigma_{2qm}$ ]:

$$= ( Ff + \Delta Fg ) / ( \pi \cdot Db \cdot eb ) + P \cdot Db / ( 4 \cdot eb )$$

$$= ( 13994 + 7330 ) / ( \pi \cdot 2530 \cdot 63.5 ) + 160 \cdot 2530 / ( 4 \cdot 63.5 )$$

$$= 159.83 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [ $\sigma_{3pm}$ ]:

$$= Fzp / ( \pi \cdot Dz \cdot ez )$$

$$= 54539 / ( \pi \cdot 2528 \cdot 28.5 )$$

$$= 2.36 \text{ N./mm}^2$$

Stress [ $\sigma_{3qm}$ ]:

$$= Fzq / ( \pi \cdot Dz \cdot ez )$$

$$= -290035 / ( \pi \cdot 2528 \cdot 28.5 )$$

$$= -12.57 \text{ N./mm}^2$$

##### 16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos( $\gamma$ )]:

$$= 1 - 0.5( Db + eb - Dz + ez ) / ( r + eb )$$

$$= 1 - 0.5( 2530 + 63.5 - 2528 + 28.5 ) / ( 1234 + 63.5 )$$

$$= 0.96$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.96) = 15.5 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{ ( eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma) ) }$$

$$= 0.5 \cdot \sqrt{ ( 63.5^2 + 28.5^2 + 2 \cdot 63.5 \cdot 28.5 \cdot 0.96 ) }$$

$$= 45.6 \text{ mm.}$$



Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 45.64 \cdot 54539.2 = 2489.23 \text{ Kg-m.} \\ M_q &= a \cdot F_{zq} = 45.64 \cdot -290035.5 = -13237.56 \text{ Kg-m.} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left( \frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left( \frac{63.5}{28.5} \right)^2 \\ &= 0.347 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [ $\sigma_{1pb(o)}$  &  $\sigma_{2pb(o)}$ ]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= 0.35 \cdot 6 \cdot 2489 / (\pi \cdot 2530 \cdot 63.5^2) \\ &= 1.59 \text{ N./mm}^2 \end{aligned}$$

Stress [ $\sigma_{1qb(o)}$  &  $\sigma_{2qb(o)}$ ]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= 0.35 \cdot 6 \cdot -13238 / (\pi \cdot 2530 \cdot 63.5^2) \\ &= -8.43 \text{ N./mm}^2 \end{aligned}$$

Stress [ $\sigma_{3pb(o)}$ ]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= 0.35 \cdot 6 \cdot 2489 / (\pi \cdot 2528 \cdot 28.5^2) \\ &= 7.88 \text{ N./mm}^2 \end{aligned}$$

Stress [ $\sigma_{3qb(o)}$ ]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= 0.35 \cdot 6 \cdot -13238 / (\pi \cdot 2528 \cdot 28.5^2) \\ &= -41.90 \text{ N./mm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} eb/D_b &= 2.5/99.6 = 0.025 \\ y &= 125 \cdot eb/D_b = 125 \cdot 63.5/2530 = 3.14 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [ $\sigma_{1b(p)}$  &  $\sigma_{2b(p)}$ ]:

$$= 0.0 \text{ N./mm}^2 \text{ (16.12-31)}$$

#### 16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [ $\sigma_{1pitot}$ ]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 161 - 1.59 + 0 \\ &= 159.30 \text{ N./mm}^2 \end{aligned}$$

Stress [ $\sigma_{1potot}$ ]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 161 + 1.59 - 0 \\ &= 162.47 \text{ N./mm}^2 \end{aligned}$$

Stress [ $\sigma_{1qitot}$ ]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 154 - 8.43 + 0 \\ &= 162.62 \text{ N./mm}^2 \end{aligned}$$

Stress [ $\sigma_{1qotot}$ ]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 154 + 8.43 - 0 \\ &= 145.76 \text{ N./mm}^2 \end{aligned}$$

Stress [ $\sigma_{2pitot}$ ]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 160 + 1.59 + 0 \\ &= 161.41 \text{ N./mm}^2 \end{aligned}$$

Stress [ $\sigma_{2potot}$ ]:



$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 160 - 1.59 - 0 \\ &= 158.24 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qitot}]: \\ &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 160 + -8.43 + 0 \\ &= 151.39 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qotot}]: \\ &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 160 - -8.43 - 0 \\ &= 168.26 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pitot}]: \\ &= \sigma_{3pm} - \sigma_{3pb} \\ &= 2.36 - 7.88 \\ &= -5.52 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3potot}]: \\ &= \sigma_{3pm} + \sigma_{3pb} \\ &= 2.36 + 7.88 \\ &= 10.24 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qitot}]: \\ &= \sigma_{3qm} - \sigma_{3qb} \\ &= -12.6 - -41.9 \\ &= 29.34 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qotot}]: \\ &= \sigma_{3qm} + \sigma_{3qb} \\ &= -12.6 + -41.9 \\ &= -54.47 \text{ N./mm}^2 \end{aligned}$$

Allowable stresses:

$$\begin{aligned} \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\ &= 162 (3 - (1/1.5)) \cdot (161/162)^2 \\ &= 378.95 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\ &= 162 (3 - (1/1.5)) \cdot (154/162)^2 \\ &= 387.64 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\ &= 162 (3 - (1/1.5)) \cdot (160/162)^2 \\ &= 380.35 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\ &= 162 (3 - (1/1.5)) \cdot (160/162)^2 \\ &= 380.35 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\ &= 162 (3 - (1/1.5)) \cdot (2.36/162)^2 \\ &= 485.54 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\ &= 162 (3 - (1/1.5)) \cdot (-12.6/162)^2 \\ &= 484.91 \text{ N./mm}^2 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

**Summary of Stresses: (N./mm<sup>2</sup>)**



| Stress  | Actual | Allowed | Result | Equation |
|---------|--------|---------|--------|----------|
| Stot1pi | 159.30 | 378.95  | Pass   | 46       |
| Stot1po | 162.47 | 378.95  | Pass   | 47       |
| Stot1qi | 162.62 | 387.64  | Pass   | 48       |
| Stot1qo | 145.76 | 387.64  | Pass   | 49       |
| Stot2pi | 161.41 | 380.35  | Pass   | 50       |
| Stot2po | 158.24 | 380.35  | Pass   | 51       |
| Stot2qi | 151.39 | 380.35  | Pass   | 52       |
| Stot2qo | 168.26 | 380.35  | Pass   | 53       |
| Stot3pi | -5.52  | 485.54  | Pass   | 54       |
| Stot3po | 10.24  | 485.54  | Pass   | 55       |
| Stot3qi | 29.34  | 484.91  | Pass   | 56       |
| Stot3qo | -54.47 | 484.91  | Pass   | 57       |

The stresses are satisfactory in the region without openings.

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### Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.  
 Packing is installed in the Field after being lifted.  
 Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

|                                                    |              |
|----------------------------------------------------|--------------|
| Center of Gravity of the Platforms                 | 7375.032 mm. |
| Center of Gravity of the Packing                   | 9575.000 mm. |
| Center of Gravity of the Liquid                    | 4534.549 mm. |
| Center of Gravity of the Insulation                | 6399.991 mm. |
| Center of Gravity of the Nozzles                   | 8520.923 mm. |
| Center of Gravity of the Added Weights (Operating) | 6857.859 mm. |
| Center of Gravity of the Added Weights (Empty)     | 6857.859 mm. |
| Center of Gravity of Bare Shell New and Cold       | 5973.606 mm. |
| Center of Gravity of Bare Shell Corroded           | 5986.278 mm. |
| Vessel CG in the Operating Condition               | 6214.923 mm. |
| Vessel CG in the Fabricated (Shop/Empty) Condition | 6302.495 mm. |
| Vessel CG in the Test Condition                    | 6330.627 mm. |

### Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Total Effective Length of Vessel for this analysis | 9800.00 mm.       |
| Total vessel weight (No Liquid)                    | Twt 108271.62 Kgf |
| Impact weight multiplication factor                | Imp 1.50          |
| Design lifting weight, DWT = Imp * Twt             | 162407.42 Kgf     |
| Elevation of the Tailing Lug (bottom)              | 200.00 mm.        |
| Elevation of the Lifting Lug (top )                | 8800.00 mm.       |
| Design Reaction force at the tailing lug           | 47164.34 Kgf      |
| Design Reaction force at the lifting lug           | 115243.08 Kgf     |
| CG distance during field lifting operation         | 6302.50 mm.       |
| CG distance from tailing lug                       | 6102.50 mm.       |
| CG distance from the nearer lifting lug/trunnion   | 2497.51 mm.       |
| Compressive Allowable Increase Factor              | 1.00              |

### Critical Values: (N./mm<sup>2</sup>, mm.)

|         | Max Stress | Elevation | Allowables    |
|---------|------------|-----------|---------------|
| Bending | 6.85       | 2900.00   | 111.50        |
| Shear   | 1.23       | 400.00    | 96.60 (0.7*S) |

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

| Distance<br>mm. | Bending Moment<br>Kg-m. | Bending Stress<br>N./mm <sup>2</sup> | Shear Force<br>Kgf | Shear Stress<br>N./mm <sup>2</sup> |
|-----------------|-------------------------|--------------------------------------|--------------------|------------------------------------|
| 0.00            | 0.0                     | 0.0                                  | 22410.0            | 0.9                                |
| 2200.00         | 79047.6                 | 5.2                                  | 22837.8            | 0.9                                |
| 2900.00         | 103915.6                | 6.8                                  | 20630.8            | 0.8                                |
| 3400.00         | 110947.5                | 1.8                                  | 13254.4            | 0.1                                |
| 6500.00         | 101630.2                | 1.6                                  | 39245.7            | 0.4                                |
| 9800.00         | 6503.9                  | 0.2                                  | 1834.3             | 0.0                                |

### Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.061 at elevation 2900 mm. - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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#### Skirt Data :

|                                    |         |           |     |
|------------------------------------|---------|-----------|-----|
| Skirt Outside Diameter at Base     | SOD, D6 | 2557.9998 | mm. |
| Skirt Thickness                    | STHK    | 30.0000   | mm. |
| Skirt Internal Corrosion Allowance | SCA     | 0.7500    | mm. |
| Skirt External Corrosion Allowance |         | 0.7500    | mm. |
| Skirt Material                     |         | SA-285 C  |     |

#### Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

|                                              |            |             |                |
|----------------------------------------------|------------|-------------|----------------|
| Thickness of Basing                          | TBA, ea4   | 28.0000     | mm.            |
| Design Temperature of the Basing             |            | 38.00       | °C             |
| Basing Material                              |            | SA-285 C    |                |
| Basing Ope. All. Stress                      | BASOPE, f4 | 138.00      | N./mm2         |
| Basing Yield Stress                          |            | 207.00      | N./mm2         |
| Inside Diameter of Basing                    | DI, D4     | 2458.0000   | mm.            |
| Outside Diameter of Basing                   | DOU, D5    | 2794.0000   | mm.            |
| Nominal Diameter of Bolts                    | BND, d(b0) | 36.0000     | mm.            |
| Bolt Corrosion Allowance                     | BCA        | 1.5000      | mm.            |
| Root Area of a Single Bolt                   | Area       | 7.8800      | cm2            |
| Bolt Material                                |            | SA-36       |                |
| Bolt Operating Allowable Stress              | SA, fB     | 137.30      | N./mm2         |
| Number of Bolts                              | RN, nb     | 20          |                |
| Diameter of Bolt Circle                      | DC, DBC    | 2680.0000   | mm.            |
| Bolt Allowable Shear Stress                  |            | 107.873     | N./mm2         |
| Ultimate Comp. Strength of Concrete          | FPC        | 35.0        | N./mm2         |
| Allowable Comp. Strength of Concrete         | FC         | 8.8         | N./mm2         |
| Modular ratio Steel/Concrete                 |            | 9.960       |                |
| Thickness of Gusset Plates                   | TGA, e7    | 12.0000     | mm.            |
| Width of Gussets at Top Plate                | TWDT       | 102.0000    | mm.            |
| Width of Gussets at Base Plate               | BWDT       | 102.0000    | mm.            |
| Gusset Plate Elastic Modulus                 | E          | 201734000.0 | KPa.           |
| Gusset Plate Yield Stress                    | SY         | 207.0       | N./mm2         |
| Height of Chair                              | h1         | 230.0000    | mm.            |
| Distance between Gussets                     | RG, b6     | 90.0000     | mm.            |
| Dist. from Bolt Center to Gusset (Rg/2)      | CG         | 45.0000     | mm.            |
| Number of Gussets per bolt                   | NG         | 2           |                |
| Thickness of Top Plate or Ring               | TTA, e5    | 28.0000     | mm.            |
| Radial Width of the Top Plate                | TOPWTH, b5 | 130.0000    | mm.            |
| Anchor Bolt Hole Dia. in Top Plate           | dia,H      | 60.0000     | mm.            |
| External Corrosion Allowance                 | CA         | 1.5000      | mm.            |
| Dead Weight of Vessel                        | DW         | 109899.5    | Kgf            |
| Operating Weight of Vessel                   | ROW        | 123893.0    | Kgf            |
| Earthquake Moment on Basing                  | EQMOM      | 139146.8    | Kg-m.          |
| Wind Moment on Basing                        | WIMOM      | 78012.0     | Kg-m.          |
| User Moment at the Base [Earthquake Case]    |            | 40705.2     | Kg-m. (UEQMOM) |
| User Moment at the Base [WindLoad Case]      |            | 40705.2     | Kg-m. (UWIMOM) |
| Percent Bolt Preload                         | ppl        | 100.0       |                |
| Use AISC A5.2 Increase in Fc and Bolt Stress |            | No          |                |
| Use Allowable Weld Stress per AISC J2.5      |            | No          |                |
| Factor for Increase of Allowables            | Fact       | 1.0000      |                |

#### Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

#### Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:  

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 179852/2680) - 123893)/20$$

$$= 7226.9 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 118717/2680) - 109899)/20$$

$$= 3364.3 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/2680) - 123893)/20$$

$$= -6194.7 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abs : 7.8800 cm<sup>2</sup>  
 Area Available in all the Bolts Abs \* RN : 157.6004 cm<sup>2</sup>

| Trial# | k     | knew  | Cc    | Ct    | z     | j     | Ft      | Fc       | Fs     |
|--------|-------|-------|-------|-------|-------|-------|---------|----------|--------|
| 2      | 0.674 | 0.511 | 2.386 | 1.578 | 0.350 | 0.782 | 30326.8 | 154219.9 | 4943.7 |
| 4      | 0.430 | 0.471 | 1.837 | 2.158 | 0.409 | 0.785 | 20963.0 | 144856.1 | 2498.9 |
| 6      | 0.450 | 0.440 | 1.885 | 2.112 | 0.404 | 0.785 | 21688.7 | 145581.8 | 2640.9 |
| 8      | 0.445 | 0.443 | 1.873 | 2.124 | 0.405 | 0.785 | 21506.6 | 145399.6 | 2604.7 |
| 10     | 0.441 | 0.441 | 1.864 | 2.132 | 0.406 | 0.785 | 21370.3 | 145263.3 | 2577.9 |
| 12     | 0.440 | 0.441 | 1.862 | 2.134 | 0.406 | 0.785 | 21336.3 | 145229.3 | 2571.2 |

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 21336 / (1.87 \cdot 2680 \cdot 2.13)$$

$$= 39.1 \text{ N./mm}^2, \text{ Should be } \leq 137 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 7.88 / (\pi \cdot 2680)$$

$$= 1.87 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 3.09[(2 \cdot 0.44 \cdot 2680 + 168) / (2 \cdot 0.44 \cdot 2680)]$$

$$= 3.31 \text{ N./mm}^2, \text{ Should be } \leq 8.75 \text{ N./mm}^2$$

Values for table 10.3, l = 118, b = 210, l/b = 0.56060576

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(582, 1342) = 1342 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 1342/137)} + 1.5$$

$$= 25.5 \text{ mm.}$$

Nomenclature:

|      |                     |                               |
|------|---------------------|-------------------------------|
| a    | = (Dc-Ds)/2         | Skirt Distance to Bolt Circle |
| P    | = Sa * Abss         | Maximum Load on one Bolt      |
| l    |                     | Radial Top Plate Width        |
| lavg |                     | Average Gusset Plate Width    |
| g1   | = Gamma 1           | Constant Term f( b/l )        |
| g2   | = Gamma 2           | Constant Term f( b/l )        |
| g    | = Flat distance / 2 | Nut 1/2 Dimension (from Tema) |
| Fb   |                     | Allowable Bending Stress      |

Values for table 10.6, l = 130, b = 90, b/l = 0.69230771

As b/l (0.69) is less than 1, inverting b/l = 1.44 0.

Moment Term, based on geometry [Mo]:

$$= P/(4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l))/(\pi \cdot g))) + 1] - [(0.7 - g2)P/(4\pi)]$$

$$= 11032/(4 \cdot \pi) [1.3(\ln((2 \cdot 130 \cdot \sin(\pi \cdot 61/130))/(\pi \cdot 30.2))) + 1] - [(0.7 - 0.077) \cdot 11032/(4 \cdot \pi)]$$

$$= 1477.7347 \text{ Kgf}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb})} + \text{Ca}$$

$$= \sqrt{(6 \cdot \text{Abs}(1478) / 138)} + 1.5$$

$$= 26.6017 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 11032 / (124 \cdot 102) + 1.5$$

$$= 11 \text{ (not less than } 9.52 + 1.5 \text{ ) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 2680 / 20 = 421 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba)) \cdot 1.5 \cdot \text{Sktope})^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (11032 \cdot 61 / (421 \cdot 202 \cdot 207))^{2/3} \cdot 12791^{1/3} + \text{Ca}$$

$$= 9.95 + 1.5 = 11.4 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 20416 / (7.88 \cdot 20)$$

$$= 12.7 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

#### **Summary of Basing Thickness Calculations**

|                                              |         |     |
|----------------------------------------------|---------|-----|
| Required Basing Thickness (tension)          | 25.5411 | mm. |
| Actual Basing Thickness as entered by user   | 28.0000 | mm. |
| Required Thickness of Chair Cap              | 26.6017 | mm. |
| Actual Top Ring Thickness as entered by user | 28.0000 | mm. |
| Required Gusset thickness, + CA              | 11.0250 | mm. |
| Actual Gusset Thickness as entered by user   | 12.0000 | mm. |
| Required Thickness of Skirt for Local Stress | 11.4490 | mm. |
| Given Thickness of Skirt                     | 30.0000 | mm. |
| Required Gusset Height to meet local stress  | 13.6631 | mm. |

#### **Weld Size Calculations per Steel Plate Engineering Data - Vol. 2**

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 17.9 \cdot (30 - 1.5)$$

$$= 52 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 52 / [(0.4 \cdot 207) \cdot 2 \cdot 0.707]$$

$$= 4.36 \text{ mm.}$$

#### **Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size**

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 11032 / (160 + 2 \cdot (174 + 28))$$

$$= 19.6 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 11032 \cdot 61 / (160 \cdot (202) + 0.6667 \cdot (202)^2)$$

$$= 11.3 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{19.6^2 + 11.3^2}$$

$$= 22.6 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$



$$= 22.6 / [(0.4 \cdot 207) \cdot 2 \cdot 0.707]$$
$$= 1.9 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 11032 / (2 \cdot 102)$$
$$= 54.1 \text{ Kg/mm.}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 11032 \cdot 61 / (2 \cdot 202 \cdot 102)$$
$$= 16.3 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{54.1^2 + 16.3^2}$$
$$= 56.5 \text{ Kg/mm.}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 56.5 / [(0.4 \cdot 207) \cdot 2 \cdot 0.707]$$
$$= 4.74 \text{ mm.}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

**Summary of Required Weld Sizes:**

|                                                      |        |     |
|------------------------------------------------------|--------|-----|
| Required Basing to Skirt Double Fillet Weld Size     | 6.3500 | mm. |
| Required Gusset to Skirt Double Fillet Weld Size     | 4.7625 | mm. |
| Required Top Plate to Skirt Weld Size                | 6.3500 | mm. |
| Required Gusset to Top Plate Double Fillet Weld Size | 4.7430 | mm. |

Review notes about nozzle loadings and supports in the Warnings report.

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**Input, Nozzle Desc: N42"**

**From: 40**

|                                               |      |           |        |
|-----------------------------------------------|------|-----------|--------|
| Pressure for Reinforcement Calculations       | P    | 162.135   | bars   |
| Temperature for Internal Pressure             | Temp | 80        | °C     |
| Parent Material                               |      | SA-516 70 | +IT    |
| Parent Allowable Stress at Temperature        | Sv   | 161.85    | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm2 |
| Inside Diameter of Hemispherical Head         | D    | 2464.00   | mm.    |
| Head Finished (Minimum) Thickness             | t    | 65.0000   | mm.    |
| Head Internal Corrosion Allowance             | cas  | 1.5000    | mm.    |
| Head External Corrosion Allowance             | co   | 0.0000    | mm.    |
| Distance from Head Centerline                 | L1   | 0.0000    | mm.    |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C     |

**Type of Element Connected to the Parent : Nozzle**

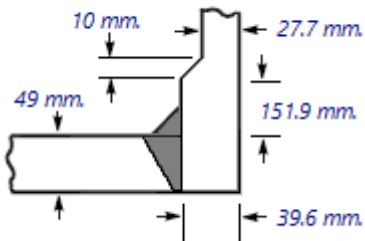
|                                                 |      |              |        |
|-------------------------------------------------|------|--------------|--------|
| Material                                        |      | SA-350 LF2   | +IT    |
| Material UNS Number                             |      | K03011       |        |
| Material Specification/Type                     |      | Forgings     |        |
| Allowable Stress at Temperature                 | Sn   | 153.86       | N./mm2 |
| Allowable Stress At Ambient                     | Sna  | 165.00       | N./mm2 |
| Diameter Basis (for tr calc only)               |      | Inside       |        |
| Layout Angle                                    |      | 0.00         | deg    |
| Diameter                                        |      | 1.9400       | in.    |
| Size and Thickness Basis                        |      | Actual       |        |
| Actual Thickness                                | tn   | 27.6860      | mm.    |
| Hub Height of Integral Nozzle                   | h    | 151.9080     | mm.    |
| Height of Beveled Transition                    | L'   | 10.0172      | mm.    |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 39.6240      | mm.    |
| Style of FVC Nozzle                             |      | Heavy Barrel |        |
| Corrosion Allowance                             | can  | 1.5000       | mm.    |
| Outside Projection                              | ho   | 200.0000     | mm.    |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000      | mm.    |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 66.0000      | mm.    |
| Inside Projection                               | h    | 0.0000       | mm.    |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000       | mm.    |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000      | deg.   |
| Piping Exit Angle [North Clockwise reference]   |      | 270.0        | deg    |
| Bend Radius Multiplier                          |      | 1.50         |        |
| Horizontal Run Length                           |      | 1500.0       | mm.    |
| Centerline Distance to Tangent                  |      | 1800.0       | mm.    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N42"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.940 | in. |
| Actual Thickness Used in Calculation       | 1.090 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 52.3 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.9 + 1.5 + 0 = 4.4 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 26.2 + 52.3) / 52.3) - 0$$

$$= 1068 - 0 = 1068 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.770 mm  
 Min. required shell thickness, internal p + c : 64.872 mm.

Nozzle neck minimum required thickness : 6.770 mm.

Minimum nozzle neck thickness 27.686 mm., is greater than required thickness 6.770 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 39.6 - 1.5$$

$$= 38.124 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 27.7 - 1.5$$

$$= 26.186 \text{ mm.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2464/2 + 1.5$$

$$= 1233.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1234 \cdot 63.5}, 2 \cdot 26.1)$$

$$= 52.276 \text{ mm.}$$

Intermediate value [Cn]:  

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((63.5 + 0)/38.1)^{0.35}, 1)$$

$$= \min(1.2, 1)$$

$$= 1.000$$

Intermediate value [Cp]:  

$$= 9999999.000$$



Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\ &= \min(63.5 + 0 + 1 \cdot \sqrt{(26.1 \cdot 38.1)}, 200 + 63.5) \\ &= 95.067 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot f_{rp}) / LR) \\ &= 63.5 + ((0 \cdot 1) / 52.3) \\ &= 63.500 \text{ mm.} \end{aligned}$$

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 63.5 \cdot 52.3 \\ &= 33.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 38.1 \cdot 95.1 \\ &= 36.243 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43} \\ &= 33.2 + 0.95(36.2 + 0) + 0.5 + 0 + 0 \\ &= 68.147 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 38.1 / \ln[1 + 38.1/26.1] \\ &= 42.380 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 63.5 / \ln[1 + 63.5/1234] \\ &= 1264.983 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 162 \cdot 1 \\ &= 242.8 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 162 \cdot 42.4 \cdot 95.1 \\ &= 6661.3 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (LR + t_n)/2 \\ &= 162 \cdot 1265 \cdot (52.3 + 38.1)/2 \\ &= 94535.6 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc})/2 \\ &= (162 \cdot 1265 \cdot 26.1)/2 \\ &= 27333.8 \text{ Kg} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (94536 + 27334 + 6661)/162 \\ &= 777.4 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/(2 \cdot t_{eff}))$$



$$= 243 / ( 2 \cdot 777 / 68.1 - 1265 / (2 \cdot 63.5) )$$
$$= 188.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$
$$= 2 \cdot 162 [63.5/1265]$$
$$= 162.5 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min( Pmax1, Pmax2 )$$
$$= \min( 189, 162 )$$
$$= 162.487 \text{ bars}$$

Average Primary Membrane Stress [ $\sigma_{Avg}$ ]:

$$= ( fN + fS + fY ) / AT$$
$$= ( 6661 + 94536 + 27334 ) / 68.1$$
$$= 184.964 \text{ N./mm}^2$$

General Primary Membrane Stress [ $\sigma_{Circ}$ ]:

$$= P \cdot Rxs / ( 2 \cdot teff )$$
$$= 162 \cdot 1265 / ( 2 \cdot 63.5 )$$
$$= 161.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max( 2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ} )$$
$$= \max( 2 \cdot 185 - 162, 162 )$$
$$= 208.4 \text{ N./mm}^2$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.78 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 208.42 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.49 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= ( Rnc + tn ) / Rnc$$
$$= ( 26.1 + 38.1 ) / 26.1$$
$$= 2.459 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot ( Rn + tn )$$
$$= \pi / 2 \cdot ( 26.1 + 38.1 )$$
$$= 100.943 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 38.124, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2 )$$
$$= \min( 27334 \cdot 2.46, 1.5 \cdot 154(36.2+0), \pi / 4 \cdot 162 \cdot 26.1^2 \cdot 2.46^2 )$$
$$= 5362.541 \text{ Kgf}$$

Weld Stress Value [ $\tau$ ]:

$$= fwelds / ( Lt( 0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T ) )$$
$$= 5363 / ( 101( 0.49 \cdot 7.07 + 0.6 \cdot 63.5 + 0.49 \cdot 0 ) )$$
$$= 12.534 < \text{ or } = \text{ to } 161.855 \text{ Weld Stress Passed}$$

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification

-46 °C



The Drop for this Nozzle is : 1.6771 mm.  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 266.6771 mm.

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**Input, Nozzle Desc: N32"**

**From: 50**

|                                               |      |           |        |
|-----------------------------------------------|------|-----------|--------|
| Pressure for Reinforcement Calculations       | P    | 162.018   | bars   |
| Temperature for Internal Pressure             | Temp | 80        | °C     |
| Parent Material                               |      | SA-516 70 | +IT    |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm2 |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.    |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.    |
| Distance from Bottom/Left Tangent             |      | 630.00    | mm.    |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C     |

**Type of Element Connected to the Parent : Nozzle**

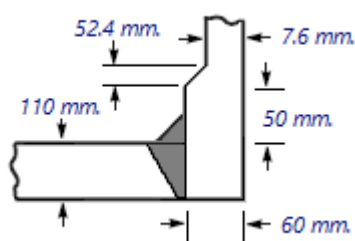
|                                                 |      |            |        |
|-------------------------------------------------|------|------------|--------|
| Material                                        |      | SA-350 LF2 | +N+IT  |
| Material UNS Number                             |      | K03011     |        |
| Material Specification/Type                     |      | Forgings   |        |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm2 |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm2 |
| Diameter Basis (for tr calc only)               |      | Inside     |        |
| Layout Angle                                    |      | 0.00       | deg    |
| Diameter                                        |      | 2.0000     | in.    |
| Size and Thickness Basis                        |      | Minimum    |        |
| Nominal Thickness                               |      | 160        |        |
| Hub Height of Integral Nozzle                   | h    | 35.0000    | mm.    |
| Height of Beveled Transition                    | L'   | 33.0000    | mm.    |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 40.0000    | mm.    |
| Corrosion Allowance                             | can  | 1.5000     | mm.    |
| Outside Projection                              | ho   | 150.0000   | mm.    |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.    |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.    |
| Inside Projection                               | h    | 0.0000     | mm.    |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.    |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.   |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N32"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.



### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0 \\ &= 2.67 + 1.5 + 0 = 4.17 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0 \\ &= 351 - 0 = 351 \text{ bars} \end{aligned}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
Min. required shell thickness, internal p + c : 128.079 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 74.8, | Allowable | : | 153.9 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 323.8 N./mm <sup>2</sup> | Passed |
| Occasional | : | 28.1, | Allowable | : | 204.7 N./mm <sup>2</sup> | Passed |
| Shear      | : | 34.6, | Allowable | : | 107.7 N./mm <sup>2</sup> | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[154/162, 1] \\ &= 0.951 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 40 - 1.5 \\ &= 38.500 \text{ mm.} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 7.65 - 1.5 \\ &= 6.145 \text{ mm.} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 2403/126 \\ &= 18.996 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 2400/2 + 1.5 \\ &= 1201.500 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24) \\ &= 48.034 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 38.5} \\ &= 30.408 \text{ mm.} \end{aligned}$$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 150 \\ &= 150.000 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(126 + 0) \\ &= 1012.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[30.4, 150, 1012] + 126 \\ &= 156.908 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 126 + ((0 \cdot 1) / 48) \\ &= 126.500 \text{ mm.} \end{aligned}$$

Parameter [ $\lambda$ ]:

$$\begin{aligned} &= \min\left(12, \frac{(2 \cdot R_n + t_n)}{\sqrt{((D_i + t_{eff}) \cdot t_{eff})}}\right) \\ &= \min\left(12, \frac{(2 \cdot 24 + 38.5)}{\sqrt{((2403 + 126) \cdot 126)}}\right) \\ &= 0.153 \end{aligned}$$

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 126 \cdot 48 \cdot \max((0.15/5)^{0.85}, 1) \\ &= 60.763 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 38.5 \cdot 157 \\ &= 60.410 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 60.8 + 0.95(60.4 + 0) + 0.5 + 0 + 0 \\ &= 118.688 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 38.5 / \ln[1 + 38.5/24] \\ &= 40.244 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 126 / \ln[1 + 126/1202] \\ &= 1263.695 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 162 \cdot 1 \\ &= 242.8 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 162 \cdot 40.2 \cdot 157 \\ &= 10432.9 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 162 \cdot 1264 (48 + 38.5) \end{aligned}$$



= 180672.1 Kgf

Discontinuity Force from Internal Pressure [fY]:

=  $P \cdot R_{xs} \cdot R_{nc}$   
=  $162 \cdot 1264 \cdot 24$   
= 50144.6 Kgf

Area Resisting Internal Pressure [Ap]:

=  $(fS + fY + fN) / P$   
=  $(180672 + 50145 + 10433) / 162$   
= 1460.2 cm<sup>2</sup>

Maximum Allowable Working Pressure Candidate [Pmax1]:

=  $Sallow / (2 \cdot Ap / AT - R_{xs} / teff)$   
=  $243 / (2 \cdot 1460 / 119 - 1264 / 126)$   
= 166.1 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

=  $S[t / R_{xs}]$   
=  $162 [126 / 1264]$   
= 162.1 bars

Maximum Allowable Working Pressure [Pmax]:

=  $\min(P_{max1}, P_{max2})$   
=  $\min(166, 162)$   
= 162.058 bars

Average Primary Membrane Stress [ $\sigma_{Avg}$ ]:

=  $(fN + fS + fY) / AT$   
=  $(10433 + 180672 + 50145) / 119$   
= 199.337 N./mm<sup>2</sup>

General Primary Membrane Stress [ $\sigma_{Circ}$ ]:

=  $P \cdot R_{xs} / teff$   
=  $162 \cdot 1264 / 126$   
= 161.9 N./mm<sup>2</sup>

Maximum Local Primary Membrane Stress [PL]:

=  $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$   
=  $\max(2 \cdot 199 - 162, 162)$   
= 236.8 N./mm<sup>2</sup>

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |                    |
|---------------------------------------|--------|--------|--------------------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm <sup>2</sup> |
| Local Primary Membrane Stress         | PL     | 236.81 | N./mm <sup>2</sup> |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars               |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

=  $(R_{nc} + t_n) / R_{nc}$   
=  $(24 + 38.5) / 24$   
= 2.603 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

=  $\pi / 2 \cdot (R_n + t_n)$   
=  $\pi / 2 \cdot (24 + 38.5)$   
= 98.202 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be  $\geq \min(0.7 \cdot 38.500, 6.000)$  Passed

Weld Load Value [fwelds]:

=  $\min(fy \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot ky^2)$   
=  $\min(50145 \cdot 2.6, 1.5 \cdot 154(60.4 + 0), \pi / 4 \cdot 162 \cdot 24^2 \cdot 2.6^2)$   
= 5071.644 Kgf



Weld Stress Value [τ]:  
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$   
 $= 5072 / (98.2 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$   
 $= 6.382 < \text{or} = \text{to } 161.900 \text{ Weld Stress Passed}$

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

##### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 1.6296 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 279.6297 mm.

#### Input Echo, WRC297 Item 1, Description: N32" :

|                                      |            |        |                    |
|--------------------------------------|------------|--------|--------------------|
| Diameter Basis for Cylindrical Shell | ID         |        |                    |
| Shell Internal Corrosion Allowance   | 1.5000     | mm.    |                    |
| Shell External Corrosion Allowance   | 0.0000     | mm.    |                    |
| Shell Diameter                       | 2400.000   | mm.    |                    |
| Shell Thickness                      | 128.0000   | mm.    |                    |
| Shell Stress Concentration Factor    | 1.000      |        |                    |
| Vessel Material                      | SA-516 70  |        |                    |
| Vessel Cold S.I. Allowable           | Smc        | 175.00 | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh        | 161.90 | N./mm <sup>2</sup> |
| Diameter Basis for Nozzle            | ID         |        |                    |
| Nozzle Corrosion Allowance           | 1.5000     | mm.    |                    |
| Nozzle Diameter                      | 45.034     | mm.    |                    |
| Nozzle Thickness                     | 40.0000    | mm.    |                    |
| Nozzle Stress Concentration Factor   | 1.000      |        |                    |
| Nozzle Material                      | SA-350 LF2 |        |                    |
| Nozzle Cold S.I. Allowable           | SNmc       | 165.00 | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable            | SNmh       | 153.90 | N./mm <sup>2</sup> |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

##### Note:

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUS tained loads.*

|                          |    |        |       |
|--------------------------|----|--------|-------|
| Design Internal Pressure | Dp | 162.02 | bars  |
| Radial Load              | P  | 163.16 | Kgf   |
| Circumferential Shear    | Vc | 254.95 | Kgf   |
| Longitudinal Shear       | Vl | 254.95 | Kgf   |
| Circumferential Moment   | Mc | 30.58  | Kg-m. |
| Longitudinal Moment      | Ml | 50.97  | Kg-m. |
| Torsional Moment         | Mt | 61.16  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tv, (.977) is < 5  
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.126) is < 10  
 check the Limitations of WRC bulletin 297.*

#### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$   
 $= 125.034 + 2 \cdot 1.65 \cdot \sqrt{(1264.75(128.0 - 1.5))}$   
 $= 1444.997 \text{ mm.}$



WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2529.500 mm.  
 Nozzle Outside Diameter (d) = 125.034 mm.  
 Vessel Thickness used (T) = 126.500 mm.  
 Nozzle Thickness used (t) = 38.500 mm.  
 T / t = 3.286  
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.221$$

Nr/P = 0.250  
 Mr/P = 0.216  
 M0/P = 0.240  
 N0/P = 0.202  
 MrD/Mc = 0.181  
 NrDL/Mc = 0.182  
 M0d/Mc = 0.234  
 N0DL/Mc = -0.130  
 MrD/Ml = 0.179  
 NrDL/Ml = 0.186  
 M0D/Ml = 0.231  
 N0DL/Ml = -0.128

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (P )                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Ml)                                            | 0                    | 0                   | 0                       | 0                      |
| Normal Pressure Stress                                           | 146                  | 162                 | 146                     | 162                    |
| Outplane Stress Summary                                          | 145                  | 162                 | 146                     | 162                    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (P )                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (Ml)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Ml)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Pressure Stress                                            | 73                   | 73                  | 73                      | 73                     |
| Inplane Stress Summary                                             | 72                   | 73                  | 73                      | 72                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 0                    | 0                   | 0                       | 0                      |
| Shear Stress Summary                                                 | 0                    | 0                   | 0                       | 0                      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|                                            |                      |                     |                         |                        |



Two \* Max Shear Stress 145 162 146 162

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu      | Cl     | Du      | Dl     |
|------------------------------------------------------------------------|---------|--------|---------|--------|
|                                                                        | Left    | Left   | Right   | Right  |
|                                                                        | Outside | Inside | Outside | Inside |
| Outplane Membrane (P )                                                 | 0       | 0      | 0       | 0      |
| Outplane Bending (P )                                                  | 0       | 0      | 0       | 0      |
| Outplane Membrane (Mc)                                                 | 0       | 0      | 0       | 0      |
| Outplane Bending (Mc)                                                  | 0       | 0      | 0       | 0      |
| Outplane Membrane (ML)                                                 | 0       | 0      | 0       | 0      |
| Outplane Bending (ML)                                                  | 0       | 0      | 0       | 0      |
| Normal Pressure Stress                                                 | 73      | 73     | 73      | 73     |
| Outplane Stress Summary                                                | 72      | 73     | 73      | 72     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu      | Cl     | Du      | Dl     |
|--------------------------------------------------------------------------|---------|--------|---------|--------|
|                                                                          | Left    | Left   | Right   | Right  |
|                                                                          | Outside | Inside | Outside | Inside |
| Inplane Membrane (P )                                                    | 0       | 0      | 0       | 0      |
| Inplane Bending (P )                                                     | 0       | 0      | 0       | 0      |
| Inplane Membrane (Mc)                                                    | 0       | 0      | 0       | 0      |
| Inplane Bending (Mc)                                                     | 0       | 0      | 0       | 0      |
| Inplane Membrane (ML)                                                    | 0       | 0      | 0       | 0      |
| Inplane Bending (ML)                                                     | 0       | 0      | 0       | 0      |
| Inplane Pressure Stress                                                  | 146     | 162    | 146     | 162    |
| Inplane Stress Summary                                                   | 145     | 162    | 146     | 162    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu      | Cl     | Du      | Dl     |
|----------------------------------------------------------------------------|---------|--------|---------|--------|
|                                                                            | Left    | Left   | Right   | Right  |
|                                                                            | Outside | Inside | Outside | Inside |
| Outplane Shear (Vc)                                                        | 0       | 0      | 0       | 0      |
| Outplane Shear (Vl)                                                        | 0       | 0      | 0       | 0      |
| Torsional Shear (Mt)                                                       | 0       | 0      | 0       | 0      |
| Shear Stress Summary                                                       | 0       | 0      | 0       | 0      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------------|---------|--------|---------|--------|
|                                               | Left    | Left   | Right   | Right  |
|                                               | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress                        | 145     | 162    | 146     | 162    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au      | Al     | Bu      | Bl     |
|-----------------------------------------------------------|---------|--------|---------|--------|
|                                                           | Top     | Top    | Bottom  | Bottom |
|                                                           | Outside | Inside | Outside | Inside |
| Hoop Membrane (P )                                        | 0       | 0      | 0       | 0      |
| Hoop Bending (P )                                         | 0       | 0      | 0       | 0      |
| Hoop Membrane (Mc)                                        | 0       | 0      | 0       | 0      |
| Hoop Bending (Mc)                                         | 0       | 0      | 0       | 0      |
| Hoop Membrane (ML)                                        | 0       | 0      | 0       | 0      |
| Hoop Bending (ML)                                         | 0       | 0      | 0       | 0      |
| Hoop Pressure Stress                                      | 6       | 26     | 6       | 26     |
| Hoop Stress Summary                                       | 6       | 26     | 6       | 26     |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au      | Al     | Bu      | Bl     |
|------------------------------------------------------------------|---------|--------|---------|--------|
|                                                                  | Top     | Top    | Bottom  | Bottom |
|                                                                  | Outside | Inside | Outside | Inside |



|                       |    |    |   |    |
|-----------------------|----|----|---|----|
| Axial Membrane (P )   | 0  | 0  | 0 | 0  |
| Axial Bending (P )    | 0  | 0  | 0 | 0  |
| Axial Membrane (Mc)   | 0  | 0  | 0 | 0  |
| Axial Bending (Mc)    | 0  | 0  | 0 | 0  |
| Axial Membrane (ML)   | -2 | -2 | 2 | 2  |
| Axial Bending (ML)    | -1 | 1  | 1 | -1 |
| Axial Pressure Stress | 3  | 3  | 3 | 3  |

|                      |    |   |   |   |
|----------------------|----|---|---|---|
| Axial Stress Summary | -1 | 2 | 6 | 5 |
|----------------------|----|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                      |         |        |         |        |
|----------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE   | Au      | Al     | Bu      | Bl     |
| (Shear stress)       | Outside | Inside | Outside | Inside |
| Shear due to (Vc)    | 0       | 0      | 0       | 0      |
| Shear due to (Vl)    | 0       | 0      | 0       | 0      |
| Shear due to Torsion | 0       | 0      | 0       | 0      |
| Shear Stress Summary | 0       | 0      | 0       | 0      |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                        |         |        |         |        |
|------------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE     | Au      | Al     | Bu      | Bl     |
| (Stress Intensities    | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress | 8       | 26     | 6       | 26     |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stresses in the      | Left    | Left   | Right   | Right  |
| hoop direction)       | Outside | Inside | Outside | Inside |
| Hoop Membrane (P )    | 0       | 0      | 0       | 0      |
| Hoop Bending (P )     | 0       | 0      | 0       | 0      |
| Hoop Membrane (Mc)    | 0       | 0      | 0       | 0      |
| Hoop Bending (Mc)     | 0       | 0      | 0       | 0      |
| Hoop Membrane (ML)    | 0       | 0      | 0       | 0      |
| Hoop Bending (ML)     | 0       | 0      | 0       | 0      |
| Hoop Pressure Stress  | 6       | 26     | 6       | 26     |
| Hoop Stress Summary   | 6       | 26     | 6       | 26     |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stresses Normal to   | Left    | Left   | Right   | Right  |
| pipe cross-section)   | Outside | Inside | Outside | Inside |
| Axial Membrane (P )   | 0       | 0      | 0       | 0      |
| Axial Bending (P )    | 0       | 0      | 0       | 0      |
| Axial Membrane (Mc)   | -1      | -1     | 1       | 1      |
| Axial Bending (Mc)    | 0       | 0      | 0       | 0      |
| Axial Membrane (ML)   | 0       | 0      | 0       | 0      |
| Axial Bending (ML)    | 0       | 0      | 0       | 0      |
| Axial Pressure Stress | 3       | 3      | 3       | 3      |
| Axial Stress Summary  | 0       | 3      | 5       | 4      |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Shear stress)        | Outside | Inside | Outside | Inside |
| Shear due to (Vc)     | 0       | 0      | 0       | 0      |
| Shear due to (Vl)     | 0       | 0      | 0       | 0      |
| Shear due to Torsion  | 0       | 0      | 0       | 0      |
| Shear Stress Summary  | 0       | 0      | 0       | 0      |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stress Intensities   | Outside | Inside | Outside | Inside |



Two \* Max Shear Stress 6 26 7 26

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Vessel Stress Summation at Vessel-Nozzle Junction (N./mm2):

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 146              | 162   | 146   | 162   | 146   | 162   | 146   | 162   |
| Circ. Pl (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Circ. Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)               | 73               | 73    | 73    | 73    | 73    | 73    | 73    | 73    |
| Long. Pl (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Pm (SUS)                     | 146.0            | 162.0 | 146.0 | 162.0 | 146.0 | 162.0 | 146.0 | 162.0 |
| Pm+Pl (SUS)                  | 146.0            | 162.0 | 146.0 | 162.0 | 146.0 | 162.0 | 146.0 | 162.0 |
| Pm+Pl+Q (Total)              | 146.0            | 162.0 | 146.0 | 162.0 | 146.0 | 162.0 | 146.0 | 162.0 |

##### Vessel Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 162.00    | 161.90         | Failed |
| Pm+Pl (SUS)         | 162.00    | 242.85         | Passed |
| Pm+Pl+Q (TOTAL)     | 162.00    | 485.70         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

| Type of Stress Int. Location | Stress Values at |      |     |      |     |      |     |      |
|------------------------------|------------------|------|-----|------|-----|------|-----|------|
|                              | Au               | Al   | Bu  | Bl   | Cu  | Cl   | Du  | Dl   |
| Circ. Pm (SUS)               | 6                | 26   | 6   | 26   | 6   | 26   | 6   | 26   |
| Circ. Pl (SUS)               | 0                | 0    | 0   | 0    | 0   | 0    | 0   | 0    |
| Circ. Q (SUS)                | 0                | 0    | 0   | 0    | 0   | 0    | 0   | 0    |
| Long. Pm (SUS)               | 3                | 3    | 3   | 3    | 3   | 3    | 3   | 3    |
| Long. Pl (SUS)               | -2               | -2   | 2   | 2    | -1  | -1   | 1   | 1    |
| Long. Q (SUS)                | -1               | 1    | 1   | -1   | 0   | 0    | 0   | 0    |
| Shear Pm (SUS)               | 0                | 0    | 0   | 0    | 0   | 0    | 0   | 0    |
| Shear Pl (SUS)               | 0                | 0    | 0   | 0    | 0   | 0    | 0   | 0    |
| Shear Q (SUS)                | 0                | 0    | 0   | 0    | 0   | 0    | 0   | 0    |
| Pm (SUS)                     | 6.0              | 26.0 | 6.0 | 26.0 | 6.0 | 26.0 | 6.0 | 26.0 |
| Pm+Pl (SUS)                  | 6.0              | 26.0 | 6.0 | 26.0 | 6.0 | 26.0 | 6.0 | 26.0 |
| Pm+Pl+Q (Total)              | 6.0              | 26.0 | 6.0 | 26.0 | 6.0 | 26.0 | 6.0 | 26.0 |

##### Nozzle Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
|---------------------|-----------|----------------|--------|



|                 |       |        |        |
|-----------------|-------|--------|--------|
| Pm (SUS)        | 26.00 | 153.90 | Passed |
| Pm+Pl (SUS)     | 26.00 | 230.85 | Passed |
| Pm+Pl+Q (TOTAL) | 26.00 | 461.70 | Passed |

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.*

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**Input, Nozzle Desc: K1B2"**

**From: 50**

|                                               |      |           |        |
|-----------------------------------------------|------|-----------|--------|
| Pressure for Reinforcement Calculations       | P    | 162.042   | bars   |
| Temperature for Internal Pressure             | Temp | 80        | °C     |
| Parent Material                               |      | SA-516 70 | +IT    |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm2 |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.    |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.    |
| Distance from Bottom/Left Tangent             |      | 250.00    | mm.    |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C     |

**Type of Element Connected to the Parent : Nozzle**

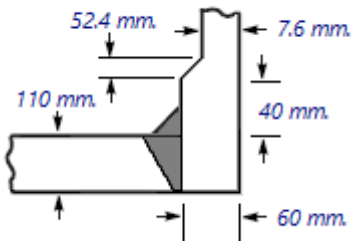
|                                                 |      |            |        |
|-------------------------------------------------|------|------------|--------|
| Material                                        |      | SA-350 LF2 | +N+IT  |
| Material UNS Number                             |      | K03011     |        |
| Material Specification/Type                     |      | Forgings   |        |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm2 |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm2 |
| Diameter Basis (for tr calc only)               |      | Inside     |        |
| Layout Angle                                    |      | 180.00     | deg    |
| Diameter                                        |      | 2.0000     | in.    |
| Size and Thickness Basis                        |      | Minimum    |        |
| Nominal Thickness                               |      | 160        |        |
| Flange Material                                 |      | SA-350 LF2 | +N+IT  |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.    |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.    |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.    |
| Corrosion Allowance                             | can  | 1.5000     | mm.    |
| Outside Projection                              | ho   | 400.0000   | mm.    |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.    |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.    |
| Inside Projection                               | h    | 0.0000     | mm.    |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.    |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.   |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K1B2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.67 + 1.5 + 0 = 4.17 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 128.098 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

=  $8(t + t_e)$   
=  $8(126 + 0)$   
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

=  $\min[LH1, LH2, LH3] + t$  for set-in nozzles  
=  $\min[34.1, 400, 1012] + 126$   
= 160.630 mm.

Effective Vessel Thickness [teff]:

=  $t + ((A5 \cdot frp) / LR)$   
=  $126 + ((0 \cdot 1) / 48)$   
= 126.500 mm.

Parameter [ $\lambda$ ]:

=  $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$   
=  $\min(12, (2 \cdot 24 + 48.5) / (\sqrt{(2403 + 126) \cdot 126}))$   
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

=  $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$   
=  $126 \cdot 48 \cdot \max((0.17/5)^{0.85}, 1)$   
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

=  $tn \cdot LH$   
=  $48.5 \cdot 161$   
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

=  $0.5 \cdot Leg^2$   
=  $0.5 \cdot 10^2$   
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

=  $A1 + frn(A2 + A3) + A41 + A42 + A43$   
=  $60.8 + 0.95(77.9 + 0) + 0.5 + 0 + 0$   
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

=  $tn / \ln[1 + tn/Rn]$   
=  $48.5 / \ln[1 + 48.5/24]$   
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

=  $teff / \ln[1 + teff/Reff]$   
=  $126 / \ln[1 + 126/1202]$   
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

=  $1.5 \cdot S \cdot E$   
=  $1.5 \cdot 162 \cdot 1$   
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

=  $P \cdot Rxn \cdot LH$   
=  $162 \cdot 43.9 \cdot 161$   
= 11649.5 Kgf

Force acting on the Shell [fS]:



$$= P \cdot Rxs( LR + tn )$$

$$= 162 \cdot 1264 ( 48 + 48.5 )$$

$$= 201580.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 162 \cdot 1264 \cdot 24$$

$$= 50151.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= ( fS + fY + fN )/P$$

$$= ( 201580 + 50152 + 11650 )/162$$

$$= 1593.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / ( 2 \cdot Ap/AT - Rxs/teff )$$

$$= 243 / ( 2 \cdot 1594/135 - 1264/126 )$$

$$= 179.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 162 [126/1264]$$

$$= 162.1 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min( Pmax1, Pmax2 )$$

$$= \min( 179, 162 )$$

$$= 162.058 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= ( fN + fS + fY ) / AT$$

$$= ( 11650 + 201580 + 50152 )/135$$

$$= 190.878 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 162 \cdot 1264/126$$

$$= 161.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ )$$

$$= \max( 2 \cdot 191 - 162, 162 )$$

$$= 219.9 \text{ N./mm}^2$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.87 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= ( Rnc + tn ) / Rnc$$

$$= ( 24 + 48.5 )/24$$

$$= 3.019 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot ( Rn + tn )$$

$$= \pi/2 \cdot ( 24 + 48.5 )$$

$$= 113.910 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 )$$

$$= \min( 50152 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 )$$



= 6824.885 Kgf

Weld Stress Value [T]:

=  $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$

=  $6825 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.403 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K3B2"**

**From: 50**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 162.042   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 250.00    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

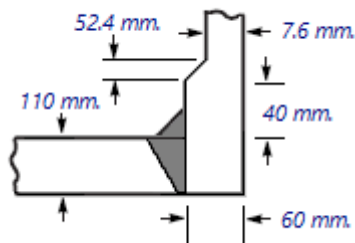
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 90.00      | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K3B2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.67 + 1.5 + 0 = 4.17 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 128.098 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

= 8( t + te )  
= 8( 126 + 0 )  
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[ LH1, LH2, LH3 ] + t for set-in nozzles  
= min[ 34.1, 400, 1012 ] + 126  
= 160.630 mm.

Effective Vessel Thickness [teff]:

= t + (( A5 • frp ) / LR )  
= 126+((0•1)/48)  
= 126.500 mm.

Parameter [λ]:

= min( 12, ( 2•Rn + tn )/( √( ( Di + teff ) • teff ) ) )  
= min( 12, (2•24 + 48.5 )/( √((2403 + 126 ) • 126 ) ) )  
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

= t • LR • max( (λ/5)<sup>0.85</sup>, 1 )  
= 126 • 48 • max( (0.17/5)<sup>0.85</sup>, 1 )  
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= tn • LH  
= 48.5 • 161  
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

= 0.5 • Leg<sup>4/3</sup>  
= 0.5 • 10<sup>2</sup>  
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

= A1 + frn( A2 + A3 ) + A41 + A42 + A43  
= 60.8+0.95(77.9+0)+0.5+0+0  
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

= tn / ln[ 1 + tn/Rn ]  
= 48.5/ln[ 1 + 48.5/24 ]  
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

= teff / ln[ 1 + teff/Reff ]  
= 126/ln[ 1 + 126/1202 ]  
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= 1.5 • S • E  
= 1.5 • 162 • 1  
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

= P • Rxn • LH  
= 162 • 43.9 • 161  
= 11649.5 Kgf

Force acting on the Shell [fS]:



$$= P \cdot Rxs( LR + tn )$$

$$= 162 \cdot 1264 ( 48 + 48.5 )$$

$$= 201580.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 162 \cdot 1264 \cdot 24$$

$$= 50151.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= ( fS + fY + fN )/P$$

$$= ( 201580 + 50152 + 11650 )/162$$

$$= 1593.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / ( 2 \cdot Ap/AT - Rxs/teff )$$

$$= 243 / ( 2 \cdot 1594/135 - 1264/126 )$$

$$= 179.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 162 [126/1264]$$

$$= 162.1 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min( Pmax1, Pmax2 )$$

$$= \min( 179, 162 )$$

$$= 162.058 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= ( fN + fS + fY ) / AT$$

$$= ( 11650 + 201580 + 50152 )/135$$

$$= 190.878 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 162 \cdot 1264/126$$

$$= 161.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ )$$

$$= \max( 2 \cdot 191 - 162, 162 )$$

$$= 219.9 \text{ N./mm}^2$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.87 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= ( Rnc + tn ) / Rnc$$

$$= ( 24 + 48.5 )/24$$

$$= 3.019 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot ( Rn + tn )$$

$$= \pi/2 \cdot ( 24 + 48.5 )$$

$$= 113.910 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 )$$

$$= \min( 50152 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 )$$



= 6824.885 Kgf

Weld Stress Value [T]:

=  $\text{fwelds}/(\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$

=  $6825/(114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.403 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification      -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K3D2'**

**From: 50**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 162.042   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 250.00    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

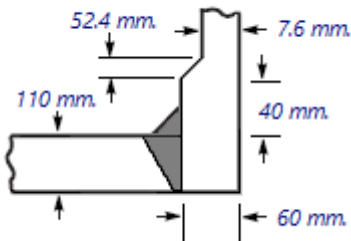
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 115.00     | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K3D2'**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$   
 $= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$   
 $= 2.67 + 1.5 + 0 = 4.17 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:  
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$   
 $= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$   
 $= 351 - 0 = 351 \text{ bars}$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 128.098 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  
 $= \min[S_n/S, 1]$   
 $= \min[154/162, 1]$   
 $= 0.951$

Thickness of Nozzle at Shell [tn]:  
 $= \text{hub thickness} - \text{corrosion allowance}$   
 $= 50 - 1.5$   
 $= 48.500 \text{ mm.}$

Thickness of Nozzle at Top [tn2]:  
 $= \text{thickness} - \text{corrosion allowance}$   
 $= 7.65 - 1.5$   
 $= 6.145 \text{ mm.}$

Shell Diameter to Thickness ratio [D/t]:  
 $= D_i/t$   
 $= 2403/126$   
 $= 18.996 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:  
 $= D_i/2 + \text{corrosion allowance}$   
 $= 2400/2 + 1.5$   
 $= 1201.500 \text{ mm.}$

Effective Length of Vessel Wall [LR]:  
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$   
 $= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$   
 $= 48.034 \text{ mm.}$

Thickness Limit Candidate [LH1]:  
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$   
 $= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

=  $8(t + t_e)$   
=  $8(126 + 0)$   
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

=  $\min[LH1, LH2, LH3] + t$  for set-in nozzles  
=  $\min[34.1, 400, 1012] + 126$   
= 160.630 mm.

Effective Vessel Thickness [teff]:

=  $t + ((A5 \cdot frp) / LR)$   
=  $126 + ((0 \cdot 1) / 48)$   
= 126.500 mm.

Parameter [ $\lambda$ ]:

=  $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$   
=  $\min(12, (2 \cdot 24 + 48.5) / (\sqrt{(2403 + 126) \cdot 126}))$   
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

=  $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$   
=  $126 \cdot 48 \cdot \max((0.17/5)^{0.85}, 1)$   
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

=  $tn \cdot LH$   
=  $48.5 \cdot 161$   
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

=  $0.5 \cdot Leg_1^2$   
=  $0.5 \cdot 10^2$   
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

=  $A1 + frn(A2 + A3) + A41 + A42 + A43$   
=  $60.8 + 0.95(77.9 + 0) + 0.5 + 0 + 0$   
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

=  $tn / \ln[1 + tn/Rn]$   
=  $48.5 / \ln[1 + 48.5/24]$   
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

=  $teff / \ln[1 + teff/Reff]$   
=  $126 / \ln[1 + 126/1202]$   
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

=  $1.5 \cdot S \cdot E$   
=  $1.5 \cdot 162 \cdot 1$   
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

=  $P \cdot Rxn \cdot LH$   
=  $162 \cdot 43.9 \cdot 161$   
= 11649.5 Kgf

Force acting on the Shell [fS]:



$$= P \cdot Rxs( LR + tn )$$

$$= 162 \cdot 1264 ( 48 + 48.5 )$$

$$= 201580.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 162 \cdot 1264 \cdot 24$$

$$= 50151.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= ( fS + fY + fN )/P$$

$$= ( 201580 + 50152 + 11650 )/162$$

$$= 1593.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / ( 2 \cdot Ap/AT - Rxs/teff )$$

$$= 243 / ( 2 \cdot 1594/135 - 1264/126 )$$

$$= 179.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 162 [126/1264]$$

$$= 162.1 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min( Pmax1, Pmax2 )$$

$$= \min( 179, 162 )$$

$$= 162.058 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= ( fN + fS + fY ) / AT$$

$$= ( 11650 + 201580 + 50152 )/135$$

$$= 190.878 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 162 \cdot 1264/126$$

$$= 161.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ )$$

$$= \max( 2 \cdot 191 - 162, 162 )$$

$$= 219.9 \text{ N./mm}^2$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.87 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= ( Rnc + tn ) / Rnc$$

$$= ( 24 + 48.5 )/24$$

$$= 3.019 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot ( Rn + tn )$$

$$= \pi/2 \cdot ( 24 + 48.5 )$$

$$= 113.910 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 )$$

$$= \min( 50152 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 )$$



= 6824.885 Kgf

Weld Stress Value [T]:

=  $\text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$

=  $6825 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.403 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K3F2"**

**From: 50**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 162.042   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 250.00    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

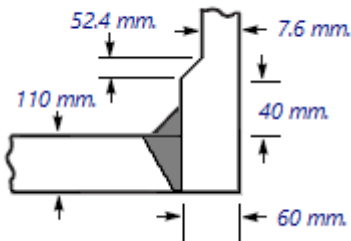
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 0.00       | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K3F2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.67 + 1.5 + 0 = 4.17 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 128.098 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

=  $8(t + t_e)$   
=  $8(126 + 0)$   
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

=  $\min[LH1, LH2, LH3] + t$  for set-in nozzles  
=  $\min[34.1, 400, 1012] + 126$   
= 160.630 mm.

Effective Vessel Thickness [teff]:

=  $t + ((A5 \cdot frp) / LR)$   
=  $126 + ((0 \cdot 1) / 48)$   
= 126.500 mm.

Parameter [ $\lambda$ ]:

=  $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$   
=  $\min(12, (2 \cdot 24 + 48.5) / (\sqrt{(2403 + 126) \cdot 126}))$   
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

=  $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$   
=  $126 \cdot 48 \cdot \max((0.17/5)^{0.85}, 1)$   
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

=  $tn \cdot LH$   
=  $48.5 \cdot 161$   
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

=  $0.5 \cdot Leg_1^2$   
=  $0.5 \cdot 10^2$   
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

=  $A1 + frn(A2 + A3) + A41 + A42 + A43$   
=  $60.8 + 0.95(77.9 + 0) + 0.5 + 0 + 0$   
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

=  $tn / \ln[1 + tn/Rn]$   
=  $48.5 / \ln[1 + 48.5/24]$   
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

=  $teff / \ln[1 + teff/Reff]$   
=  $126 / \ln[1 + 126/1202]$   
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

=  $1.5 \cdot S \cdot E$   
=  $1.5 \cdot 162 \cdot 1$   
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

=  $P \cdot Rxn \cdot LH$   
=  $162 \cdot 43.9 \cdot 161$   
= 11649.5 Kgf

Force acting on the Shell [fS]:



$$\begin{aligned}
 &= P \cdot Rxs( LR + tn ) \\
 &= 162 \cdot 1264 ( 48 + 48.5 ) \\
 &= 201580.3 \text{ Kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 162 \cdot 1264 \cdot 24 \\
 &= 50151.9 \text{ Kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= ( fS + fY + fN )/P \\
 &= ( 201580 + 50152 + 11650 )/162 \\
 &= 1593.9 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / ( 2 \cdot Ap/AT - Rxs/teff ) \\
 &= 243 / ( 2 \cdot 1594/135 - 1264/126 ) \\
 &= 179.0 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 162 [126/1264 ] \\
 &= 162.1 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min( Pmax1, Pmax2 ) \\
 &= \min( 179, 162 ) \\
 &= 162.058 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= ( fN + fS + fY ) / AT \\
 &= ( 11650 + 201580 + 50152 )/135 \\
 &= 190.878 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 162 \cdot 1264/126 \\
 &= 161.9 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ ) \\
 &= \max( 2 \cdot 191 - 162, 162 ) \\
 &= 219.9 \text{ N./mm}^2
 \end{aligned}$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.87 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= ( Rnc + tn ) / Rnc \\
 &= ( 24 + 48.5 )/24 \\
 &= 3.019 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot ( Rn + tn ) \\
 &= \pi/2 \cdot ( 24 + 48.5 ) \\
 &= 113.910 \text{ mm.}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 ) \\
 &= \min( 50152 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 )
 \end{aligned}$$



= 6824.885 Kgf

Weld Stress Value [T]:

=  $fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$

=  $6825 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.403 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification      -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K2B2"**

**From: 50**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 162.042   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 250.00    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

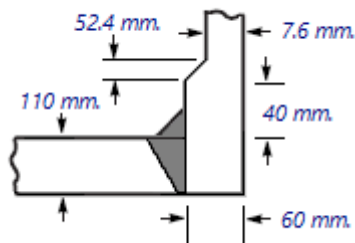
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 270.00     | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K2B2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.67 + 1.5 + 0 = 4.17 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 128.098 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

=  $8(t + t_e)$   
=  $8(126 + 0)$   
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

=  $\min[LH1, LH2, LH3] + t$  for set-in nozzles  
=  $\min[34.1, 400, 1012] + 126$   
= 160.630 mm.

Effective Vessel Thickness [teff]:

=  $t + ((A5 \cdot frp) / LR)$   
=  $126 + ((0 \cdot 1) / 48)$   
= 126.500 mm.

Parameter [ $\lambda$ ]:

=  $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$   
=  $\min(12, (2 \cdot 24 + 48.5) / (\sqrt{(2403 + 126) \cdot 126}))$   
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

=  $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$   
=  $126 \cdot 48 \cdot \max((0.17/5)^{0.85}, 1)$   
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

=  $tn \cdot LH$   
=  $48.5 \cdot 161$   
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

=  $0.5 \cdot Leg^2$   
=  $0.5 \cdot 10^2$   
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

=  $A1 + frn(A2 + A3) + A41 + A42 + A43$   
=  $60.8 + 0.95(77.9 + 0) + 0.5 + 0 + 0$   
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

=  $tn / \ln[1 + tn/Rn]$   
=  $48.5 / \ln[1 + 48.5/24]$   
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

=  $teff / \ln[1 + teff/Reff]$   
=  $126 / \ln[1 + 126/1202]$   
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

=  $1.5 \cdot S \cdot E$   
=  $1.5 \cdot 162 \cdot 1$   
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

=  $P \cdot Rxn \cdot LH$   
=  $162 \cdot 43.9 \cdot 161$   
= 11649.5 Kgf

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs( LR + tn ) \\ &= 162 \cdot 1264 ( 48 + 48.5 ) \\ &= 201580.3 \text{ Kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 162 \cdot 1264 \cdot 24 \\ &= 50151.9 \text{ Kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= ( fS + fY + fN )/P \\ &= ( 201580 + 50152 + 11650 )/162 \\ &= 1593.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / ( 2 \cdot Ap/AT - Rxs/teff ) \\ &= 243 / ( 2 \cdot 1594/135 - 1264/126 ) \\ &= 179.0 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 162 [126/1264 ] \\ &= 162.1 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min( Pmax1, Pmax2 ) \\ &= \min( 179, 162 ) \\ &= 162.058 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [ $\sigma$ Avg]:

$$\begin{aligned} &= ( fN + fS + fY ) / AT \\ &= ( 11650 + 201580 + 50152 )/135 \\ &= 190.878 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [ $\sigma$ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 162 \cdot 1264/126 \\ &= 161.9 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ ) \\ &= \max( 2 \cdot 191 - 162, 162 ) \\ &= 219.9 \text{ N./mm}^2 \end{aligned}$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.87 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= ( Rnc + tn ) / Rnc \\ &= ( 24 + 48.5 )/24 \\ &= 3.019 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot ( Rn + tn ) \\ &= \pi/2 \cdot ( 24 + 48.5 ) \\ &= 113.910 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 ) \\ &= \min( 50152 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 ) \end{aligned}$$



= 6824.885 Kgf

Weld Stress Value [T]:

=  $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$

=  $6825 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.403 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K1A2"**

**From: 60**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 161.808   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 4149.97   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

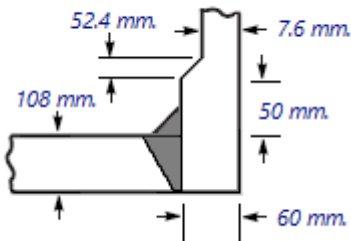
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 180.00     | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K1A2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.66 + 1.5 + 0 = 4.16 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 127.906 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

=  $8(t + t_e)$   
=  $8(126 + 0)$   
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

=  $\min[LH1, LH2, LH3] + t$  for set-in nozzles  
=  $\min[34.1, 400, 1012] + 126$   
= 160.630 mm.

Effective Vessel Thickness [teff]:

=  $t + ((A5 \cdot frp) / LR)$   
=  $126 + ((0 \cdot 1) / 48)$   
= 126.500 mm.

Parameter [ $\lambda$ ]:

=  $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$   
=  $\min(12, (2 \cdot 24 + 48.5) / (\sqrt{(2403 + 126) \cdot 126}))$   
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

=  $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$   
=  $126 \cdot 48 \cdot \max((0.17/5)^{0.85}, 1)$   
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

=  $tn \cdot LH$   
=  $48.5 \cdot 161$   
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

=  $0.5 \cdot Leg^2$   
=  $0.5 \cdot 10^2$   
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

=  $A1 + frn(A2 + A3) + A41 + A42 + A43$   
=  $60.8 + 0.95(77.9 + 0) + 0.5 + 0 + 0$   
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

=  $tn / \ln[1 + tn/Rn]$   
=  $48.5 / \ln[1 + 48.5/24]$   
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

=  $teff / \ln[1 + teff/Reff]$   
=  $126 / \ln[1 + 126/1202]$   
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

=  $1.5 \cdot S \cdot E$   
=  $1.5 \cdot 162 \cdot 1$   
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

=  $P \cdot Rxn \cdot LH$   
=  $162 \cdot 43.9 \cdot 161$   
= 11632.7 Kgf

Force acting on the Shell [fS]:



$$= P \cdot Rxs( LR + tn )$$

$$= 162 \cdot 1264 ( 48 + 48.5 )$$

$$= 201289.5 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 162 \cdot 1264 \cdot 24$$

$$= 50079.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= ( fS + fY + fN )/P$$

$$= ( 201289 + 50080 + 11633 )/162$$

$$= 1593.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / ( 2 \cdot Ap/AT - Rxs/teff )$$

$$= 243 / ( 2 \cdot 1594/135 - 1264/126 )$$

$$= 179.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 162 [126/1264]$$

$$= 162.1 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min( Pmax1, Pmax2 )$$

$$= \min( 179, 162 )$$

$$= 162.058 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= ( fN + fS + fY ) / AT$$

$$= ( 11633 + 201289 + 50080 )/135$$

$$= 190.602 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 162 \cdot 1264/126$$

$$= 161.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ )$$

$$= \max( 2 \cdot 191 - 162, 162 )$$

$$= 219.6 \text{ N./mm}^2$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.55 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= ( Rnc + tn ) / Rnc$$

$$= ( 24 + 48.5 )/24$$

$$= 3.019 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot ( Rn + tn )$$

$$= \pi/2 \cdot ( 24 + 48.5 )$$

$$= 113.910 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 )$$

$$= \min( 50080 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 )$$



= 6815.039 Kgf

Weld Stress Value [T]:

=  $fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$

=  $6815 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.393 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K3A2"**

**From: 60**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 161.808   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 4149.97   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

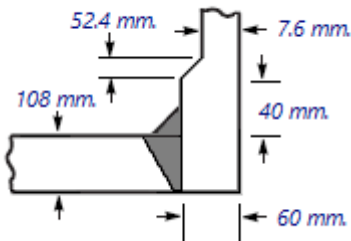
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 0.00       | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K3A2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.66 + 1.5 + 0 = 4.16 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 127.906 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

=  $8(t + t_e)$   
=  $8(126 + 0)$   
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

=  $\min[LH1, LH2, LH3] + t$  for set-in nozzles  
=  $\min[34.1, 400, 1012] + 126$   
= 160.630 mm.

Effective Vessel Thickness [teff]:

=  $t + ((A5 \cdot frp) / LR)$   
=  $126 + ((0 \cdot 1) / 48)$   
= 126.500 mm.

Parameter [ $\lambda$ ]:

=  $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$   
=  $\min(12, (2 \cdot 24 + 48.5) / (\sqrt{(2403 + 126) \cdot 126}))$   
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

=  $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$   
=  $126 \cdot 48 \cdot \max((0.17/5)^{0.85}, 1)$   
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

=  $tn \cdot LH$   
=  $48.5 \cdot 161$   
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

=  $0.5 \cdot Leg^2$   
=  $0.5 \cdot 10^2$   
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

=  $A1 + frn(A2 + A3) + A41 + A42 + A43$   
=  $60.8 + 0.95(77.9 + 0) + 0.5 + 0 + 0$   
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

=  $tn / \ln[1 + tn/Rn]$   
=  $48.5 / \ln[1 + 48.5/24]$   
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

=  $teff / \ln[1 + teff/Reff]$   
=  $126 / \ln[1 + 126/1202]$   
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

=  $1.5 \cdot S \cdot E$   
=  $1.5 \cdot 162 \cdot 1$   
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

=  $P \cdot Rxn \cdot LH$   
=  $162 \cdot 43.9 \cdot 161$   
= 11632.7 Kgf

Force acting on the Shell [fS]:



$$= P \cdot Rxs( LR + tn )$$

$$= 162 \cdot 1264 ( 48 + 48.5 )$$

$$= 201289.5 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 162 \cdot 1264 \cdot 24$$

$$= 50079.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= ( fS + fY + fN )/P$$

$$= ( 201289 + 50080 + 11633 )/162$$

$$= 1593.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / ( 2 \cdot Ap/AT - Rxs/teff )$$

$$= 243 / ( 2 \cdot 1594/135 - 1264/126 )$$

$$= 179.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 162 [126/1264]$$

$$= 162.1 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min( Pmax1, Pmax2 )$$

$$= \min( 179, 162 )$$

$$= 162.058 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= ( fN + fS + fY ) / AT$$

$$= ( 11633 + 201289 + 50080 )/135$$

$$= 190.602 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 162 \cdot 1264/126$$

$$= 161.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ )$$

$$= \max( 2 \cdot 191 - 162, 162 )$$

$$= 219.6 \text{ N./mm}^2$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.55 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= ( Rnc + tn ) / Rnc$$

$$= ( 24 + 48.5 )/24$$

$$= 3.019 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot ( Rn + tn )$$

$$= \pi/2 \cdot ( 24 + 48.5 )$$

$$= 113.910 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 )$$

$$= \min( 50080 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 )$$



= 6815.039 Kgf

Weld Stress Value [T]:

=  $fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$

=  $6815 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.393 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification      -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K3C2"**

**From: 60**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 161.836   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 3549.97   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

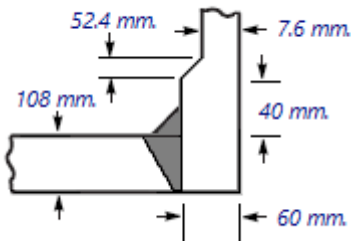
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 90.00      | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K3C2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.66 + 1.5 + 0 = 4.16 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 127.929 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

=  $8(t + t_e)$   
=  $8(126 + 0)$   
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

=  $\min[LH1, LH2, LH3] + t$  for set-in nozzles  
=  $\min[34.1, 400, 1012] + 126$   
= 160.630 mm.

Effective Vessel Thickness [teff]:

=  $t + ((A5 \cdot frp) / LR)$   
=  $126 + ((0 \cdot 1) / 48)$   
= 126.500 mm.

Parameter [ $\lambda$ ]:

=  $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$   
=  $\min(12, (2 \cdot 24 + 48.5) / (\sqrt{(2403 + 126) \cdot 126}))$   
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

=  $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$   
=  $126 \cdot 48 \cdot \max((0.17/5)^{0.85}, 1)$   
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

=  $tn \cdot LH$   
=  $48.5 \cdot 161$   
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

=  $0.5 \cdot Leg^2$   
=  $0.5 \cdot 10^2$   
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

=  $A1 + frn(A2 + A3) + A41 + A42 + A43$   
=  $60.8 + 0.95(77.9 + 0) + 0.5 + 0 + 0$   
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

=  $tn / \ln[1 + tn/Rn]$   
=  $48.5 / \ln[1 + 48.5/24]$   
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

=  $teff / \ln[1 + teff/Reff]$   
=  $126 / \ln[1 + 126/1202]$   
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

=  $1.5 \cdot S \cdot E$   
=  $1.5 \cdot 162 \cdot 1$   
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

=  $P \cdot Rxn \cdot LH$   
=  $162 \cdot 43.9 \cdot 161$   
= 11634.7 Kgf

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs( LR + tn ) \\ &= 162 \cdot 1264 ( 48 + 48.5 ) \\ &= 201324.4 \text{ Kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 162 \cdot 1264 \cdot 24 \\ &= 50088.2 \text{ Kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= ( fS + fY + fN )/P \\ &= ( 201324 + 50088 + 11635 )/162 \\ &= 1593.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / ( 2 \cdot Ap/AT - Rxs/teff ) \\ &= 243 / ( 2 \cdot 1594/135 - 1264/126 ) \\ &= 179.0 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 162 [126/1264 ] \\ &= 162.1 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min( Pmax1, Pmax2 ) \\ &= \min( 179, 162 ) \\ &= 162.058 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [ $\sigma$ Avg]:

$$\begin{aligned} &= ( fN + fS + fY ) / AT \\ &= ( 11635 + 201324 + 50088 )/135 \\ &= 190.635 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [ $\sigma$ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 162 \cdot 1264/126 \\ &= 161.7 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ ) \\ &= \max( 2 \cdot 191 - 162, 162 ) \\ &= 219.6 \text{ N./mm}^2 \end{aligned}$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.59 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= ( Rnc + tn ) / Rnc \\ &= ( 24 + 48.5 )/24 \\ &= 3.019 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot ( Rn + tn ) \\ &= \pi/2 \cdot ( 24 + 48.5 ) \\ &= 113.910 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 ) \\ &= \min( 50088 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 ) \end{aligned}$$



= 6816.220 Kgf

Weld Stress Value [T]:

=  $fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$

=  $6816 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.394 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K3E2"**

**From: 60**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 161.836   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 3549.97   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

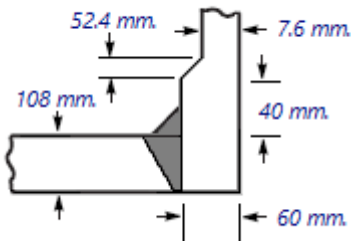
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 125.00     | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K3E2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.66 + 1.5 + 0 = 4.16 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 127.929 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$



= 34.130 mm.

Thickness Limit Candidate [LH2]:

= Lpr1  
= 400  
= 400.000 mm.

Thickness Limit Candidate [LH3]:

=  $8(t + t_e)$   
=  $8(126 + 0)$   
= 1012.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

=  $\min[LH1, LH2, LH3] + t$  for set-in nozzles  
=  $\min[34.1, 400, 1012] + 126$   
= 160.630 mm.

Effective Vessel Thickness [teff]:

=  $t + ((A5 \cdot frp) / LR)$   
=  $126 + ((0 \cdot 1) / 48)$   
= 126.500 mm.

Parameter [ $\lambda$ ]:

=  $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$   
=  $\min(12, (2 \cdot 24 + 48.5) / (\sqrt{(2403 + 126) \cdot 126}))$   
= 0.171

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

=  $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$   
=  $126 \cdot 48 \cdot \max((0.17/5)^{0.85}, 1)$   
= 60.763 cm<sup>2</sup>

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

=  $tn \cdot LH$   
=  $48.5 \cdot 161$   
= 77.905 cm<sup>2</sup>

Area Contributed by the Outside Fillet Weld [A41]:

=  $0.5 \cdot Leg_1^2$   
=  $0.5 \cdot 10^2$   
= 0.500 cm<sup>2</sup>

The total area contributed by A1 through A43 [AT]:

=  $A1 + frn(A2 + A3) + A41 + A42 + A43$   
=  $60.8 + 0.95(77.9 + 0) + 0.5 + 0 + 0$   
= 135.319 cm<sup>2</sup>

Nozzle Radius for Force Calculation [Rxn]:

=  $tn / \ln[1 + tn/Rn]$   
=  $48.5 / \ln[1 + 48.5/24]$   
= 43.889 mm.

Shell Radius for Force Calculation [Rxs]:

=  $teff / \ln[1 + teff/Reff]$   
=  $126 / \ln[1 + 126/1202]$   
= 1263.695 mm.

Allowable Local Primary Membrane Stress [Sallow]:

=  $1.5 \cdot S \cdot E$   
=  $1.5 \cdot 162 \cdot 1$   
= 242.8 N./mm<sup>2</sup>

Force acting on the Nozzle [fN]:

=  $P \cdot Rxn \cdot LH$   
=  $162 \cdot 43.9 \cdot 161$   
= 11634.7 Kgf

Force acting on the Shell [fS]:



$$\begin{aligned}
 &= P \cdot Rxs( LR + tn ) \\
 &= 162 \cdot 1264 ( 48 + 48.5 ) \\
 &= 201324.4 \text{ Kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 162 \cdot 1264 \cdot 24 \\
 &= 50088.2 \text{ Kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= ( fS + fY + fN )/P \\
 &= ( 201324 + 50088 + 11635 )/162 \\
 &= 1593.9 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / ( 2 \cdot Ap/AT - Rxs/teff ) \\
 &= 243 / ( 2 \cdot 1594/135 - 1264/126 ) \\
 &= 179.0 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 162 [126/1264 ] \\
 &= 162.1 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min( Pmax1, Pmax2 ) \\
 &= \min( 179, 162 ) \\
 &= 162.058 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= ( fN + fS + fY ) / AT \\
 &= ( 11635 + 201324 + 50088 )/135 \\
 &= 190.635 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 162 \cdot 1264/126 \\
 &= 161.7 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ ) \\
 &= \max( 2 \cdot 191 - 162, 162 ) \\
 &= 219.6 \text{ N./mm}^2
 \end{aligned}$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.59 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= ( Rnc + tn ) / Rnc \\
 &= ( 24 + 48.5 )/24 \\
 &= 3.019 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot ( Rn + tn ) \\
 &= \pi/2 \cdot ( 24 + 48.5 ) \\
 &= 113.910 \text{ mm.}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 ) \\
 &= \min( 50088 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 )
 \end{aligned}$$



= 6816.220 Kgf

Weld Stress Value [T]:

=  $fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$

=  $6816 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.394 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification      -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: K2A2"**

**From: 60**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 161.808   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 4149.97   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

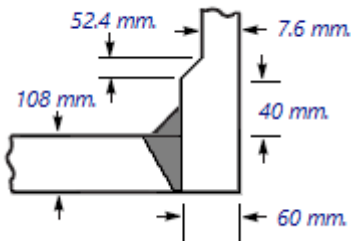
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +N+IT              |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.90     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 270.00     | deg                |
| Diameter                                        |      | 2.0000     | in.                |
| Size and Thickness Basis                        |      | Minimum    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +N+IT              |
| Hub Height of Integral Nozzle                   | h    | 40.0000    | mm.                |
| Height of Beveled Transition                    | L'   | 42.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K2A2"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 1.773 | in. |
| Actual Thickness Used in Calculation       | 0.301 | in. |



Nozzle input data check completed without errors.

### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 48 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 2.66 + 1.5 + 0 = 4.16 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:  

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 6.15 + 48) / 48) - 0$$

$$= 351 - 0 = 351 \text{ bars}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm  
 Min. required shell thickness, internal p + c : 127.906 mm.

Nozzle neck minimum required thickness : 4.920 mm.

Minimum nozzle neck thickness 7.645 mm., is greater than required thickness 4.920 mm.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:  

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:  

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.65 - 1.5$$

$$= 6.145 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:  

$$= D_i/t$$

$$= 2403/126$$

$$= 18.996 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:  

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2400/2 + 1.5$$

$$= 1201.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:  

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1202 \cdot 126}, 2 \cdot 24)$$

$$= 48.034 \text{ mm.}$$

Thickness Limit Candidate [LH1]:  

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 126, 0] + \sqrt{24 \cdot 48.5}$$

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs( LR + tn ) \\ &= 162 \cdot 1264 ( 48 + 48.5 ) \\ &= 201289.5 \text{ Kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 162 \cdot 1264 \cdot 24 \\ &= 50079.6 \text{ Kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= ( fS + fY + fN )/P \\ &= ( 201289 + 50080 + 11633 )/162 \\ &= 1593.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / ( 2 \cdot Ap/AT - Rxs/teff ) \\ &= 243 / ( 2 \cdot 1594/135 - 1264/126 ) \\ &= 179.0 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 162 [126/1264 ] \\ &= 162.1 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min( Pmax1, Pmax2 ) \\ &= \min( 179, 162 ) \\ &= 162.058 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= ( fN + fS + fY ) / AT \\ &= ( 11633 + 201289 + 50080 )/135 \\ &= 190.602 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 162 \cdot 1264/126 \\ &= 161.7 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max( 2 \cdot \sigma Avg - \sigma Circ, \sigma Circ ) \\ &= \max( 2 \cdot 191 - 162, 162 ) \\ &= 219.6 \text{ N./mm}^2 \end{aligned}$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 219.55 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= ( Rnc + tn ) / Rnc \\ &= ( 24 + 48.5 )/24 \\ &= 3.019 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot ( Rn + tn ) \\ &= \pi/2 \cdot ( 24 + 48.5 ) \\ &= 113.910 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min( 0.7 \cdot 48.500, 6.000 ) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min( fy \cdot ky, 1.5 \cdot Sn( A2 + A3 ), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2 ) \\ &= \min( 50080 \cdot 3.02, 1.5 \cdot 154( 77.9 + 0 ), \pi/4 \cdot 162 \cdot 24^2 \cdot 3.02^2 ) \end{aligned}$$



= 6815.039 Kgf

Weld Stress Value [T]:

=  $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$

=  $6815 / (114 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0))$

= 7.393 < or = to 161.900 Weld Stress Passed

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 2.1932 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1932 mm.

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**Input, Nozzle Desc: MH124"**

**From: 70**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 161.808   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 5900.00   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

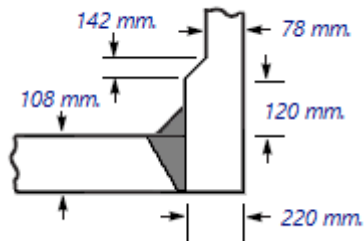
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +IT                |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.86     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 0.00       | deg                |
| Diameter                                        |      | 24.0000    | in.                |
| Size and Thickness Basis                        |      | Actual     |                    |
| Actual Thickness                                | tn   | 78.0000    | mm.                |
| Flange Material                                 |      | SA-350 LF2 | +IT                |
| Hub Height of Integral Nozzle                   | h    | 100.0000   | mm.                |
| Height of Beveled Transition                    | L'   | 142.0000   | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 220.0000   | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 400.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 128.0000   | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |
| This is a Manway, Coupling or Access Opening.   |      |            |                    |
| Flange Class                                    |      | 1500       |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: MH124"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |        |     |
|--------------------------------------------|--------|-----|
| Actual Inside Diameter Used in Calculation | 24.000 | in. |
| Actual Thickness Used in Calculation       | 3.071  | in. |



**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

**Cylindrical Shell Calculation - Section 4.3.3.1**

Computed Minimum Required Thickness [t]:  
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$   
 $= 0.5 \cdot 613 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$   
 $= 34 + 1.5 + 0 = 35.5 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:  
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$   
 $= 154 \cdot 1 \cdot \ln((2 \cdot 76.5 + 613) / 613) - 0$   
 $= 343 - 0 = 343 \text{ bars}$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

**Nozzle Calculations per Section 4.5: Internal Pressure Case:**

Nozzle Material Factor [frn]:  
 $= \min[S_n/S, 1]$   
 $= \min[154/162, 1]$   
 $= 0.950$

Thickness of Nozzle at Shell [tn]:  
 $= \text{hub thickness} - \text{corrosion allowance}$   
 $= 220 - 1.5$   
 $= 218.500 \text{ mm.}$

Thickness of Nozzle at Top [tn2]:  
 $= \text{thickness} - \text{corrosion allowance}$   
 $= 78 - 1.5$   
 $= 76.500 \text{ mm.}$

Shell Diameter to Thickness ratio [D/t]:  
 $= D_i/t$   
 $= 2403/126$   
 $= 18.996 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:  
 $= D_i/2 + \text{corrosion allowance}$   
 $= 2400/2 + 1.5$   
 $= 1201.500 \text{ mm.}$

Effective Length of Vessel Wall [LR]:  
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$   
 $= \min(\sqrt{1202 \cdot 126}, 2 \cdot 306)$   
 $= 389.859 \text{ mm.}$

Thickness Limit Candidate [LH1]:  
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$   
 $= \min[1.5 \cdot 126, 0] + \sqrt{306 \cdot 218}$   
 $= 258.702 \text{ mm.}$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 400 \\ &= 400.000 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(126 + 0) \\ &= 1012.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[259, 400, 1012] + 126 \\ &= 385.202 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 126 + ((0 \cdot 1) / 390) \\ &= 126.500 \text{ mm.} \end{aligned}$$

Parameter [ $\lambda$ ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right) \\ &= \min\left(12, \left(2 \cdot 306 + 218\right) / \left(\sqrt{\left(2403 + 126\right) \cdot 126}\right)\right) \\ &= 1.469 \end{aligned}$$

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 126 \cdot 390 \cdot \max((1.47/5)^{0.85}, 1) \\ &= 493.171 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2C \\ &= 495 + 91.3 \\ &= 586.242 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 493 + 0.95(586 + 0) + 0.5 + 0 + 0 \\ &= 1050.784 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 218 / \ln[1 + 218/306] \\ &= 405.793 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 126 / \ln[1 + 126/1202] \\ &= 1263.695 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 162 \cdot 1 \\ &= 242.8 \text{ N/mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 162 \cdot 406 \cdot 385 \\ &= 257923.0 \text{ Kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$



$$= 162 \cdot 1264 (390 + 218)$$

$$= 1268526.5 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 162 \cdot 1264 \cdot 306$$

$$= 638685.2 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (1268526 + 638685 + 257923) / 162$$

$$= 13121.6 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 243 / (2 \cdot 13122 / 1051 - 1264 / 126)$$

$$= 162.1 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 162 [126/1264]$$

$$= 162.1 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(162, 162)$$

$$= 162.050 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (257923 + 1268526 + 638685) / 1051$$

$$= 202.069 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 162 \cdot 1264 / 126$$

$$= 161.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 202 - 162, 162)$$

$$= 242.5 \text{ N./mm}^2$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 242.49 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.05 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (306 + 218) / 306$$

$$= 1.713 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (306 + 218)$$

$$= 824.354 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 218.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(638685 \cdot 1.71, 1.5 \cdot 154(586+0), \pi / 4 \cdot 162 \cdot 306^2 \cdot 1.71^2)$$

$$= 356923.531 \text{ Kgf}$$



Weld Stress Value [ $\tau$ ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw1 + 0.49 \cdot L_{43T})) \\ &= 356924 / (824 (0.49 \cdot 7.07 + 0.6 \cdot 126 + 0.49 \cdot 0)) \\ &= 53.501 < \text{or} = \text{to } 161.900 \text{ Weld Stress Passed} \end{aligned}$$

**Nozzle/Structural Discontinuity Proximity Warning:**

*This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld (Impact tested) :**

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification      -46 °C

**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification      -46 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction      -66 °C

The Drop for this Nozzle is : 120.8406 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 648.8406 mm.

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**Input, Nozzle Desc: N118"**

**From: 70**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 161.808   | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 | +IT                |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 2400.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 128.0000  | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 5249.98   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

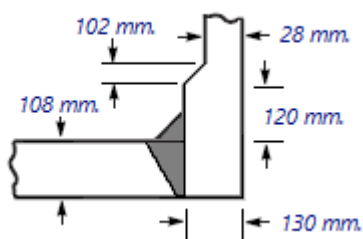
|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +IT                |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 153.86     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 180.00     | deg                |
| Diameter                                        |      | 18.0000    | in.                |
| Size and Thickness Basis                        |      | Actual     |                    |
| Actual Thickness                                | tn   | 86.0000    | mm.                |
| Hub Height of Integral Nozzle                   | h    | 100.0000   | mm.                |
| Height of Beveled Transition                    | L'   | 44.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 130.0000   | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Outside Projection                              | ho   | 350.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 40.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 40.0000    | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.               |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N118"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |        |     |
|--------------------------------------------|--------|-----|
| Actual Inside Diameter Used in Calculation | 18.000 | in. |
| Actual Thickness Used in Calculation       | 3.386  | in. |



Nozzle input data check completed without errors.



### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 460 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0 \\ &= 25.5 + 1.5 + 0 = 27 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 154 \cdot 1 \cdot \ln((2 \cdot 84.5 + 460) / 460) - 0 \\ &= 481 - 0 = 481 \text{ bars} \end{aligned}$$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm  
Min. required shell thickness, internal p + c : 127.906 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 86.004 mm., is greater than required thickness 9.840 mm.

### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |                            |        |
|------------|---------|-----------|----------------------------|--------|
| Sustained  | : 22.6, | Allowable | : 153.9 N./mm <sup>2</sup> | Passed |
| Expansion  | : 0.0,  | Allowable | : 376.0 N./mm <sup>2</sup> | Passed |
| Occasional | : 18.6, | Allowable | : 204.6 N./mm <sup>2</sup> | Passed |
| Shear      | : 2.3,  | Allowable | : 107.7 N./mm <sup>2</sup> | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[154/162, 1] \\ &= 0.950 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 130 - 1.5 \\ &= 128.500 \text{ mm.} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 86 - 1.5 \\ &= 84.500 \text{ mm.} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 2403/126 \\ &= 18.996 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 2400/2 + 1.5 \\ &= 1201.500 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1202 \cdot 126}, 2 \cdot 230) \\ &= 389.859 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 126, 0] + \sqrt{(230 \cdot 128)}$$
$$= 171.953 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 350$$
$$= 350.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(126 + 0)$$
$$= 1012.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$
$$= \min[172, 350, 1012] + 126$$
$$= 298.453 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 126 + ((0 \cdot 1) / 390)$$
$$= 126.500 \text{ mm.}$$

Parameter [ $\lambda$ ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 230 + 128) / (\sqrt{(2403 + 126) \cdot 126}))$$
$$= 1.041$$

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 126 \cdot 390 \cdot \max((1.04/5)^{0.85}, 1)$$
$$= 493.171 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$
$$= 291 + 70.5$$
$$= 361.533 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 40^2$$
$$= 8.000 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 493 + 0.95(362 + 0) + 8 + 0 + 0$$
$$= 844.740 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 128 / \ln[1 + 128/230]$$
$$= 289.614 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 126 / \ln[1 + 126/1202]$$
$$= 1263.695 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 162 \cdot 1$$
$$= 242.8 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 162 \cdot 290 \cdot 298$$
$$= 142624.3 \text{ Kg}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 162 \cdot 1264 (390 + 128) \\ &= 1080861.9 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 162 \cdot 1264 \cdot 230 \\ &= 479795.9 \text{ Kg} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (1080861 + 479796 + 142624) / 162 \\ &= 10322.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 243 / (2 \cdot 10323 / 845 - 1264 / 126) \\ &= 168.1 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 162 [126 / 1264] \\ &= 162.1 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(168, 162) \\ &= 162.058 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (142624 + 1080861 + 479796) / 845 \\ &= 197.739 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 162 \cdot 1264 / 126 \\ &= 161.7 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 198 - 162, 162) \\ &= 233.8 \text{ N./mm}^2 \end{aligned}$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 233.83 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.06 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (230 + 128) / 230 \\ &= 1.558 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (230 + 128) \\ &= 563.288 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 28.284, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 28.284, \text{ must be } \geq \min(0.7 \cdot 128.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(479796 \cdot 1.56, 1.5 \cdot 154(362+0), \pi/4 \cdot 162 \cdot 230^2 \cdot 1.56^2)$$

$$= 166651.094 \text{ Kgf}$$

Weld Stress Value [ $\tau$ ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 166651 / (563 (0.49 \cdot 28.3 + 0.6 \cdot 40 + 0.49 \cdot 0))$$

$$= 76.636 < \text{or} = \text{to } 161.900 \text{ Weld Stress Passed}$$

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 54.8336 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 532.8336 mm.

#### Input Echo, WRC297 Item 1, Description: N118" :

|                                      |            |        |                    |
|--------------------------------------|------------|--------|--------------------|
| Diameter Basis for Cylindrical Shell | ID         |        |                    |
| Shell Internal Corrosion Allowance   | 1.5000     | mm.    |                    |
| Shell External Corrosion Allowance   | 0.0000     | mm.    |                    |
| Shell Diameter                       | 2400.000   | mm.    |                    |
| Shell Thickness                      | 128.0000   | mm.    |                    |
| Shell Stress Concentration Factor    | 1.000      |        |                    |
| Vessel Material                      | SA-516 70  |        |                    |
| Vessel Cold S.I. Allowable           | Smc        | 175.00 | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh        | 161.90 | N./mm <sup>2</sup> |
| Diameter Basis for Nozzle            | ID         |        |                    |
| Nozzle Corrosion Allowance           | 1.5000     | mm.    |                    |
| Nozzle Diameter                      | 457.200    | mm.    |                    |
| Nozzle Thickness                     | 130.0000   | mm.    |                    |
| Nozzle Stress Concentration Factor   | 1.000      |        |                    |
| Nozzle Material                      | SA-350 LF2 |        |                    |
| Nozzle Cold S.I. Allowable           | SNmc       | 165.00 | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable            | SNmh       | 153.86 | N./mm <sup>2</sup> |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:*

*These loads are assumed to be SUS tained loads.*

|                          |    |          |       |
|--------------------------|----|----------|-------|
| Design Internal Pressure | Dp | 161.81   | bars  |
| Radial Load              | P  | 15398.72 | Kgf   |
| Circumferential Shear    | Vc | 2294.51  | Kgf   |
| Longitudinal Shear       | Vl | 2294.51  | Kgf   |
| Circumferential Moment   | Mc | 3088.76  | Kg-m. |
| Longitudinal Moment      | Ml | 4128.53  | Kg-m. |
| Torsional Moment         | Mt | 5158.12  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tn (5.517) is < 10  
 check the Limitations of WRC bulletin 297.*

#### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 717.2 + 2 \cdot 1.65 \cdot \sqrt{1264.75(128.0 - 1.5)}$$

$$= 2037.162 \text{ mm.}$$



WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2529.500 mm.  
 Nozzle Outside Diameter (d) = 717.200 mm.  
 Vessel Thickness used (T) = 126.500 mm.  
 Nozzle Thickness used (t) = 128.500 mm.  
 T / t = 0.984  
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.268$$

Nr/P = 0.112  
 Mr/P = 0.162  
 M0/P = 0.055  
 N0/P = 0.113  
 MrD/Mc = 0.272  
 NrDL/Mc = 0.144  
 M0d/Mc = 0.102  
 N0DL/Mc = 0.045  
 MrD/M1 = 0.169  
 NrDL/M1 = 0.100  
 M0D/M1 = 0.062  
 N0DL/M1 = 0.131

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -1                   | -1                  | -1                      | -1                     |
| Outplane Bending (P )                                            | -3                   | 3                   | -3                      | 3                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (ML)                                            | -1                   | 1                   | 1                       | -1                     |
| Normal Pressure Stress                                           | 146                  | 162                 | 146                     | 162                    |
| Outplane Stress Summary                                          | 140                  | 165                 | 143                     | 163                    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | -1                   | -1                  | -1                      | -1                     |
| Inplane Bending (P )                                               | -9                   | 9                   | -9                      | 9                      |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (ML)                                               | -3                   | 3                   | 3                       | -3                     |
| Inplane Pressure Stress                                            | 73                   | 73                  | 73                      | 73                     |
| Inplane Stress Summary                                             | 58                   | 84                  | 66                      | 77                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 0                    | 0                   | 0                       | 0                      |
| Shear Stress Summary                                                 | 0                    | 0                   | 0                       | 0                      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|



Two \* Max Shear Stress 140 165 143 163

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | -1                    | -1                   | -1                     | -1                    |
| Outplane Bending (P )                                                  | -3                    | 3                    | -3                     | 3                     |
| Outplane Membrane (Mc)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (Mc)                                                  | -1                    | 1                    | 1                      | -1                    |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 73                    | 73                   | 73                     | 73                    |
| Outplane Stress Summary                                                | 67                    | 76                   | 70                     | 73                    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | -1                    | -1                   | -1                     | -1                    |
| Inplane Bending (P )                                                     | -9                    | 9                    | -9                     | 9                     |
| Inplane Membrane (Mc)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (Mc)                                                     | -4                    | 4                    | 4                      | -4                    |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 146                   | 162                  | 146                    | 162                   |
| Inplane Stress Summary                                                   | 131                   | 174                  | 140                    | 166                   |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | 0                     | 0                    | 0                      | 0                     |
| Torsional Shear (Mt)                                                       | 0                     | 0                    | 0                      | 0                     |
| Shear Stress Summary                                                       | 0                     | 0                    | 0                      | 0                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 131                   | 174                  | 140                    | 166                   |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | -1                   | -1                  | -1                      | -1                     |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 27                   | 46                  | 27                      | 46                     |
| Hoop Stress Summary                                       | 25                   | 45                  | 26                      | 45                     |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|



|                       |    |    |    |    |
|-----------------------|----|----|----|----|
| Axial Membrane (P )   | 0  | 0  | 0  | 0  |
| Axial Bending (P )    | -5 | 5  | -5 | 5  |
| Axial Membrane (Mc)   | 0  | 0  | 0  | 0  |
| Axial Bending (Mc)    | 0  | 0  | 0  | 0  |
| Axial Membrane (ML)   | -1 | -1 | 1  | 1  |
| Axial Bending (ML)    | -2 | 2  | 2  | -2 |
| Axial Pressure Stress | 13 | 13 | 13 | 13 |

|                      |   |    |    |    |
|----------------------|---|----|----|----|
| Axial Stress Summary | 3 | 19 | 10 | 17 |
|----------------------|---|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                      |               |              |               |              |
|--------------------------------------|---------------|--------------|---------------|--------------|
| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 0             | 0            | 0             | 0            |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 0 | 0 | 0 | 0 |
|----------------------|---|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                            |               |              |               |              |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Two * Max Shear Stress                     | 25            | 45           | 26            | 45           |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                                              |                       |                      |                        |                       |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Hoop Membrane (P )                                           | -1                    | -1                   | -1                     | -1                    |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 27                    | 46                   | 27                     | 46                    |

|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 26 | 45 | 26 | 45 |
|---------------------|----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                                                     |                       |                      |                        |                       |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | -5                    | 5                    | -5                     | 5                     |
| Axial Membrane (Mc)                                                 | -1                    | -1                   | 1                      | 1                     |
| Axial Bending (Mc)                                                  | -3                    | 3                    | 3                      | -3                    |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 13                    | 13                   | 13                     | 13                    |

|                      |   |    |    |    |
|----------------------|---|----|----|----|
| Axial Stress Summary | 3 | 20 | 11 | 16 |
|----------------------|---|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |               |              |               |              |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 0             | 0            | 0             | 0            |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 0 | 0 | 0 | 0 |
|----------------------|---|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                               |               |              |               |              |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|

Two \* Max Shear Stress 26 45 26 45

### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 146              | 162   | 146   | 162   | 146   | 162   | 146   | 162   |
| Circ. Pl (SUS)               | -1               | -1    | -1    | -1    | -1    | -1    | -1    | -1    |
| Circ. Q (SUS)                | -4               | 4     | -2    | 2     | -13   | 13    | -5    | 5     |
| Long. Pm (SUS)               | 73               | 73    | 73    | 73    | 73    | 73    | 73    | 73    |
| Long. Pl (SUS)               | -1               | -1    | -1    | -1    | -1    | -1    | -1    | -1    |
| Long. Q (SUS)                | -12              | 12    | -6    | 6     | -4    | 4     | -2    | 2     |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Pm (SUS)                     | 146.0            | 162.0 | 146.0 | 162.0 | 146.0 | 162.0 | 146.0 | 162.0 |
| Pm+Pl (SUS)                  | 145.0            | 161.0 | 145.0 | 161.0 | 145.0 | 161.0 | 145.0 | 161.0 |
| Pm+Pl+Q (Total)              | 141.0            | 165.0 | 143.0 | 163.0 | 132.0 | 174.0 | 140.0 | 166.0 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 162.00    | 161.90         | Failed |
| Pm+Pl (SUS)         | 161.00    | 242.85         | Passed |
| Pm+Pl+Q (TOTAL)     | 174.00    | 485.70         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 27               | 46   | 27   | 46   | 27   | 46   | 27   | 46   |
| Circ. Pl (SUS)               | -1               | -1   | -1   | -1   | -1   | -1   | -1   | -1   |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 13               | 13   | 13   | 13   | 13   | 13   | 13   | 13   |
| Long. Pl (SUS)               | -1               | -1   | 1    | 1    | -1   | -1   | 1    | 1    |
| Long. Q (SUS)                | -7               | 7    | -3   | 3    | -8   | 8    | -2   | 2    |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Pm (SUS)                     | 27.0             | 46.0 | 27.0 | 46.0 | 27.0 | 46.0 | 27.0 | 46.0 |
| Pm+Pl (SUS)                  | 26.0             | 45.0 | 26.0 | 45.0 | 26.0 | 45.0 | 26.0 | 45.0 |
| Pm+Pl+Q (Total)              | 26.0             | 45.0 | 26.0 | 45.0 | 26.0 | 45.0 | 26.0 | 45.0 |

#### Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of | Max. S.I. | S.I. Allowable | Result |
|---------|-----------|----------------|--------|
|---------|-----------|----------------|--------|



| Stress Int.     |       |        |        |
|-----------------|-------|--------|--------|
| Pm (SUS)        | 46.00 | 153.86 | Passed |
| Pm+Pl (SUS)     | 45.00 | 230.78 | Passed |
| Pm+Pl+Q (TOTAL) | 45.00 | 461.57 | Passed |

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.*

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**Input, Nozzle Desc: N216"**

**From: 80**

|                                               |      |           |        |
|-----------------------------------------------|------|-----------|--------|
| Pressure for Reinforcement Calculations       | P    | 161.808   | bars   |
| Temperature for Internal Pressure             | Temp | 80        | °C     |
| Parent Material                               |      | SA-516 70 | +IT    |
| Parent Allowable Stress at Temperature        | Sv   | 161.90    | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 175.00    | N./mm2 |
| Inside Diameter of Hemispherical Head         | D    | 2464.00   | mm.    |
| Head Finished (Minimum) Thickness             | t    | 65.0000   | mm.    |
| Head Internal Corrosion Allowance             | cas  | 1.5000    | mm.    |
| Head External Corrosion Allowance             | co   | 0.0000    | mm.    |
| Distance from Head Centerline                 | L1   | 0.0000    | mm.    |
| User Entered Minimum Design Metal Temperature |      | -33.00    | °C     |

**Type of Element Connected to the Parent : Nozzle**

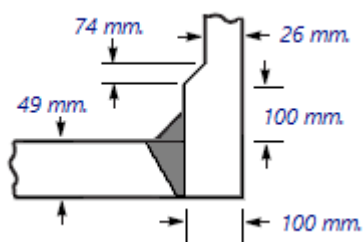
|                                                 |      |            |        |
|-------------------------------------------------|------|------------|--------|
| Material                                        |      | SA-350 LF2 | +IT    |
| Material UNS Number                             |      | K03011     |        |
| Material Specification/Type                     |      | Forgings   |        |
| Allowable Stress at Temperature                 | Sn   | 153.86     | N./mm2 |
| Allowable Stress At Ambient                     | Sna  | 165.00     | N./mm2 |
| Diameter Basis (for tr calc only)               |      | Inside     |        |
| Layout Angle                                    |      | 0.00       | deg    |
| Diameter                                        |      | 16.0000    | in.    |
| Size and Thickness Basis                        |      | Actual     |        |
| Actual Thickness                                | tn   | 30.0000    | mm.    |
| Hub Height of Integral Nozzle                   | h    | 95.0000    | mm.    |
| Height of Beveled Transition                    | L'   | 56.0000    | mm.    |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 86.0000    | mm.    |
| Corrosion Allowance                             | can  | 1.5000     | mm.    |
| Outside Projection                              | ho   | 300.0000   | mm.    |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 38.0000    | mm.    |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 38.0000    | mm.    |
| Inside Projection                               | h    | 0.0000     | mm.    |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.    |
| User Defined Nozzle/Shell Centerline Angle      |      | 90.0000    | deg.   |

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N216"**

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

|                                            |        |     |
|--------------------------------------------|--------|-----|
| Actual Inside Diameter Used in Calculation | 16.000 | in. |
| Actual Thickness Used in Calculation       | 1.181  | in. |



Nozzle input data check completed without errors.



### Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:  
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$   
 $= 0.5 \cdot 409 \cdot (\exp(162/(154 \cdot 1)) - 1) + 1.5 + 0$   
 $= 22.7 + 1.5 + 0 = 24.2 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:  
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$   
 $= 154 \cdot 1 \cdot \ln((2 \cdot 28.5 + 409) / 409) - 0$   
 $= 201 - 0 = 201 \text{ bars}$

### Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm  
Min. required shell thickness, internal p + c : 64.723 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 29.997 mm., is greater than required thickness 9.840 mm.

### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 67.3, | Allowable | : | 153.9 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 331.2 N./mm <sup>2</sup> | Passed |
| Occasional | : | 54.3, | Allowable | : | 204.6 N./mm <sup>2</sup> | Passed |
| Shear      | : | 4.7,  | Allowable | : | 107.7 N./mm <sup>2</sup> | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

### Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:  
 $= \min[S_n/S, 1]$   
 $= \min[154/162, 1]$   
 $= 0.950$

Thickness of Nozzle at Shell [tn]:  
 $= \text{hub thickness} - \text{corrosion allowance}$   
 $= 86 - 1.5$   
 $= 84.500 \text{ mm.}$

Thickness of Nozzle at Top [tn2]:  
 $= \text{thickness} - \text{corrosion allowance}$   
 $= 30 - 1.5$   
 $= 28.500 \text{ mm.}$

Effective Pressure Radius [Reff]:  
 $= D_i/2 + \text{corrosion allowance}$   
 $= 2464/2 + 1.5$   
 $= 1233.500 \text{ mm.}$

Effective Length of Vessel Wall [LR]:  
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$   
 $= \min(\sqrt{1234 \cdot 63.5}, 2 \cdot 205)$   
 $= 279.870 \text{ mm.}$

Intermediate value [Cn]:  
 $= \min(((t + t_e)/t_n)^{0.35}, 1)$   
 $= \min(((63.5 + 0)/84.5)^{0.35}, 1)$   
 $= \min(0.9, 1)$   
 $= 0.905$

Intermediate value [Cp]:  
 $= 9999999.000$

Effective Nozzle Wall Length Outside the Vessel [LH]:



$$\begin{aligned} &= \min( t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t ) \\ &= \min(63.5 + 0 + 0.9 \cdot \sqrt{(205 \cdot 84.5)}, 300 + 63.5) \\ &= 182.503 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot f_{rp}) / LR) \\ &= 63.5 + ((0 \cdot 1) / 280) \\ &= 63.500 \text{ mm.} \end{aligned}$$

**Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :**

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 63.5 \cdot 280 \\ &= 177.717 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 134 + 17.4 \\ &= 151.334 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 38^2 \\ &= 7.220 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 \\ &= 178 + 0.95(151 + 0) + 7.22 + 0 + 0 \\ &= 328.752 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 84.5 / \ln[1 + 84.5/205] \\ &= 244.521 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 63.5 / \ln[1 + 63.5/1234] \\ &= 1264.983 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 162 \cdot 1 \\ &= 242.8 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 162 \cdot 245 \cdot 183 \\ &= 73635.1 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (LR + t_n) / 2 \\ &= 162 \cdot 1265 \cdot (280 + 84.5) / 2 \\ &= 380272.7 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (162 \cdot 1265 \cdot 205) / 2 \\ &= 213634.0 \text{ Kg} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (380273 + 213634 + 73635) / 162 \\ &= 4045.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff})) \\ &= 243 / (2 \cdot 4046 / 329 - 1265 / (2 \cdot 63.5)) \\ &= 165.7 \text{ bars} \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S[t/Rxs] \\ &= 2 \cdot 162 [63.5/1265] \\ &= 162.5 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(166, 163) \\ &= 162.533 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [ $\sigma_{Avg}$ ]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (73635 + 380273 + 213634) / 329 \\ &= 199.131 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [ $\sigma_{Circ}$ ]:

$$\begin{aligned} &= P \cdot Rxs / (2 \cdot teff) \\ &= 162 \cdot 1265 / (2 \cdot 63.5) \\ &= 161.2 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 199 - 161, 161) \\ &= 237.1 \text{ N./mm}^2 \end{aligned}$$

#### Summary of Nozzle Pressure/Stress Results:

|                                       |        |        |        |
|---------------------------------------|--------|--------|--------|
| Allowed Local Primary Membrane Stress | Sallow | 242.85 | N./mm2 |
| Local Primary Membrane Stress         | PL     | 237.08 | N./mm2 |
| Maximum Allowable Working Pressure    | Pmax   | 162.53 | bars   |

#### Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (205 + 84.5) / 205 \\ &= 1.413 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (205 + 84.5) \\ &= 454.274 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 26.870, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 26.870, \text{ must be } \geq \min(0.7 \cdot 84.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(213634 \cdot 1.41, 1.5 \cdot 154(158+0), \pi/4 \cdot 162 \cdot 205^2 \cdot 1.41^2) \\ &= 108388.734 \text{ Kg} \end{aligned}$$

Weld Stress Value [ $\tau$ ]:

$$\begin{aligned} &= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 108389 / (454(0.49 \cdot 26.9 + 0.6 \cdot 38 + 0.49 \cdot 0)) \\ &= 65.058 < \text{ or } = \text{ to } 161.900 \text{ Weld Stress Passed} \end{aligned}$$

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 34.4244 mm.



The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.4243 mm.

**Input Echo, WRC107/537 Item 1, Description: N216 :**

|                                 |        |           |        |
|---------------------------------|--------|-----------|--------|
| Diameter Basis for Vessel       | Vbasis | ID        |        |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |        |
| Internal Corrosion Allowance    | Cas    | 1.5000    | mm.    |
| Vessel Diameter                 | Dv     | 2464.000  | mm.    |
| Vessel Thickness                | Tv     | 65.000    | mm.    |
| Design Temperature              | T1     | 80.0      | C      |
| Vessel Material                 |        | SA-516 70 |        |
| Vessel UNS Number               |        | K02700    |        |
| Vessel Cold S.I. Allowable      | Smc    | 175.00    | N./mm2 |
| Vessel Hot S.I. Allowable       | Smh    | 161.90    | N./mm2 |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                  |        |            |        |
|----------------------------------|--------|------------|--------|
| Attachment Type                  | Type   | Round      |        |
| WRC107 Attachment Classification | Holsol | Hollow     |        |
| Diameter Basis for Nozzle        | Nbasis | ID         |        |
| Corrosion Allowance for Nozzle   | Can    | 1.5000     | mm.    |
| Nozzle Diameter                  | Dn     | 406.400    | mm.    |
| Nozzle Thickness                 | Tn     | 86.000     | mm.    |
| Nozzle Material                  |        | SA-350 LF2 |        |
| Nozzle UNS Number                |        | K03011     |        |
| Nozzle Cold S.I. Allowable       | SNmc   | 165.00     | N./mm2 |
| Nozzle Hot S.I. Allowable        | SNmh   | 153.86     | N./mm2 |
| Design Internal Pressure         | Dp     | 161.808    | bars   |
| Include Pressure Thrust          |        | No         |        |

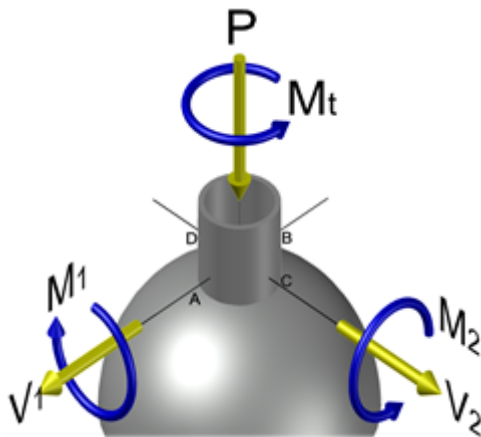
**External Forces and Moments in WRC 107/537 Convention:**

|                              |         |        |       |
|------------------------------|---------|--------|-------|
| Radial Load (SUS)            | P       | 1376.7 | Kgf   |
| Longitudinal Shear (SUS)     | (V1) V1 | 2039.6 | Kgf   |
| Circumferential Shear (SUS)  | (Vc) V2 | 2039.6 | Kgf   |
| Circumferential Moment (SUS) | (Mc) M1 | 4077.6 | Kg-m. |
| Longitudinal Moment (SUS)    | (Ml) M2 | 3262.1 | Kg-m. |
| Torsional Moment (SUS)       | Mt      | 2446.5 | Kg-m. |

|                         |                    |
|-------------------------|--------------------|
| Use Interactive Control | No                 |
| WRC107 Version          | Version March 1979 |

|                                                 |    |
|-------------------------------------------------|----|
| Include Pressure Stress Indices per Div. 2      | No |
| Compute Pressure Stress per WRC-368             | No |
| Local Loads applied at end of Nozzle/Attachment | No |

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$
  

$$= 578.4 + 2 \cdot 1.65 \cdot \sqrt{(1265.25 (65.0 - 1.5))}$$
  

$$= 1513.782 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTAINED loads:

|                        |         |        |       |
|------------------------|---------|--------|-------|
| Radial Load            | P       | 1376.7 | Kgf   |
| Circumferential Shear  | (VC) V2 | 2039.6 | Kgf   |
| Longitudinal Shear     | (VL) V1 | 2039.6 | Kgf   |
| Circumferential Moment | (MC) M1 | 4077.6 | Kg-m. |
| Longitudinal Moment    | (ML) M2 | 3262.1 | Kg-m. |
| Torsional Moment       | MT      | 2446.5 | Kg-m. |

Dimensionless Param: U = 1.02 TAU = 5.00 ( 2.92) RHO = 0.75

#### Dimensionless Loads for Spherical Shells at Attachment Junction:

| Curves read for 1979                   | Figure | Value   | Location  |
|----------------------------------------|--------|---------|-----------|
| $N(x) * T / P$                         | SP 1   | 0.03975 | (A,B,C,D) |
| $M(x) / P$                             | SP 1   | 0.03938 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(R_m * T) / MC$ | SM 1   | 0.05122 | (A,B,C,D) |
| $M(x) * \text{SQRT}(R_m * T) / MC$     | SM 1   | 0.06969 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(R_m * T) / ML$ | SM 1   | 0.05122 | (A,B,C,D) |
| $M(x) * \text{SQRT}(R_m * T) / ML$     | SM 1   | 0.06969 | (A,B,C,D) |
| $N(y) * T / P$                         | SP 1   | 0.07060 | (A,B,C,D) |
| $M(y) / P$                             | SP 1   | 0.01254 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(R_m * T) / MC$ | SM 1   | 0.06996 | (A,B,C,D) |
| $M(y) * \text{SQRT}(R_m * T) / MC$     | SM 1   | 0.02614 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(R_m * T) / ML$ | SM 1   | 0.06996 | (A,B,C,D) |
| $M(y) * \text{SQRT}(R_m * T) / ML$     | SM 1   | 0.02614 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (N./mm2)

| Type of Stress | Load | Stress Intensity Values at |      |      |       |       |      |      |       |
|----------------|------|----------------------------|------|------|-------|-------|------|------|-------|
|                |      | Au                         | Al   | Bu   | Bl    | Cu    | Cl   | Du   | Dl    |
| Rad. Memb.     | P    | -0.1                       | -0.1 | -0.1 | -0.1  | -0.1  | -0.1 | -0.1 | -0.1  |
| Rad. Bend.     | P    | -0.8                       | 0.8  | -0.8 | 0.8   | -0.8  | 0.8  | -0.8 | 0.8   |
| Rad. Memb.     | MC   | 0.0                        | 0.0  | 0.0  | 0.0   | -1.8  | -1.8 | 1.8  | 1.8   |
| Rad. Bend.     | MC   | 0.0                        | 0.0  | 0.0  | 0.0   | -14.6 | 14.6 | 14.6 | -14.6 |
| Rad. Memb.     | ML   | -1.4                       | -1.4 | 1.4  | 1.4   | 0.0   | 0.0  | 0.0  | 0.0   |
| Rad. Bend.     | ML   | -11.7                      | 11.7 | 11.7 | -11.7 | 0.0   | 0.0  | 0.0  | 0.0   |
| Tot. Rad. Str. |      | -14.1                      | 10.9 | 12.2 | -9.6  | -17.3 | 13.5 | 15.5 | -12.2 |



|                 |      |      |      |      |      |      |      |      |
|-----------------|------|------|------|------|------|------|------|------|
| Tang. Memb. P   | -0.2 | -0.2 | -0.2 | -0.2 | -0.2 | -0.2 | -0.2 | -0.2 |
| Tang. Bend. P   | -0.3 | 0.3  | -0.3 | 0.3  | -0.3 | 0.3  | -0.3 | 0.3  |
| Tang. Memb. MC  | 0.0  | 0.0  | 0.0  | 0.0  | -2.4 | -2.4 | 2.4  | 2.4  |
| Tang. Bend. MC  | 0.0  | 0.0  | 0.0  | 0.0  | -5.5 | 5.5  | 5.5  | -5.5 |
| Tang. Memb. ML  | -2.0 | -2.0 | 2.0  | 2.0  | 0.0  | 0.0  | 0.0  | 0.0  |
| Tang. Bend. ML  | -4.4 | 4.4  | 4.4  | -4.4 | 0.0  | 0.0  | 0.0  | 0.0  |
| Tot. Tang. Str. | -6.8 | 2.4  | 5.9  | -2.4 | -8.4 | 3.1  | 7.4  | -3.0 |
| Shear VC        | 0.3  | 0.3  | -0.3 | -0.3 | 0.0  | 0.0  | 0.0  | 0.0  |
| Shear VL        | 0.0  | 0.0  | 0.0  | 0.0  | -0.3 | -0.3 | 0.3  | 0.3  |
| Shear MT        | 0.7  | 0.7  | 0.7  | 0.7  | 0.7  | 0.7  | 0.7  | 0.7  |
| Tot. Shear      | 1.1  | 1.1  | 0.4  | 0.4  | 0.4  | 0.4  | 1.1  | 1.1  |
| Str. Int.       | 14.2 | 11.1 | 12.2 | 9.6  | 17.4 | 13.5 | 15.6 | 12.3 |

### WRC 107/537 Stress Summations:

#### Vessel Stress Summation at Attachment Junction (N./mm2)

| Type of Stress  | Load     | Stress Intensity Values at |       |       |       |       |       |       |       |
|-----------------|----------|----------------------------|-------|-------|-------|-------|-------|-------|-------|
|                 |          | Au                         | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Rad.            | Pm (SUS) | 161.2                      | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 |
| Rad.            | Pl (SUS) | -1.6                       | -1.6  | 1.3   | 1.3   | -1.9  | -1.9  | 1.7   | 1.7   |
| Rad.            | Q (SUS)  | -12.5                      | 12.5  | 10.9  | -10.9 | -15.4 | 15.4  | 13.8  | -13.8 |
| Long.           | Pm (SUS) | 161.2                      | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 |
| Long.           | Pl (SUS) | -2.2                       | -2.2  | 1.7   | 1.7   | -2.7  | -2.7  | 2.2   | 2.2   |
| Long.           | Q (SUS)  | -4.6                       | 4.6   | 4.1   | -4.1  | -5.7  | 5.7   | 5.2   | -5.2  |
| Shear           | Pm (SUS) | 0.0                        | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear           | Pl (SUS) | 0.3                        | 0.3   | -0.3  | -0.3  | -0.3  | -0.3  | 0.3   | 0.3   |
| Shear           | Q (SUS)  | 0.7                        | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   |
| Pm (SUS)        |          | 161.2                      | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 | 161.2 |
| Pm+Pl (SUS)     |          | 159.8                      | 159.8 | 163.1 | 163.1 | 159.4 | 159.4 | 163.6 | 163.6 |
| Pm+Pl+Q (Total) |          | 154.5                      | 172.2 | 173.4 | 158.8 | 152.8 | 174.7 | 176.8 | 158.3 |

#### Vessel Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 161.18    | 161.90         | Passed |
| Pm+Pl (SUS)         | 163.56    | 242.85         | Passed |
| Pm+Pl+Q (TOTAL)     | 176.81    | 486.02         | Passed |

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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